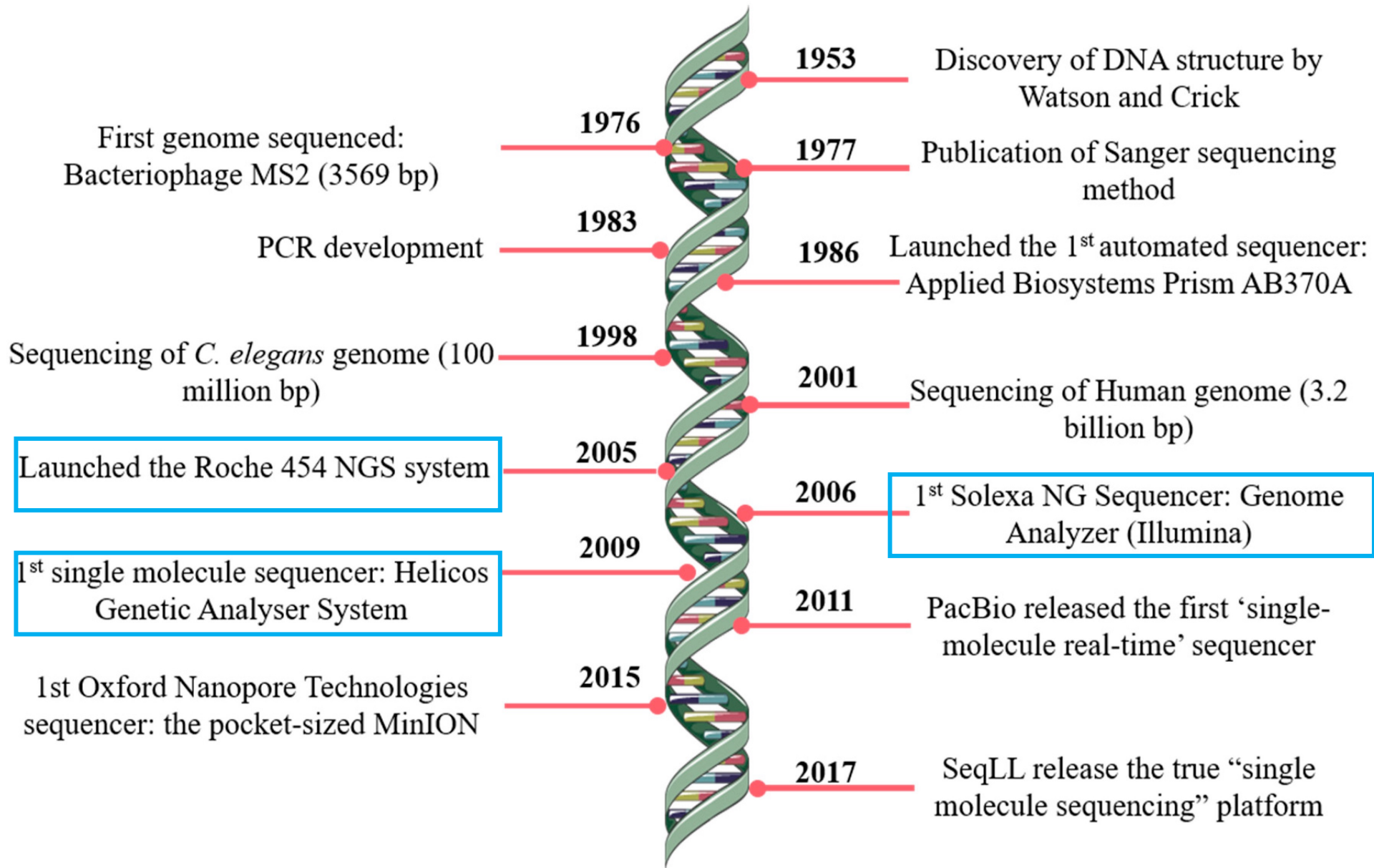


2nd generation sequencing

Chapter 3



Next-Gen Sequencing

- Millions of reactions performed in parallel
- Shorter read lengths, higher error rate
- Sample/library prep is required
- Many different approaches
 - Illumina sequencing-by-synthesis (Solexa technology)
 - Roche 454 pyrosequencing
 - AB SOLID color-based sequencing by ligation
 - Ion Torrent semiconductor sequencing

Main context

- 454 pyrosequencing



- SOLID sequencing



- Illumina sequencing

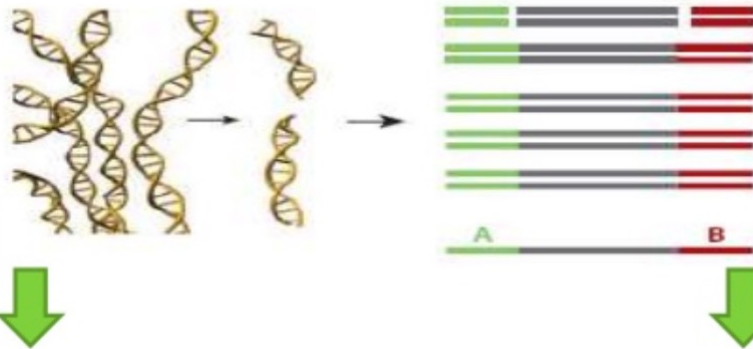


- Ion torrent sequencing

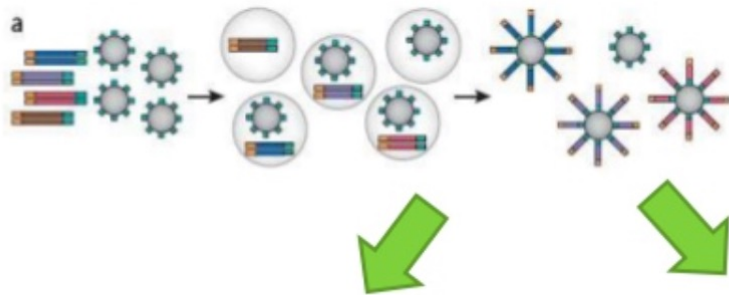


NGS WORKFLOW

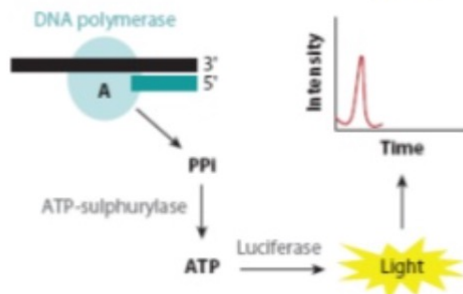
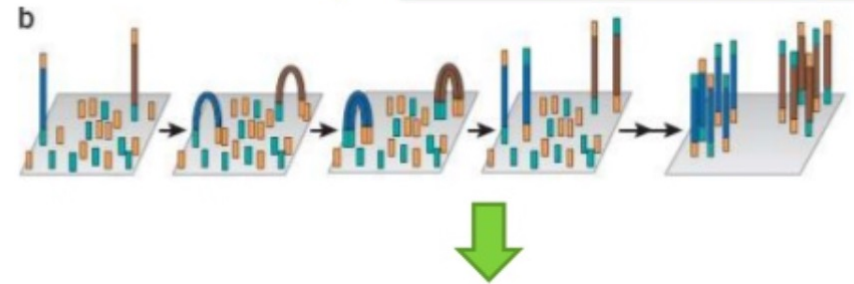
Sample Extraction , DNA fragmentation and *invitro* adapter ligation



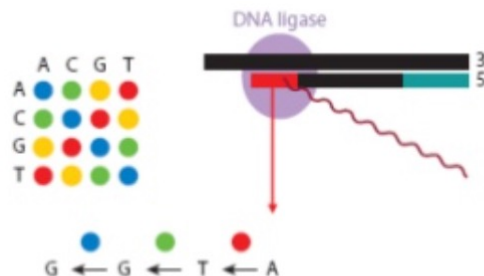
Clonal Amplification by Emulsion PCR



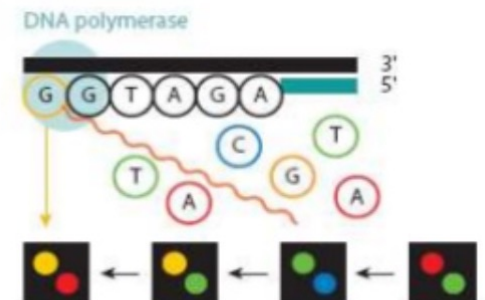
Clonal Amplification by Bridge PCR



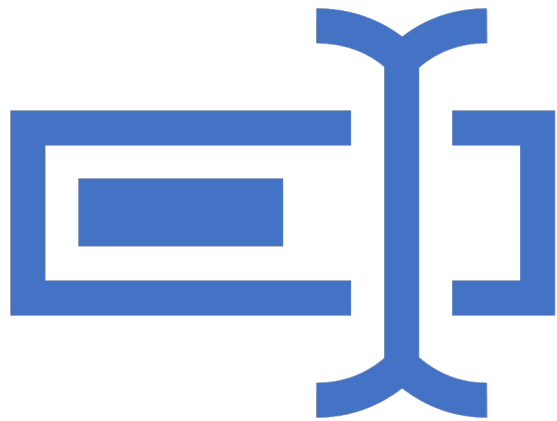
Pyrosequencing
(454 Sequencing)



Sequencing-by-ligation
(SOLiD Platform)



Sequencing-by-synthesis
(Solexa Technology)



Part One

Illumina sequencing platforms

Illumina sequencing machine

MiniSeq



MAX OUTPUT
8 Gb

MAX READ NUMBER
25 million

MAX READ LENGTH
2x150 bp

MiSeq



MAX OUTPUT
15 Gb

MAX READ NUMBER
25 million

MAX READ LENGTH
2x300 bp

NextSeq



MAX OUTPUT
120 Gb

MAX READ NUMBER
400 million

MAX READ LENGTH
2x150 bp

HiSeq 4000



MAX OUTPUT
1500 Gb

MAX READ NUMBER
5 billion

MAX READ LENGTH
2x150 bp

HiSeq X Ten



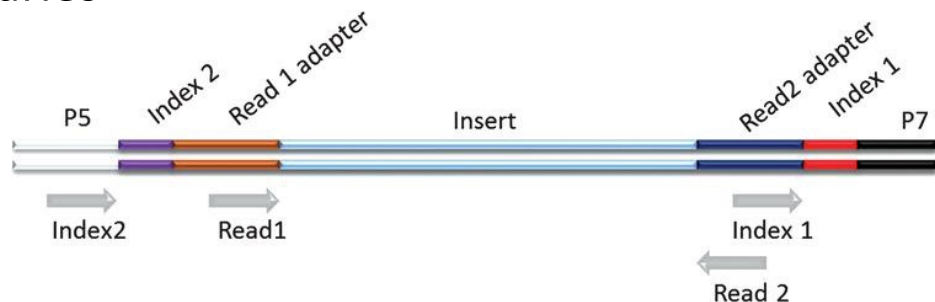
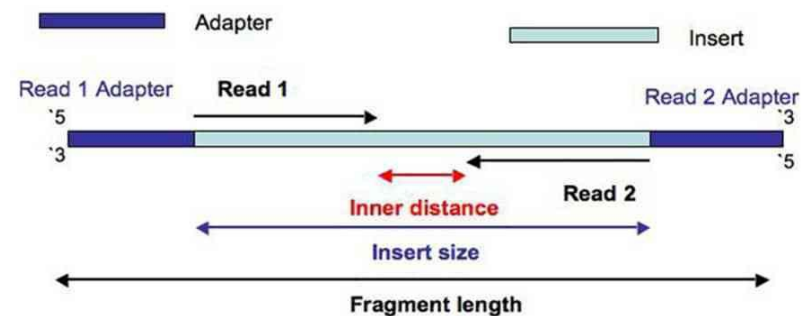
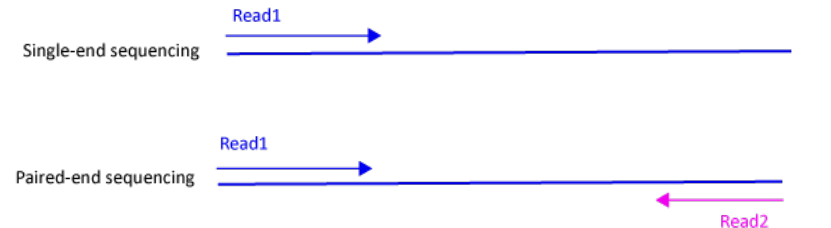
MAX OUTPUT
1800 Gb

MAX READ NUMBER
6 billion

MAX READ LENGTH
2x150 bp

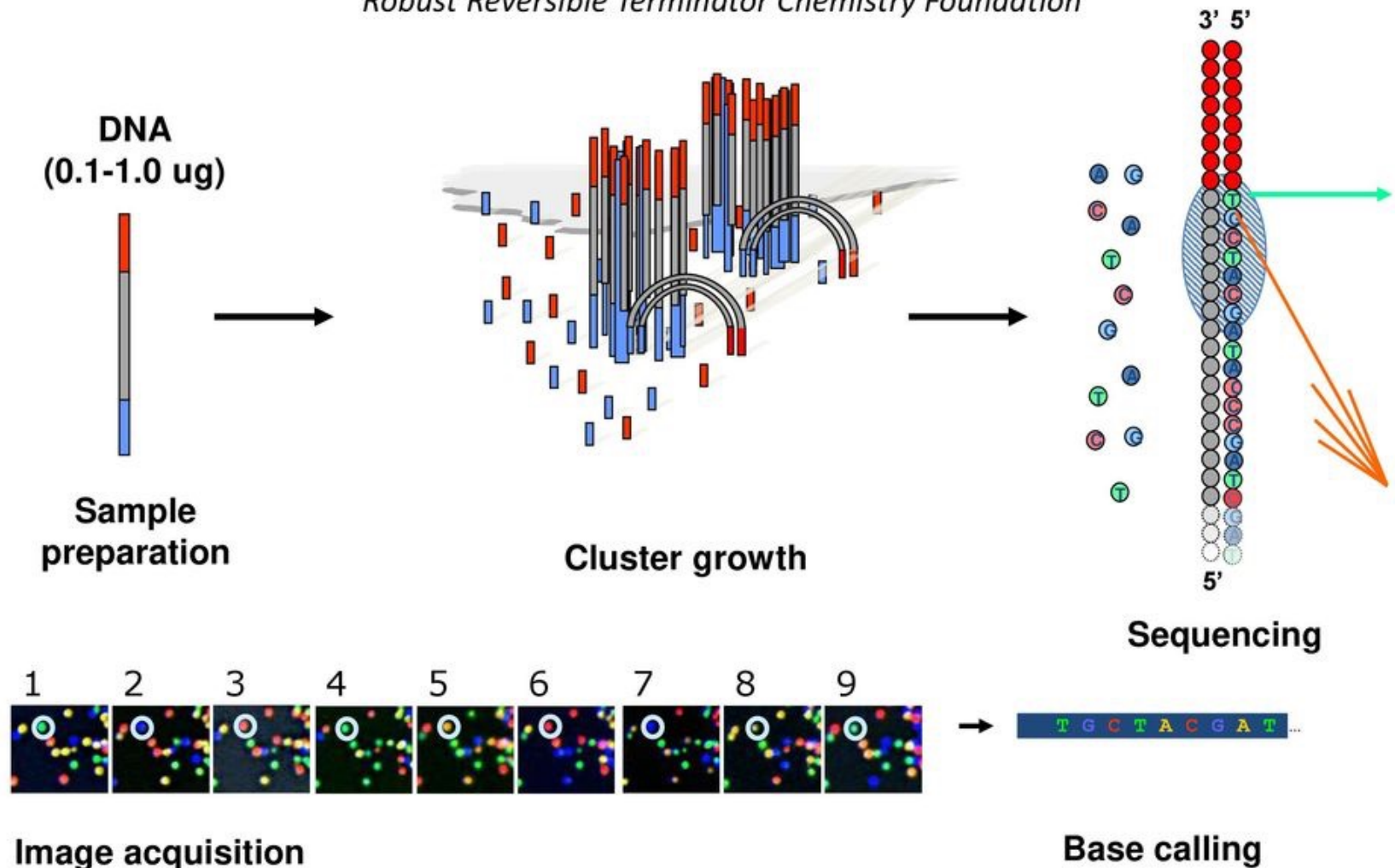
Some general terminology

- **SR**: single-read sequencing, sequence from only one end
- **PE**: paired-end sequencing, sequence from both ends
- **Adapters**: DNA added to the ends of DNA/RNA fragments to be sequenced. The adapters allow the DNA/RNA to attach to the flowcell
- **Index/barcode**: used interchangeable to indicate sequence identifier for multiplexing
- **PhiX**: commercially available genomic library of PhiX bacteriophage DNA, commonly spiked into libraries

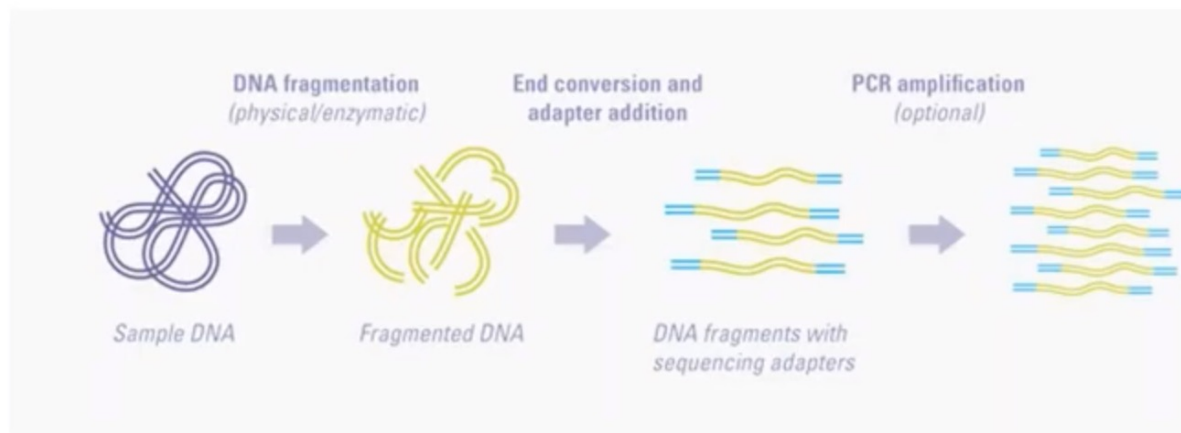


Illumina Sequencing Technology

Robust Reversible Terminator Chemistry Foundation



Library Prep is Critical for Successful Sequencing



For clustering:
Libraries must have P5 and P7 binding regions on either end of a library

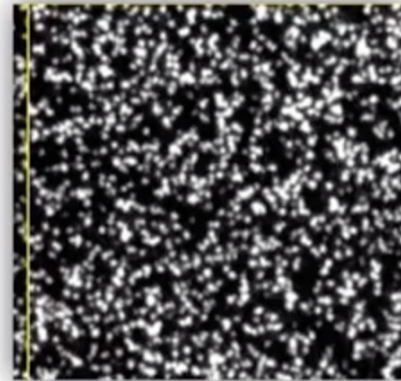
For sequencing:
Libraries must have sequencing primer binding regions

For mixing samples:
Libraries must have a unique index or barcodes sequence

What is a Cluster?

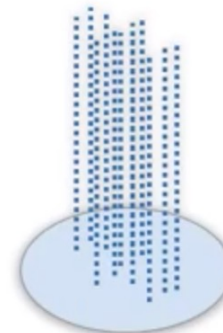
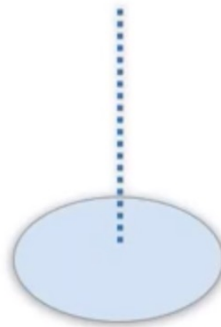
Clusters are a group of DNA strands positioned closely together

Each cluster represents thousands of copies of the same DNA strand in a 1–2 micron spot



An image of fluorescently labelled clusters on a flow cell

**Single
DNA
Library**



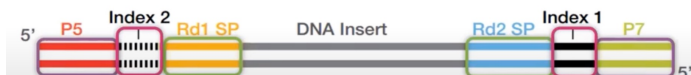
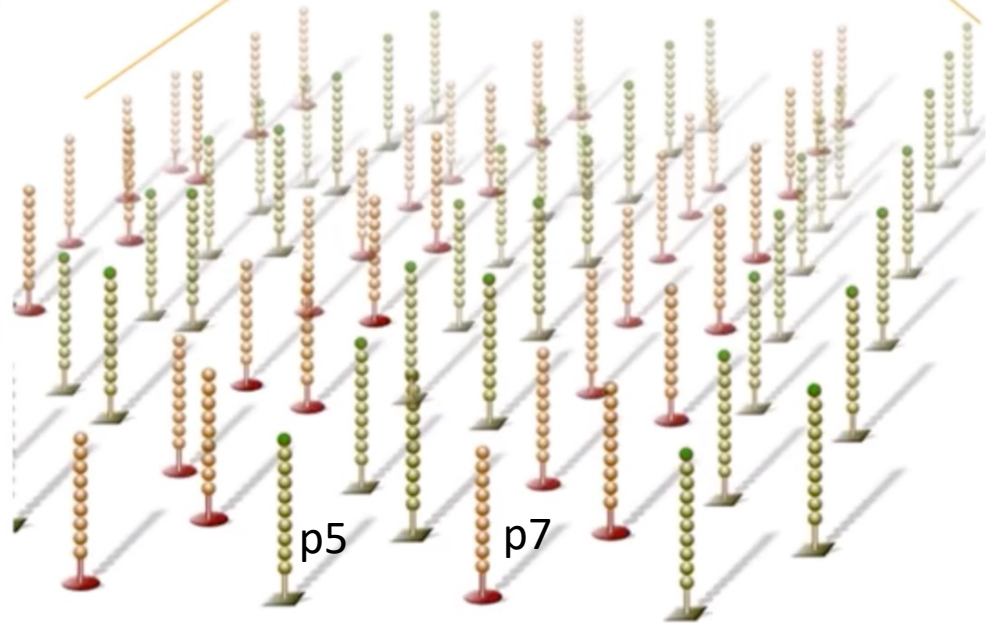
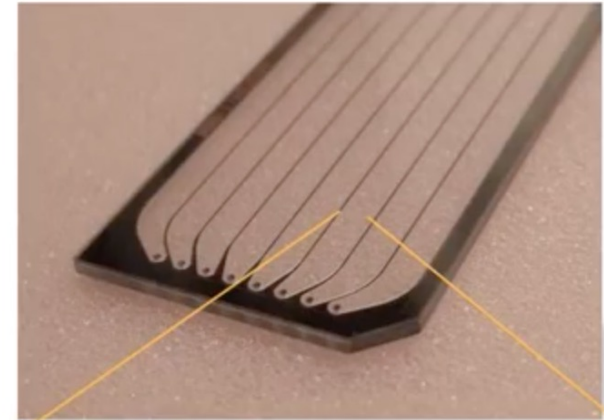
**Amplified
Clonal
Cluster**

What is a flow cell?

Cluster generation occurs on a flow cell

A flow cell is a thick glass slide with channels or lanes

Each lane is coated with a lawn of oligos complementary to library adapters



Hybridize Fragment & Extend

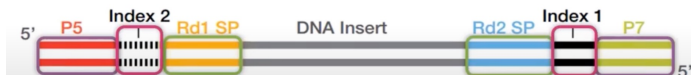
Single-stranded DNA libraries are hybridized to primer lawn

Bound libraries are then extended by polymerases

Surface of flow cell coated with a lawn of oligo pairs

Single-stranded DNA library template molecule

3' extension

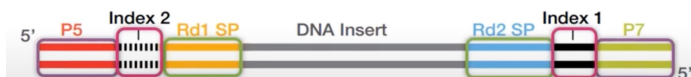
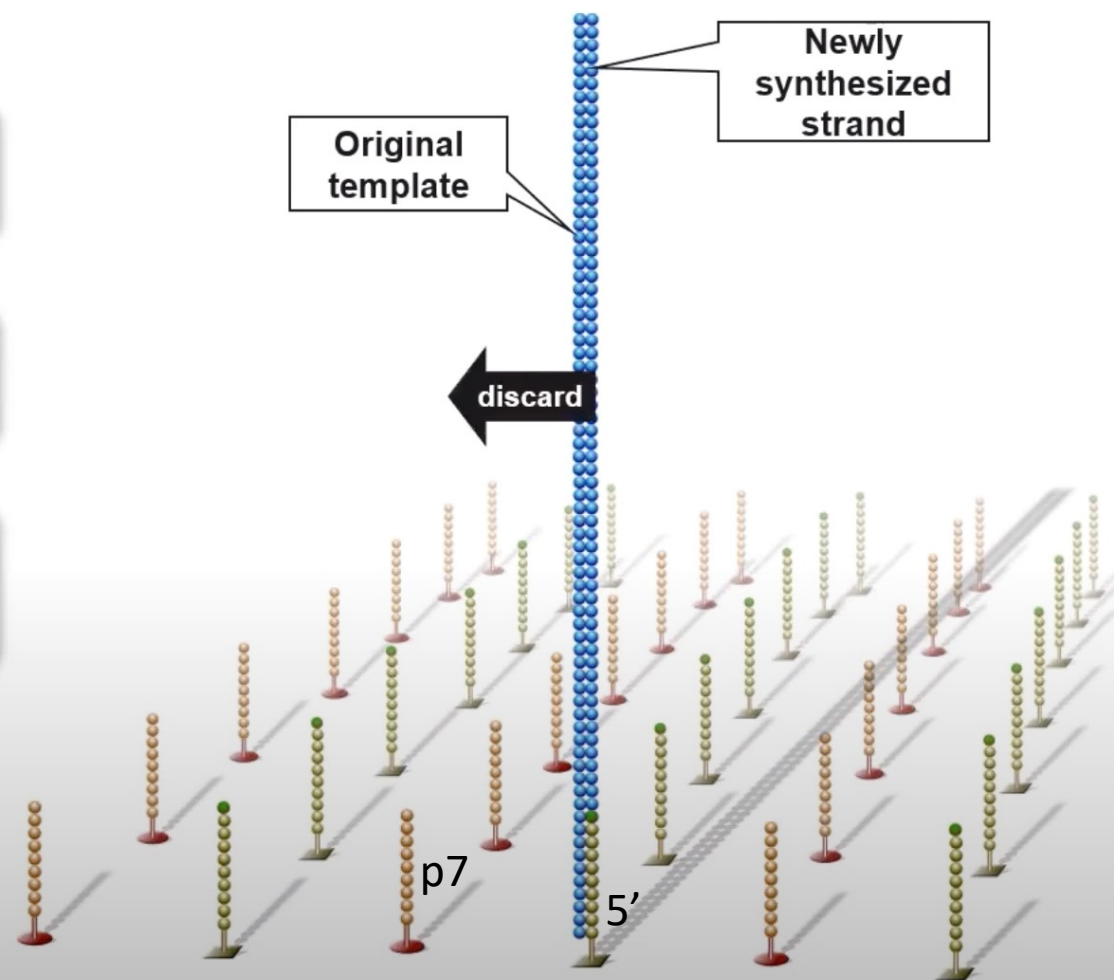


Denature Double-Stranded DNA

Double-stranded molecule is denatured

Original template washed away

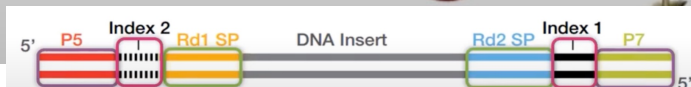
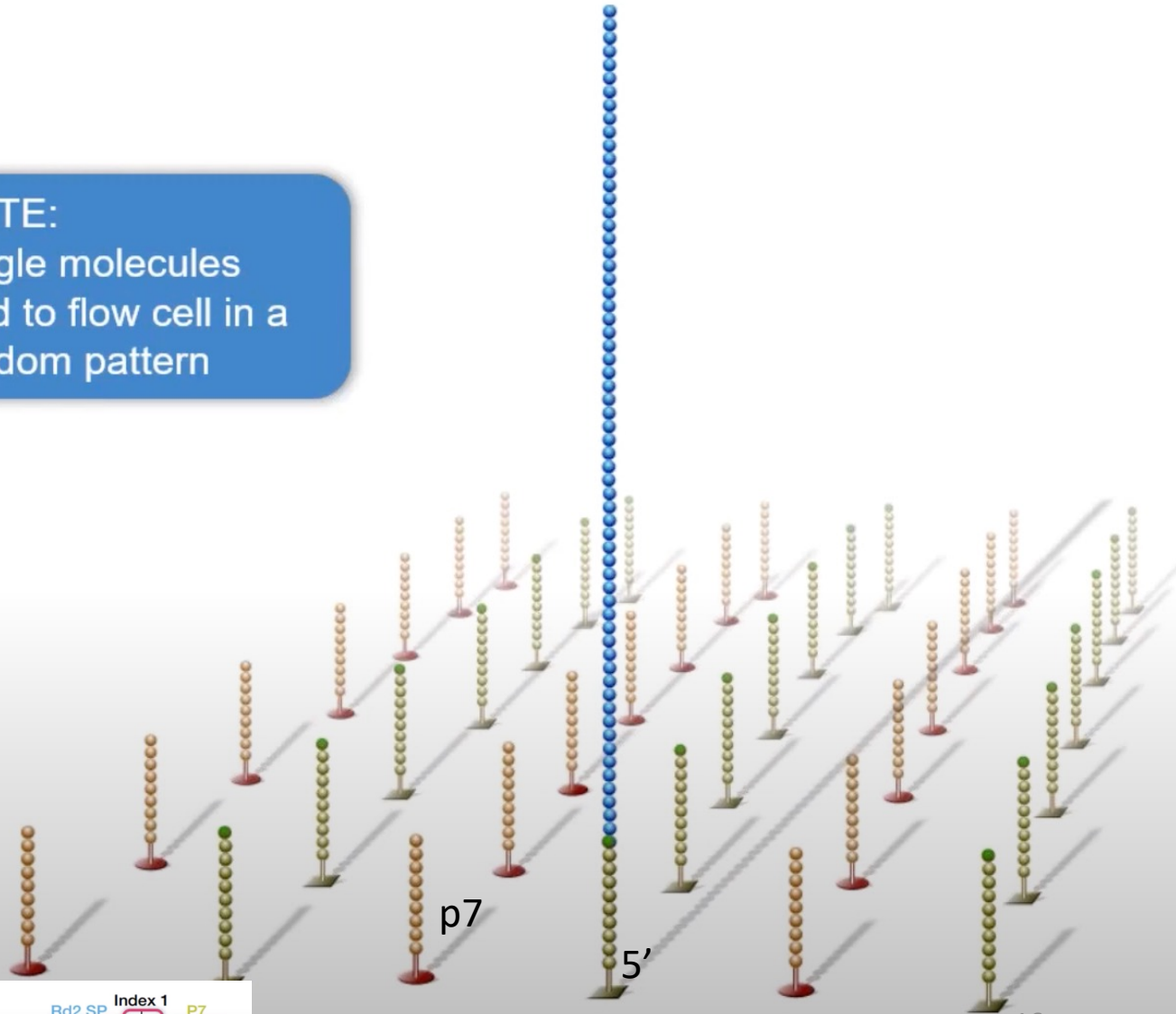
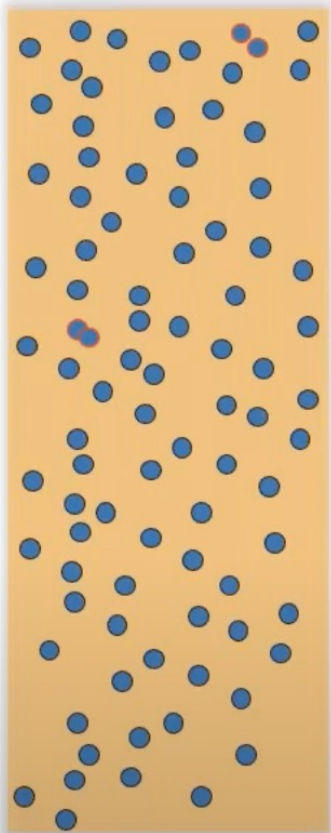
Newly synthesized strand is covalently attached to flow cell surface



Single-Stranded DNA



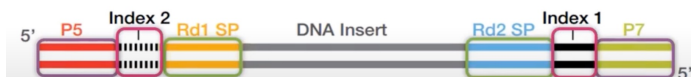
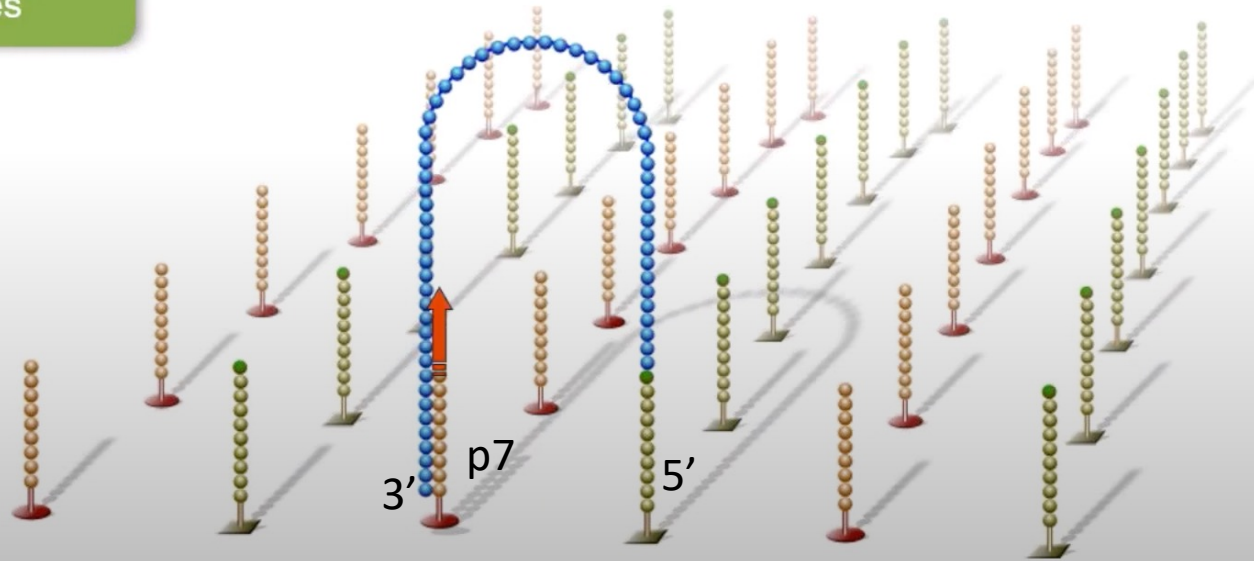
NOTE:
Single molecules
bind to flow cell in a
random pattern



Bridge Amplification

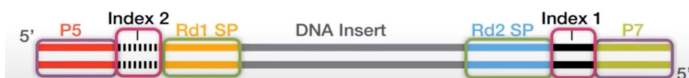
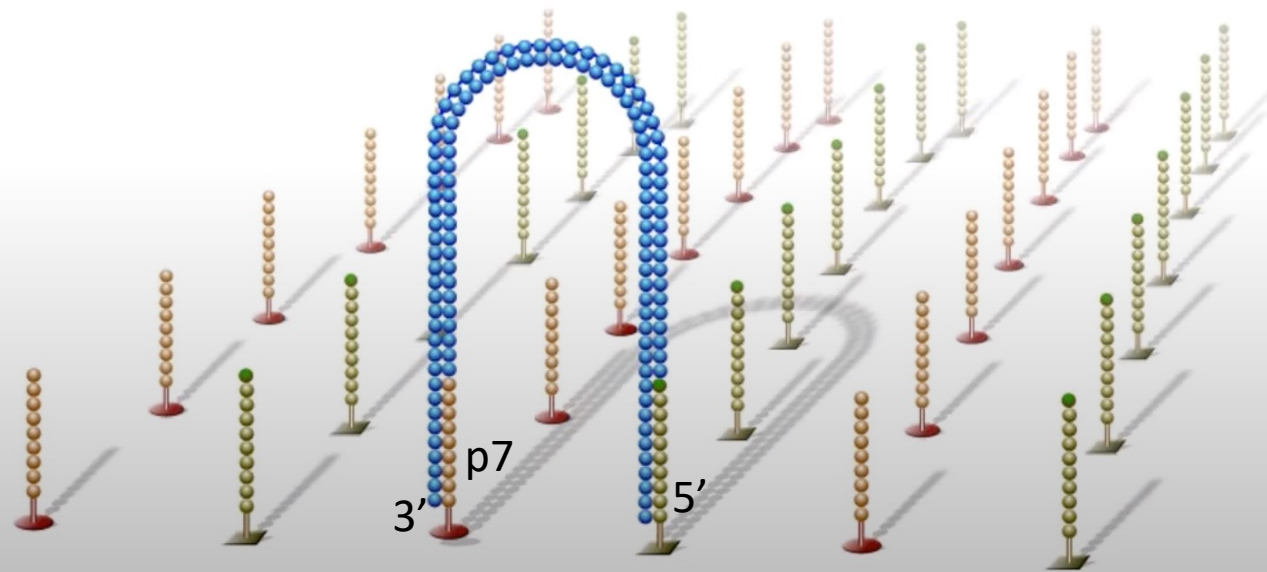
Single-stranded molecule flips over and forms a bridge by hybridizing to adjacent, complementary primer

Hybridized primer extends by polymerases



Bridge Amplification

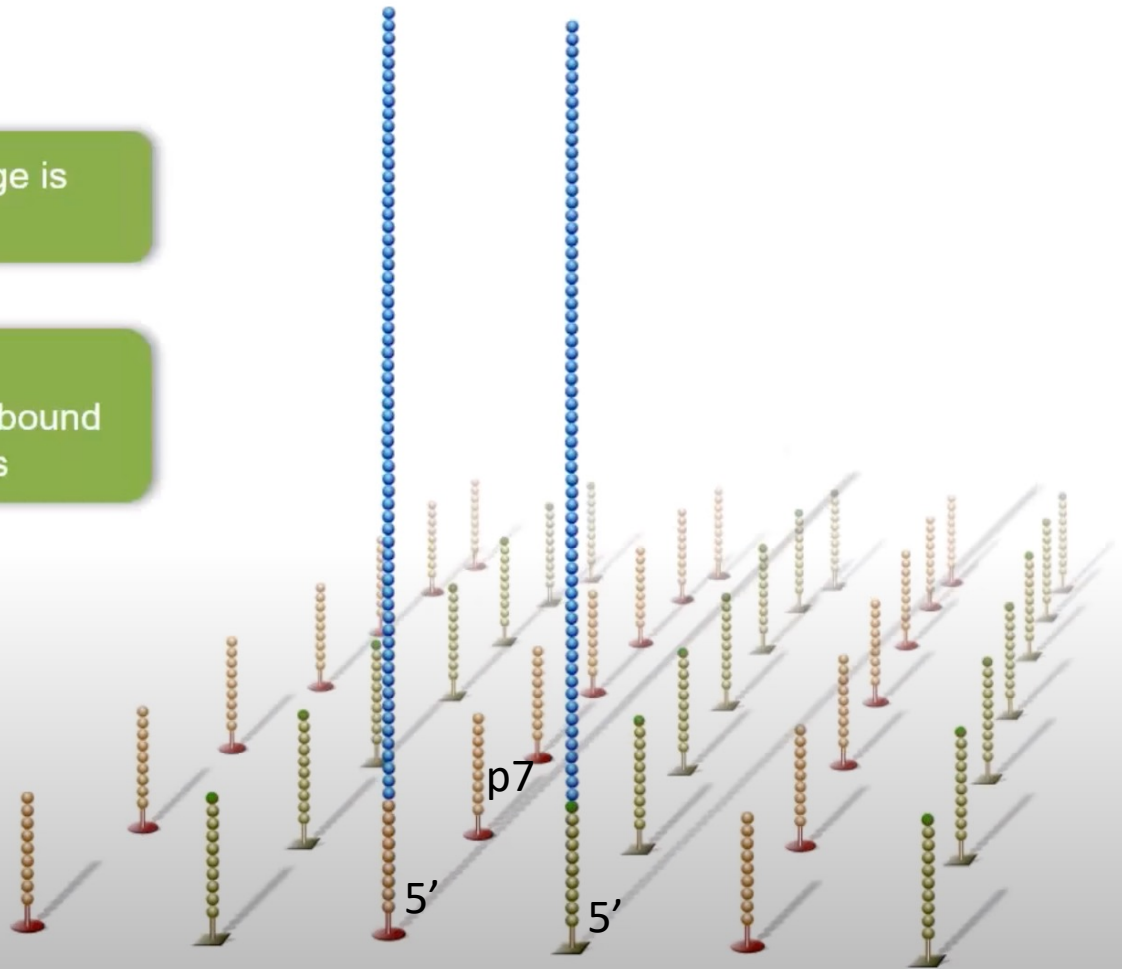
Double-stranded bridge is formed



Denature Double-Stranded Bridge

Double-stranded bridge is
denatured

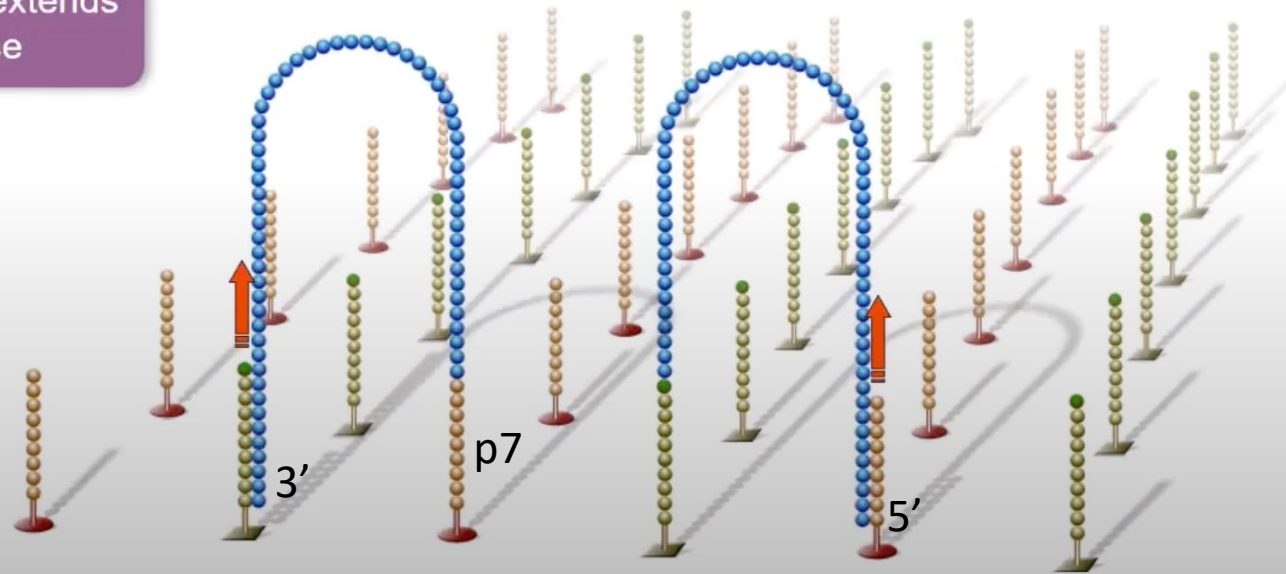
Result:
Two copies of covalently bound
single-stranded templates



Bridge Amplification

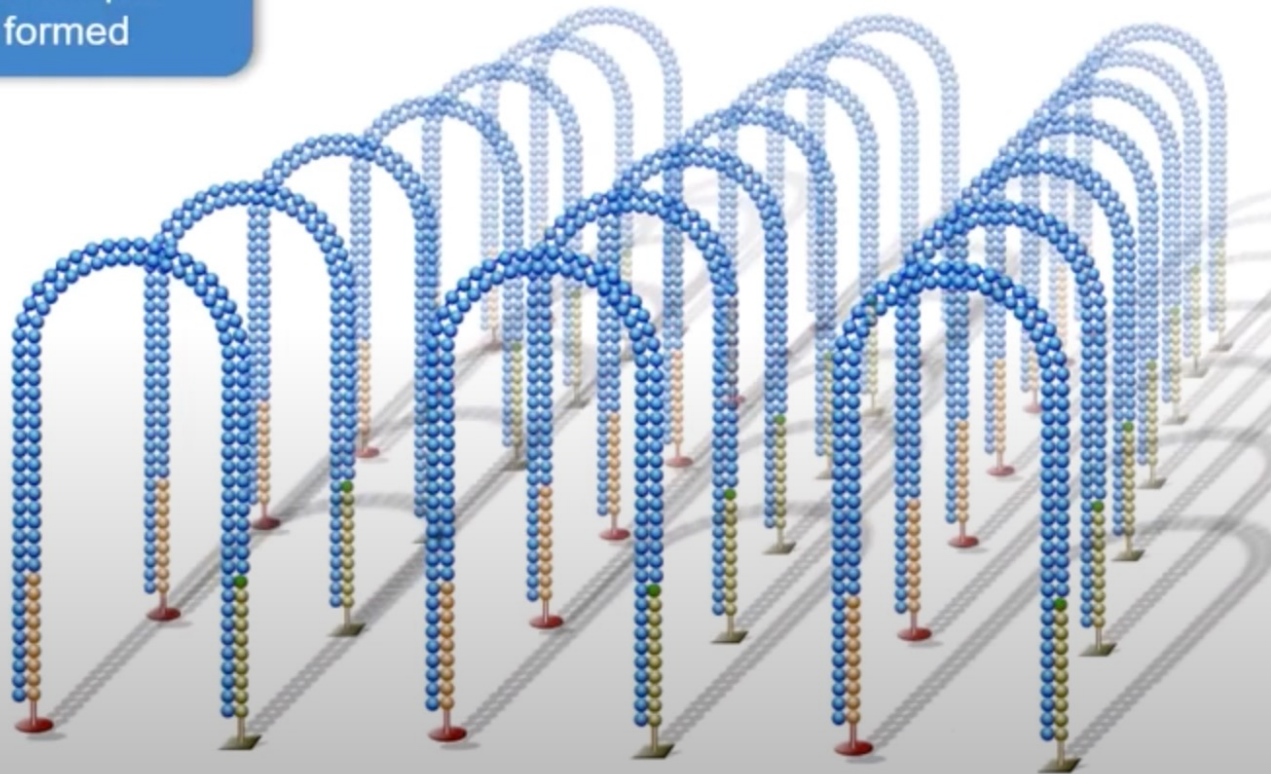
Single-stranded molecules flip over to hybridize to adjacent primers

Hybridized primer extends by polymerase



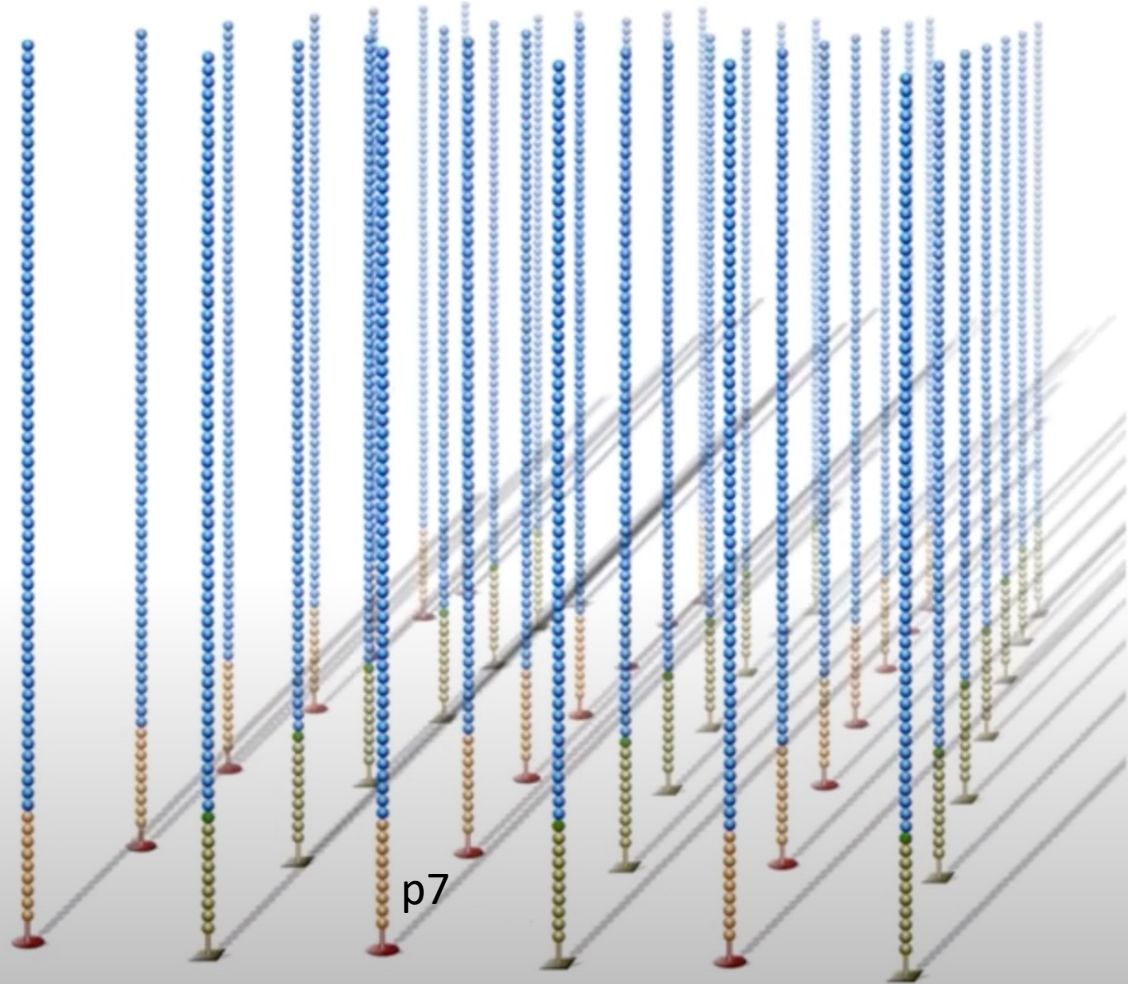
Bridge Amplification

Bridge amplification cycle is repeated until multiple bridges are formed



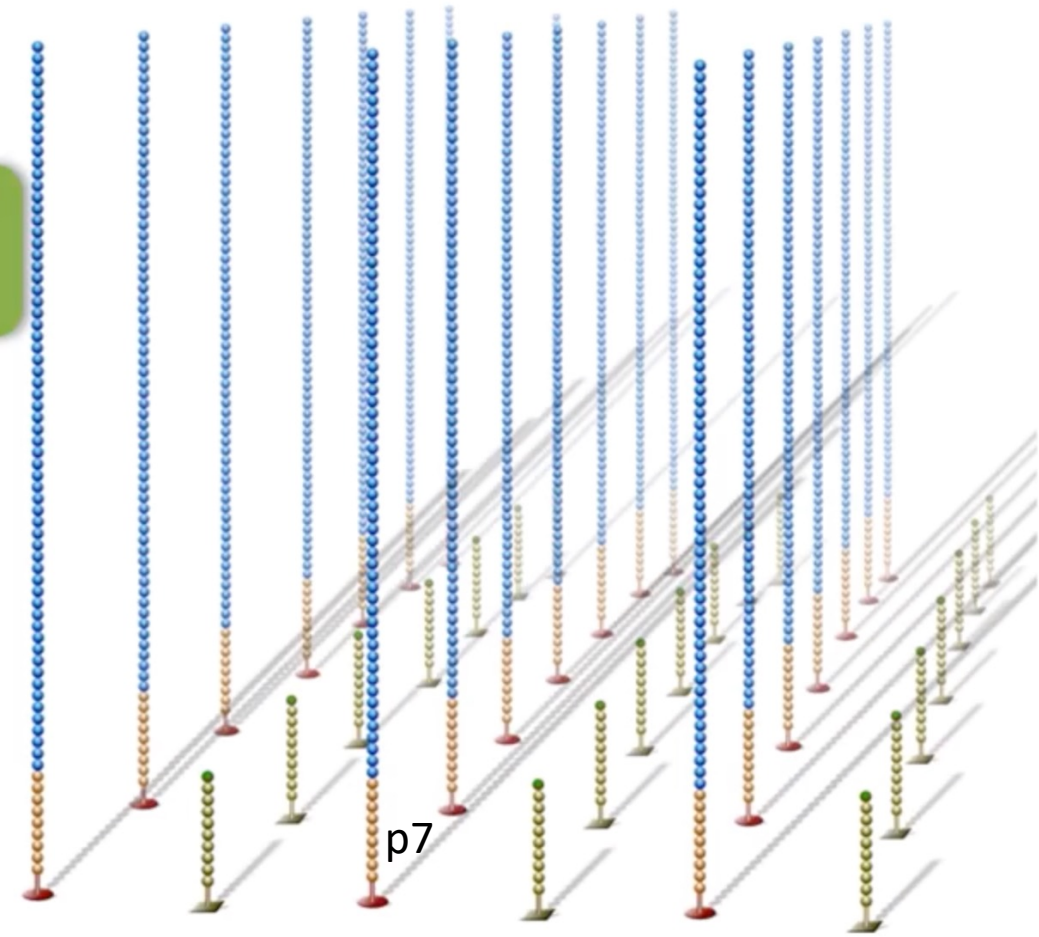
Linearization

dsDNA bridges are
denatured



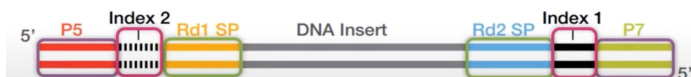
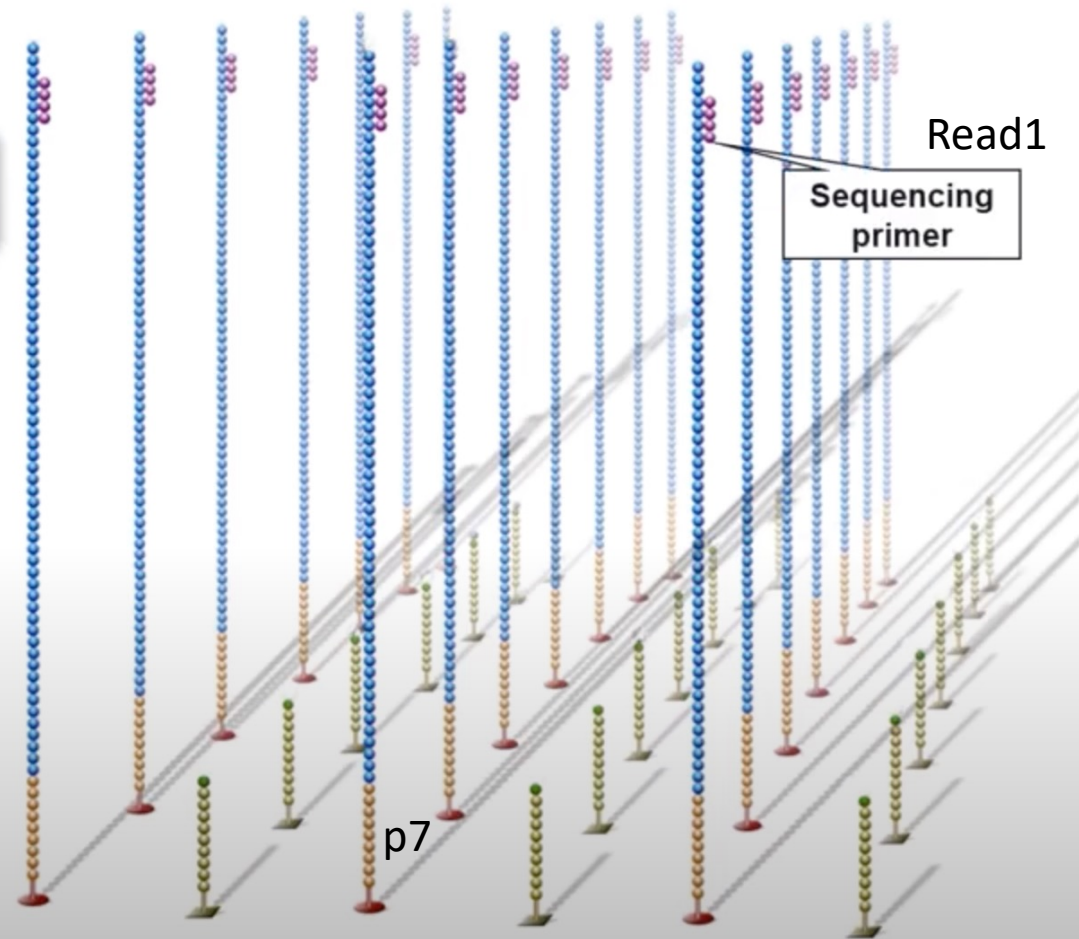
Reverse Strand Cleavage

Reverse strands are cleaved and washed away, leaving a cluster with forward strands only



Read 1 Primer Hybridization

Sequencing primer is hybridized to Read 1 sequencing primer binding site

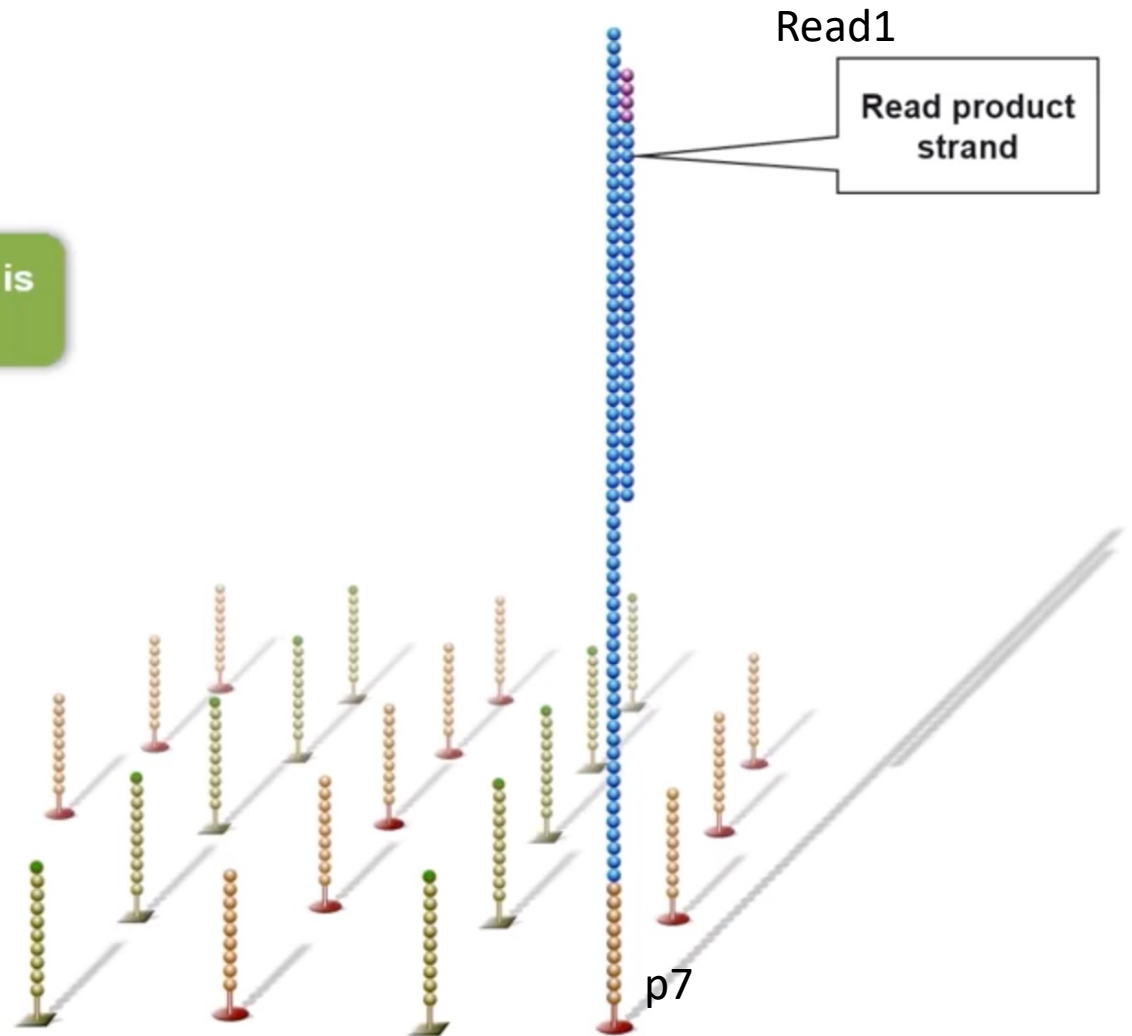


Paired-End Sequencing



Paired-End Sequencing

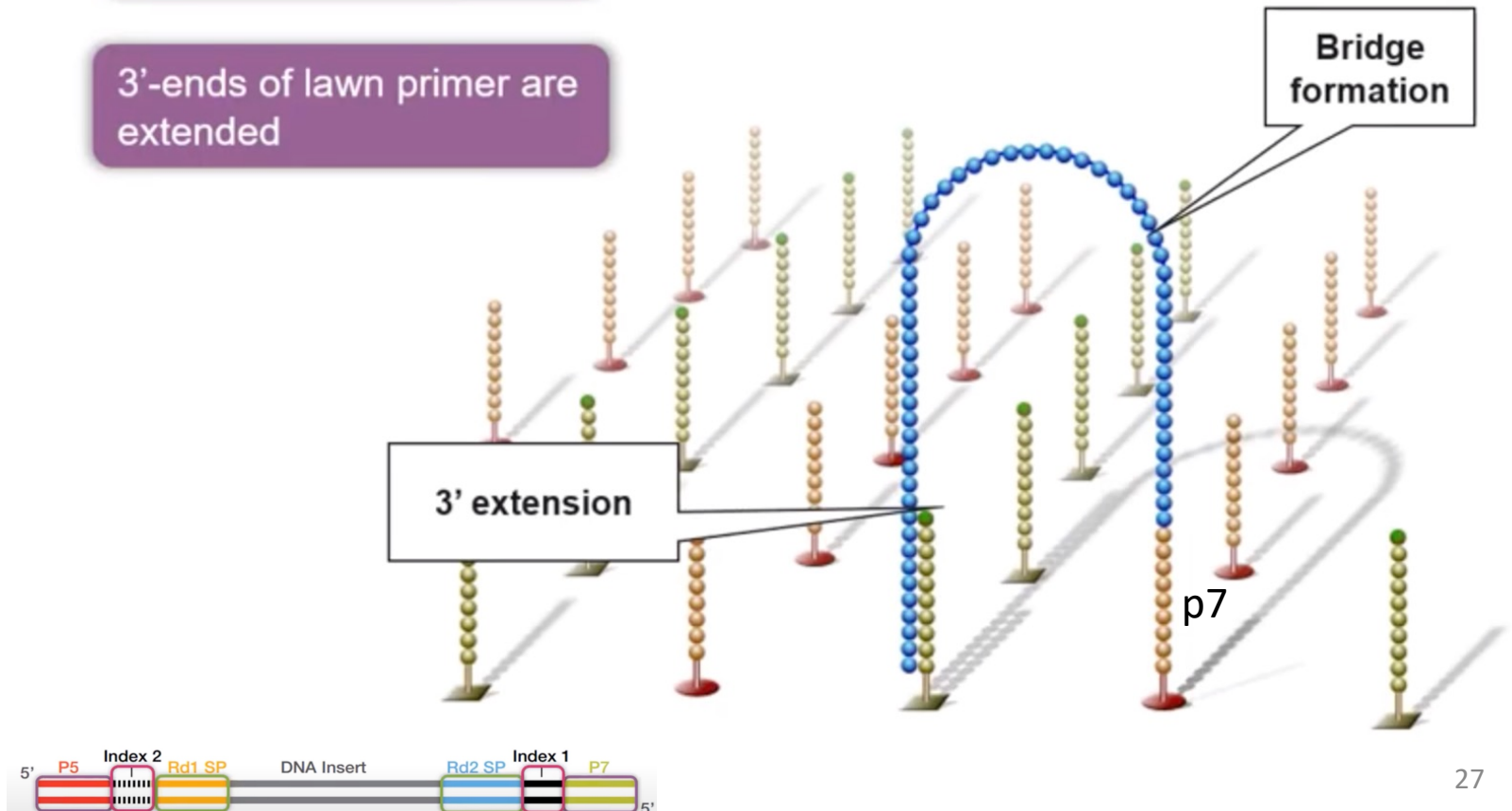
The read product is stripped off



Paired-End Sequencing

Single-stranded template loops over to form a bridge by hybridizing with a lawn primer

3'-ends of lawn primer are extended



The read product is stripped off

Read product strand

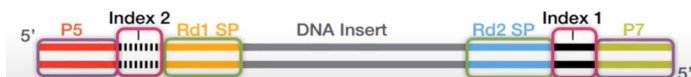
Single-stranded template loops over to form a bridge by hybridizing with a lawn primer

3'-ends of lawn primer are extended

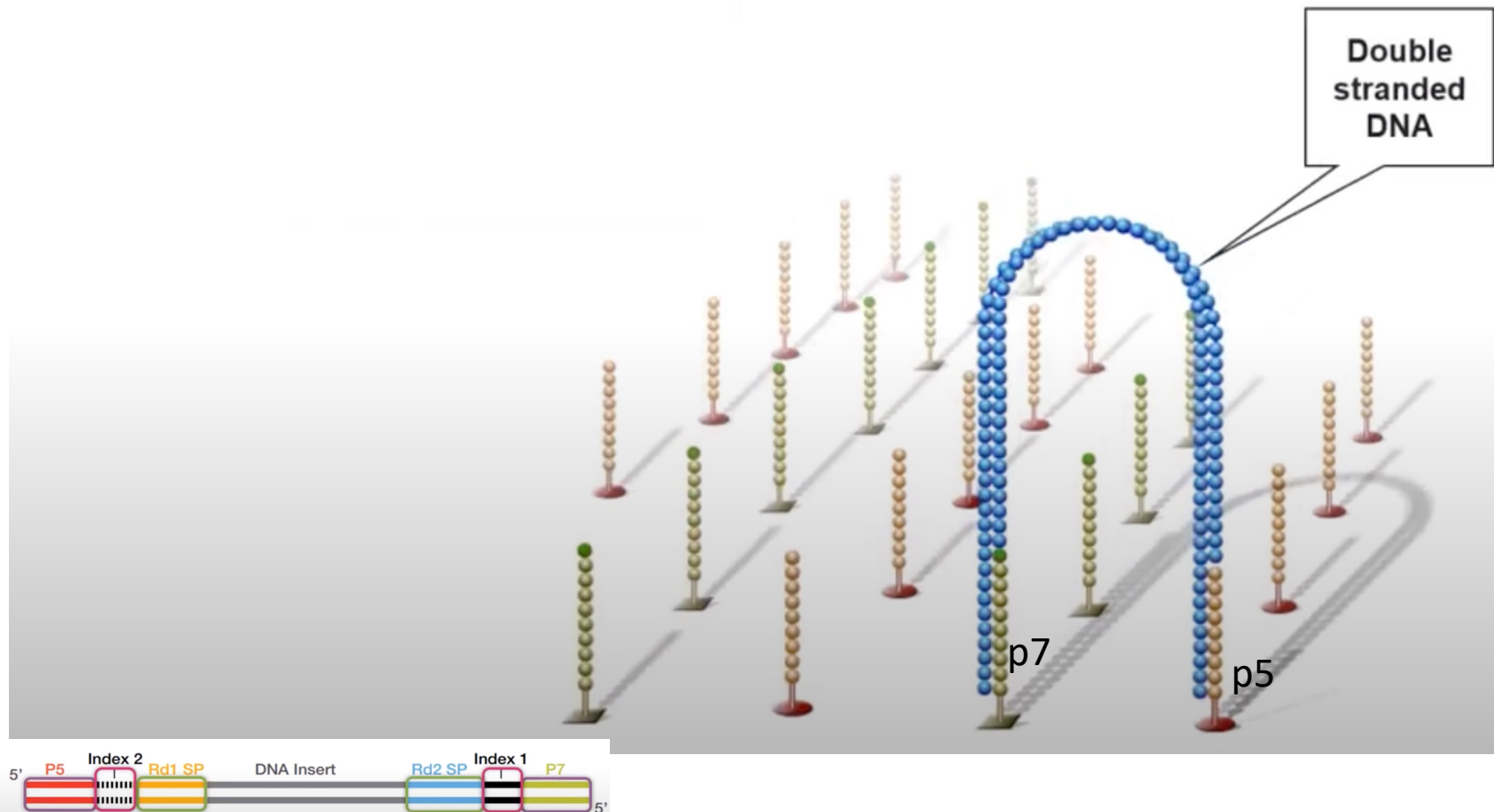
Bridge formation

3' extension

p7

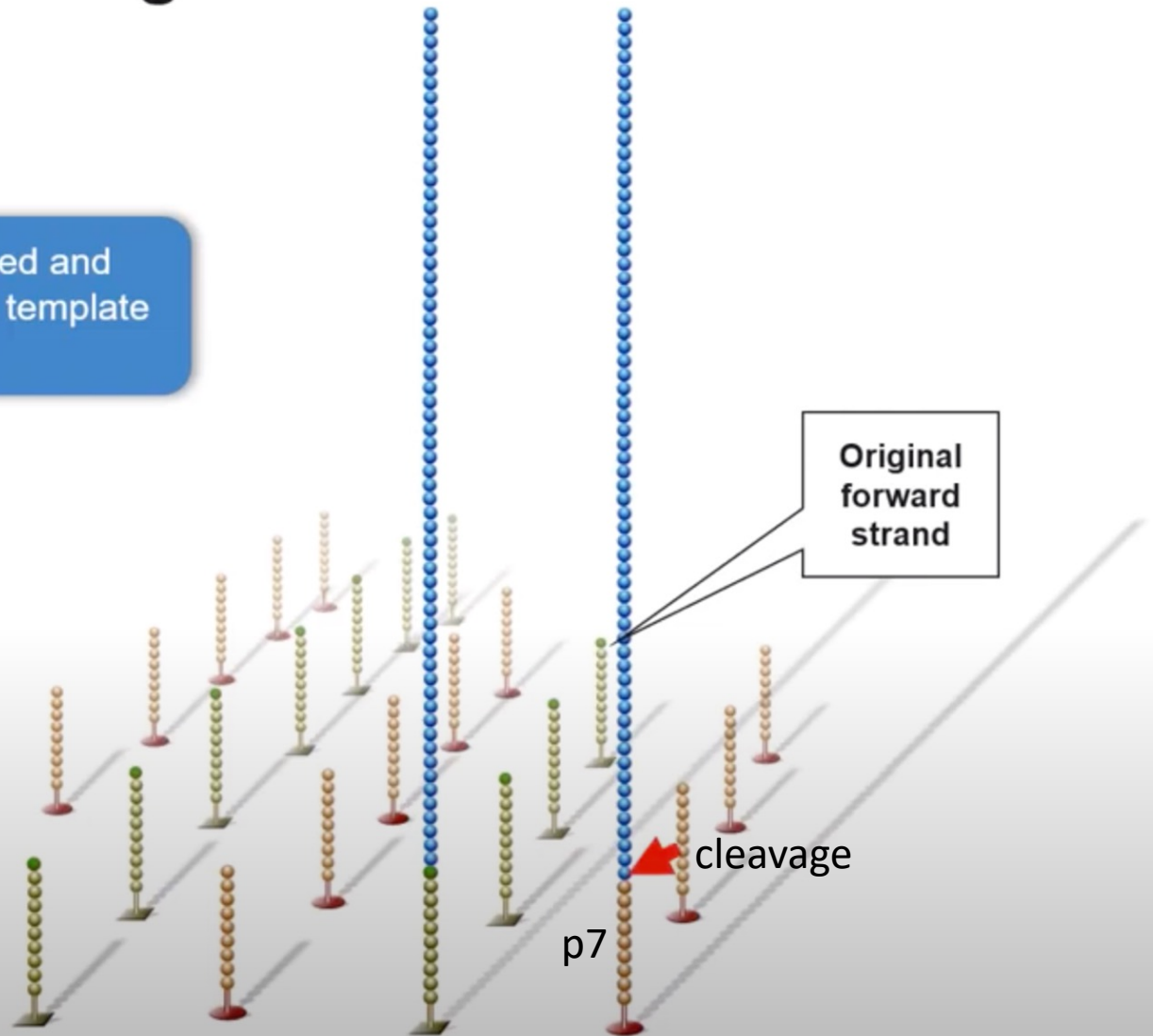


Paired-End Sequencing



Paired-End Sequencing

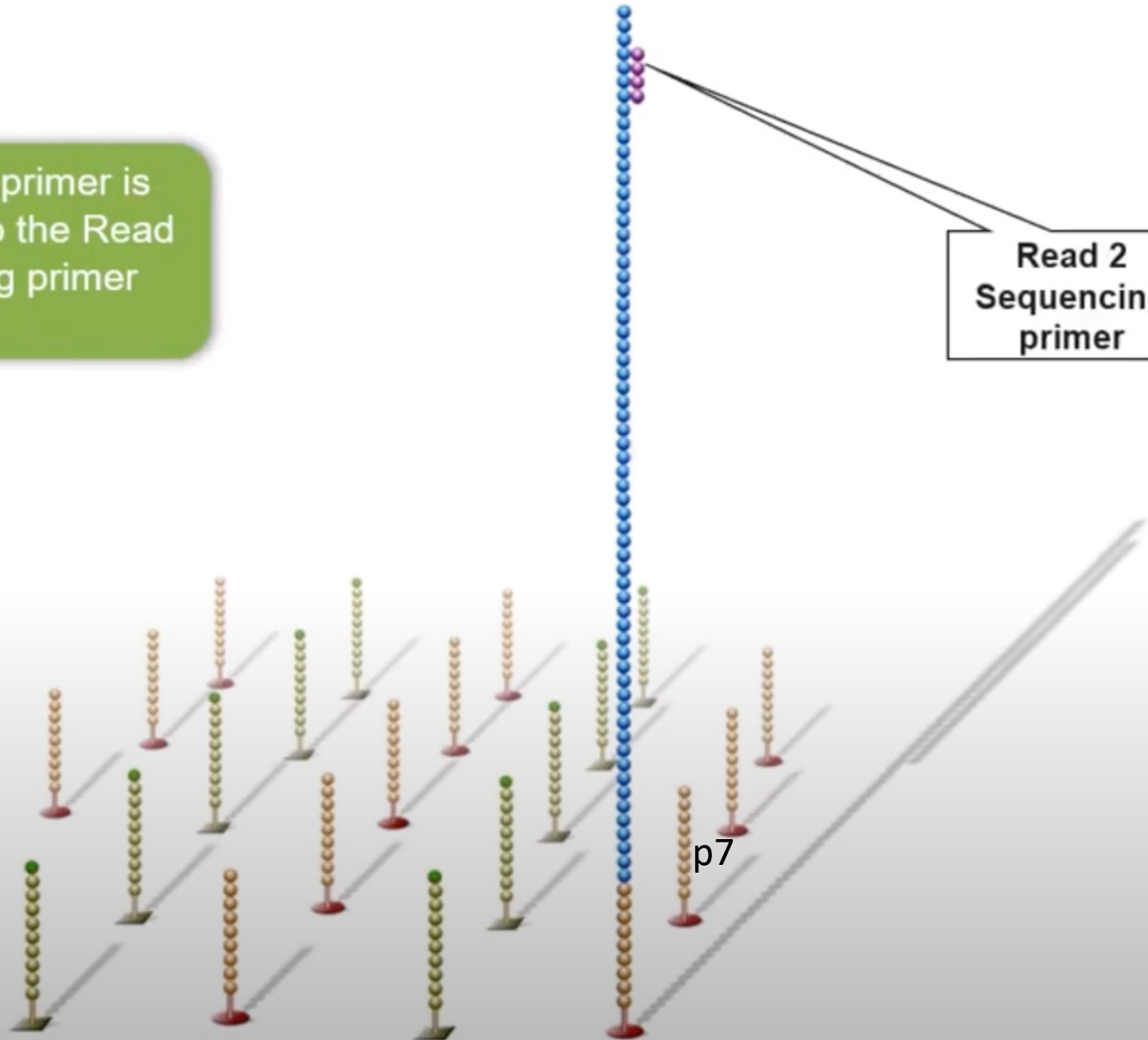
Bridges are linearized and the original forward template is cleaved



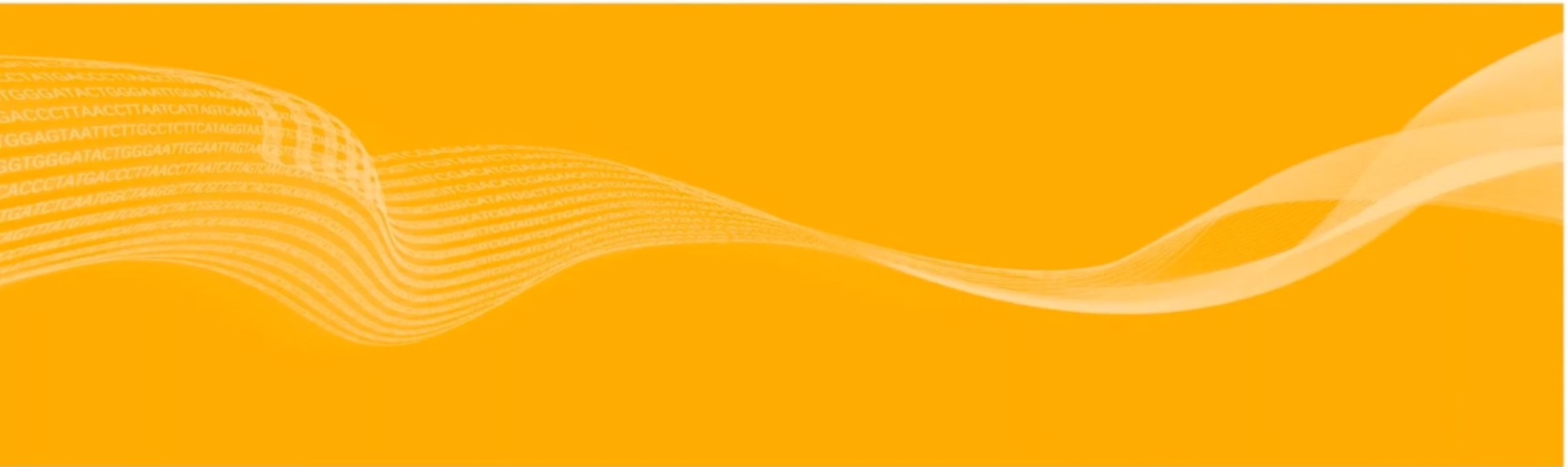
Paired-End Sequencing

Sequencing primer is hybridized to the Read 2 sequencing primer binding site

Read 2
Sequencing
primer

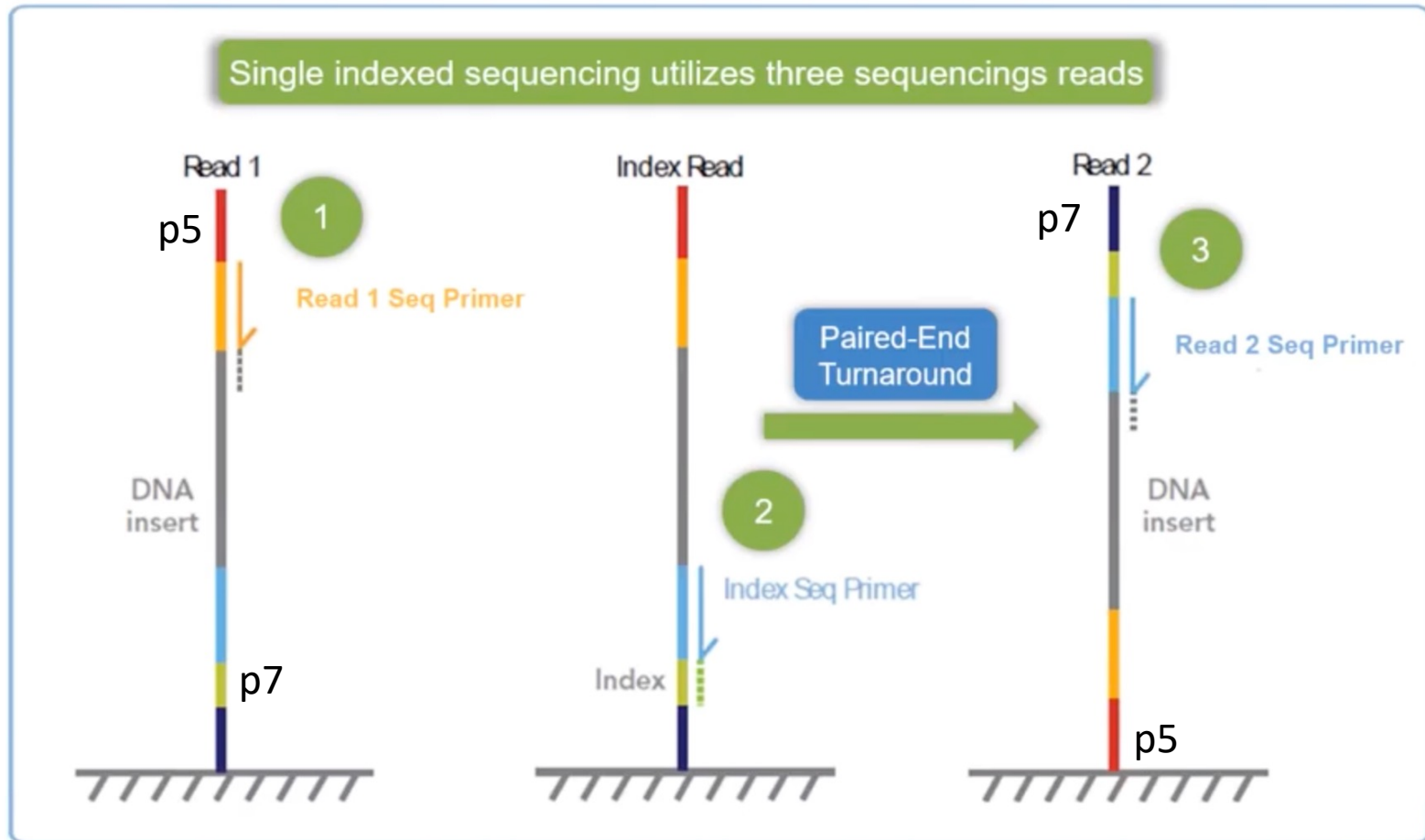


Sequencing with Index Reads



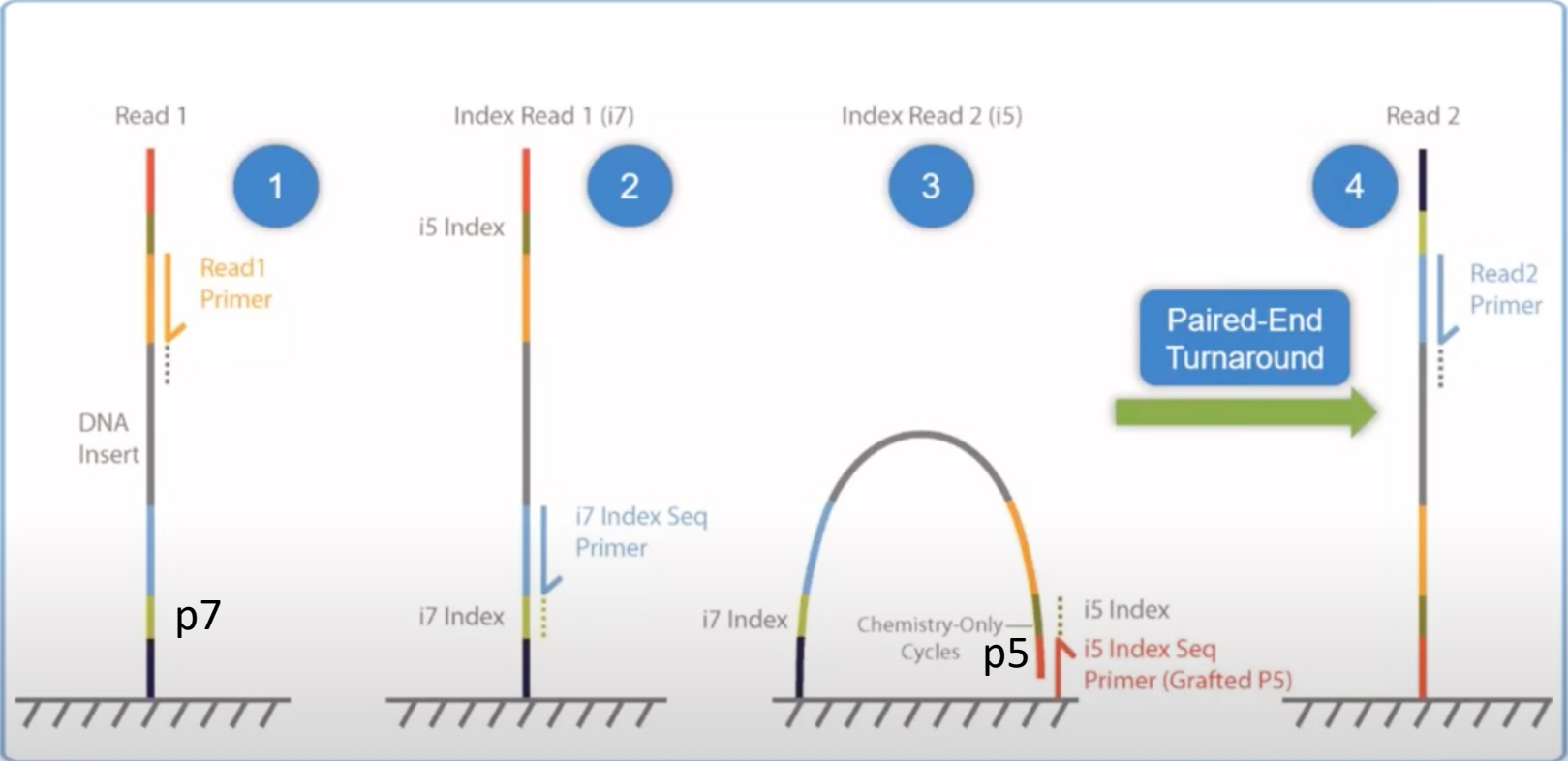
Single Index Reads

All Platforms



Dual Index Reads

MiSeq, HiSeq 2500, NovaSeq 6000

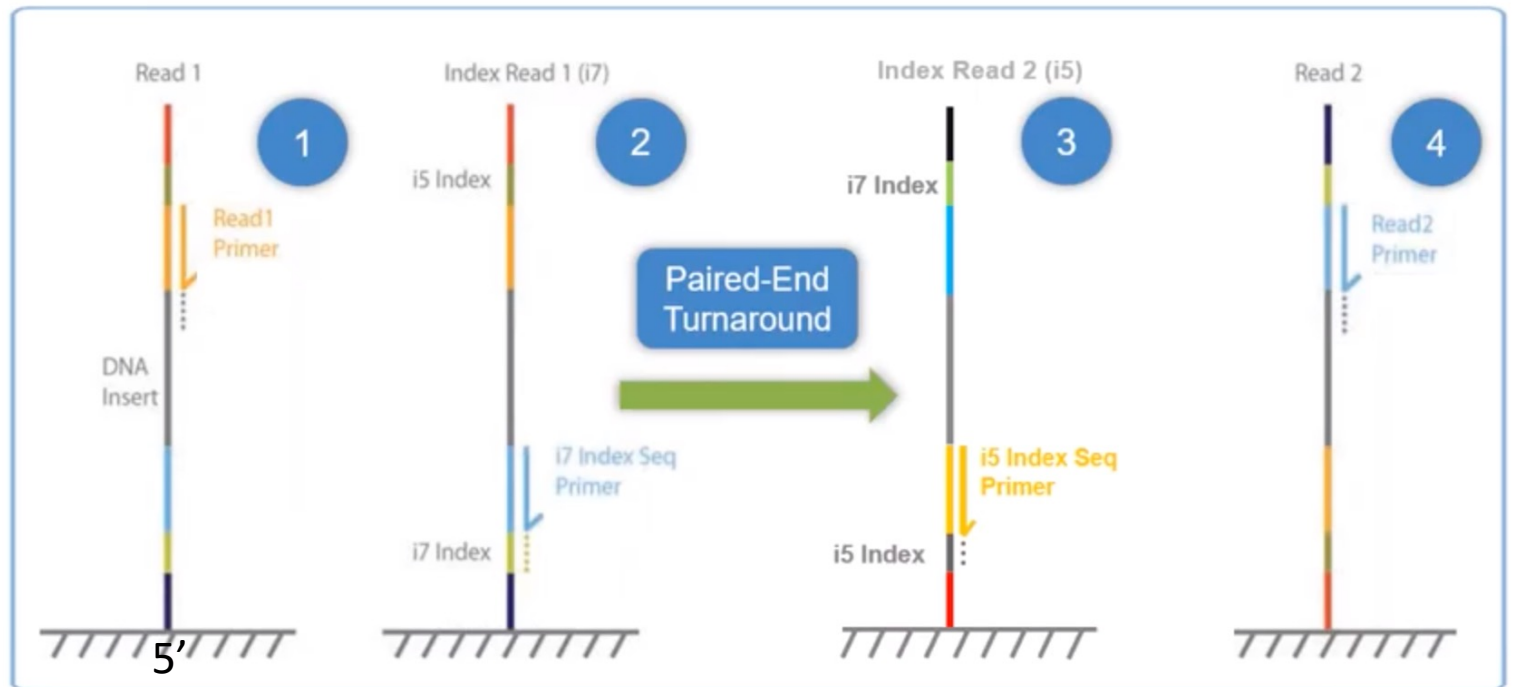


Dual indexed sequencing utilizes four sequencing reads



Dual Index Reads

*i*Seq 100, MiniSeq, NextSeq, HiSeq 3000/4000



Exclusion Amplification (ExAmp) Cluster Generation

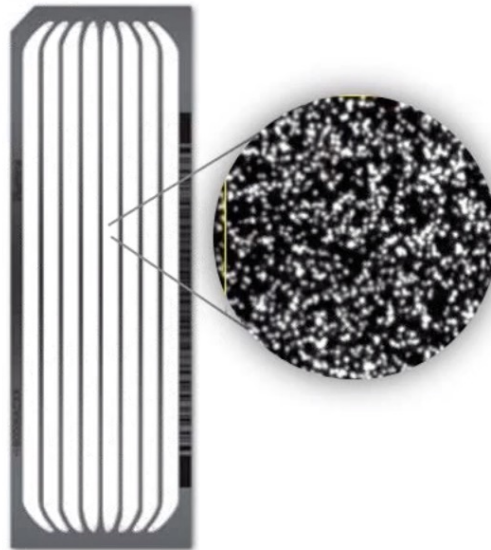


Flow cell Architecture

Random vs Patterned

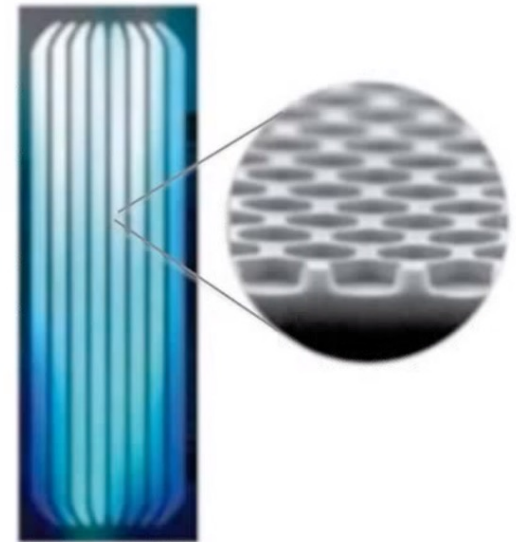
Random Flow Cell

- HiSeq™ 2500, MiSeq™, NextSeq™, MiniSeq™
- Randomly spaced clusters
- Variable Insert Sizes
- Lower Duplication Rates



Patterned Flow Cell

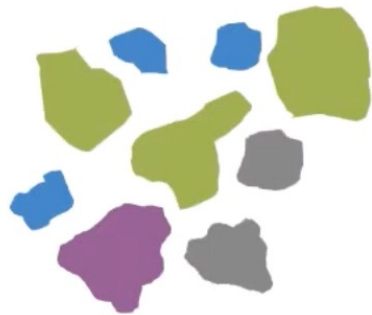
- HiSeq 3K/4K/X, NovaSeq™ 6000, iSeq™ 100
- Defined size and spacing
- Increased Cluster density
- Simplified imaging



For Research Use Only. Not for use in diagnostic procedures.

Patterned flow cells

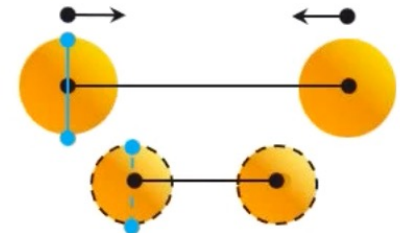
Complete control of pitch & feature size



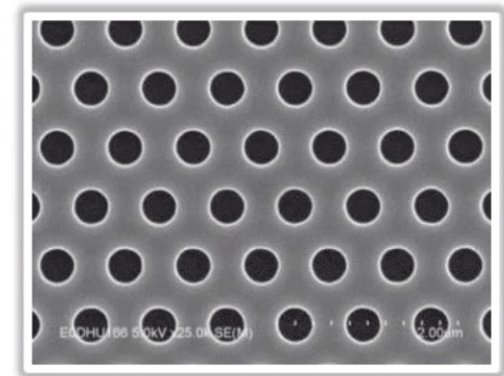
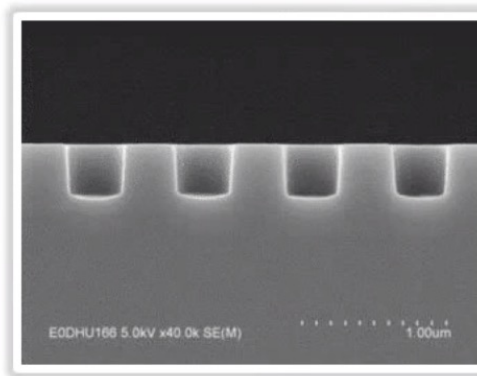
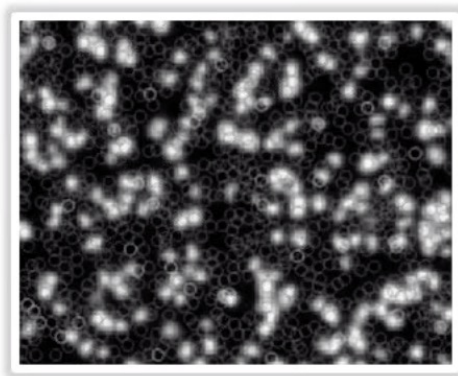
Pitch (Center to Center)



Decrease Pitch



Decrease Feature Size



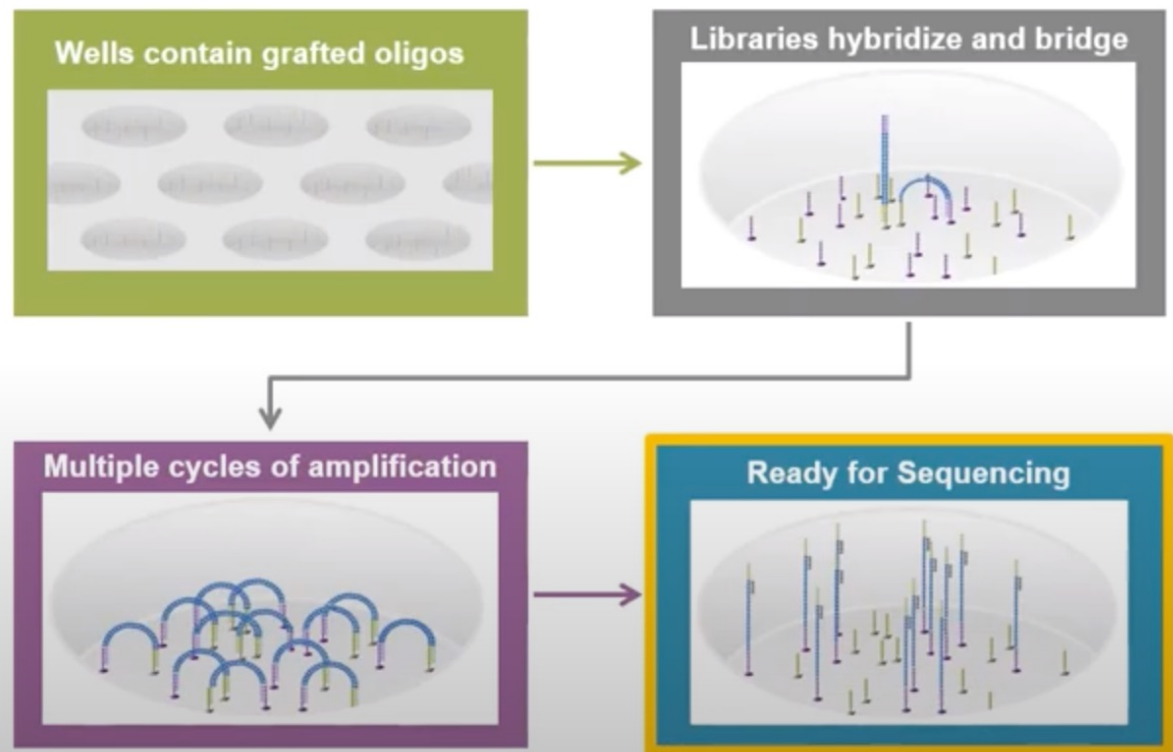
Random

Patterned

Rigid registration reduces time by skipping template generation

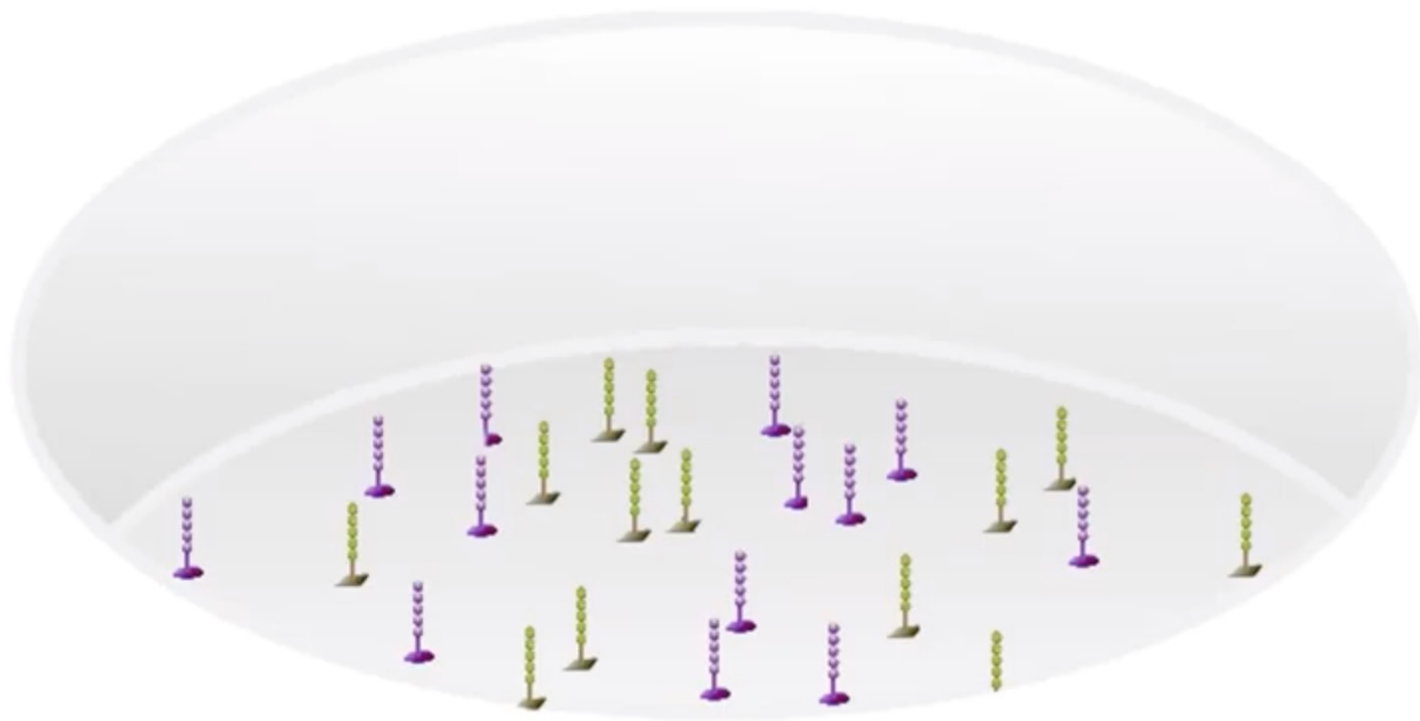
Patterned Flow Cell and ExAmp Technology

- ExAmp technology creates clonal clusters in each well from individual library molecules

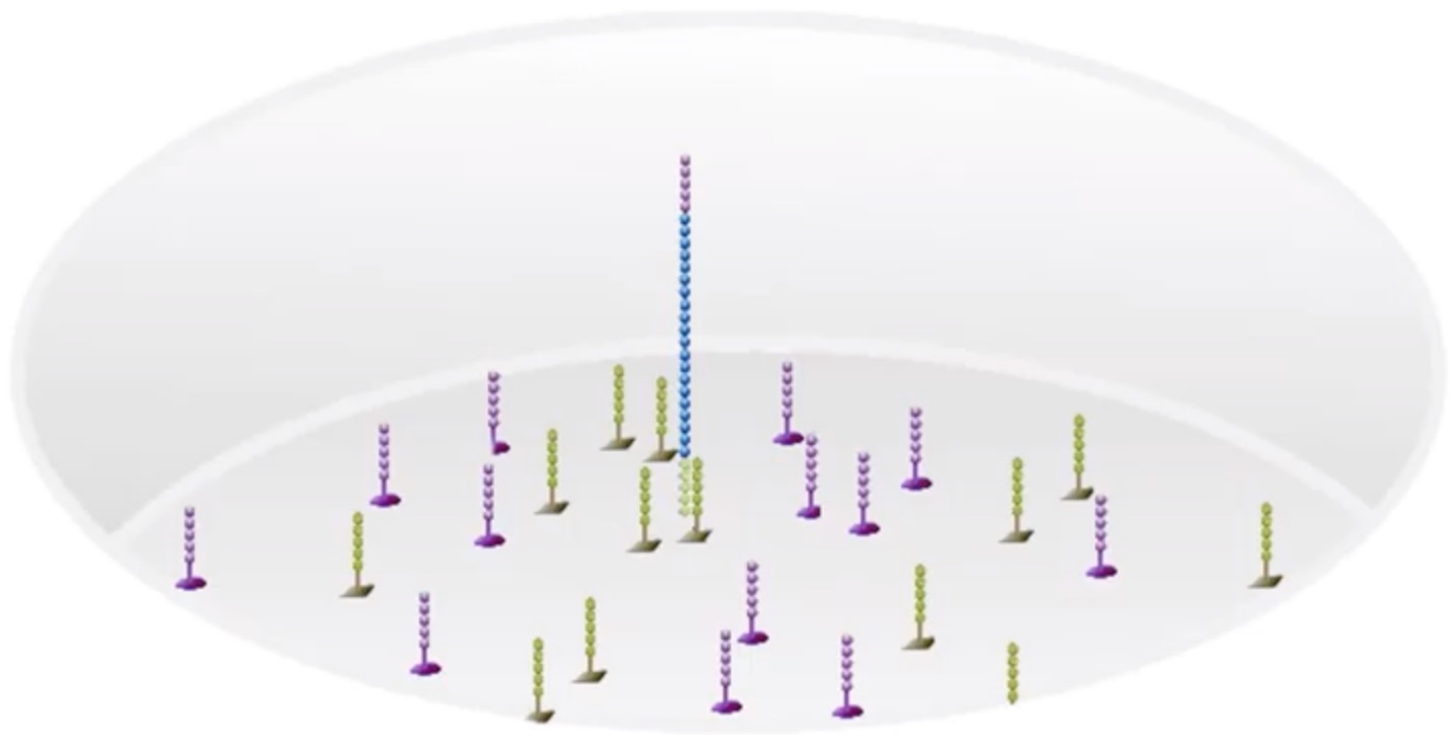


The background of the slide is a repeating pattern of light gray oval shapes, each representing a microwell. Inside each oval, there are several short, vertical lines of different colors (yellow, orange, and purple) of varying heights, representing oligonucleotides (oligos) attached to the surface of the microwell.

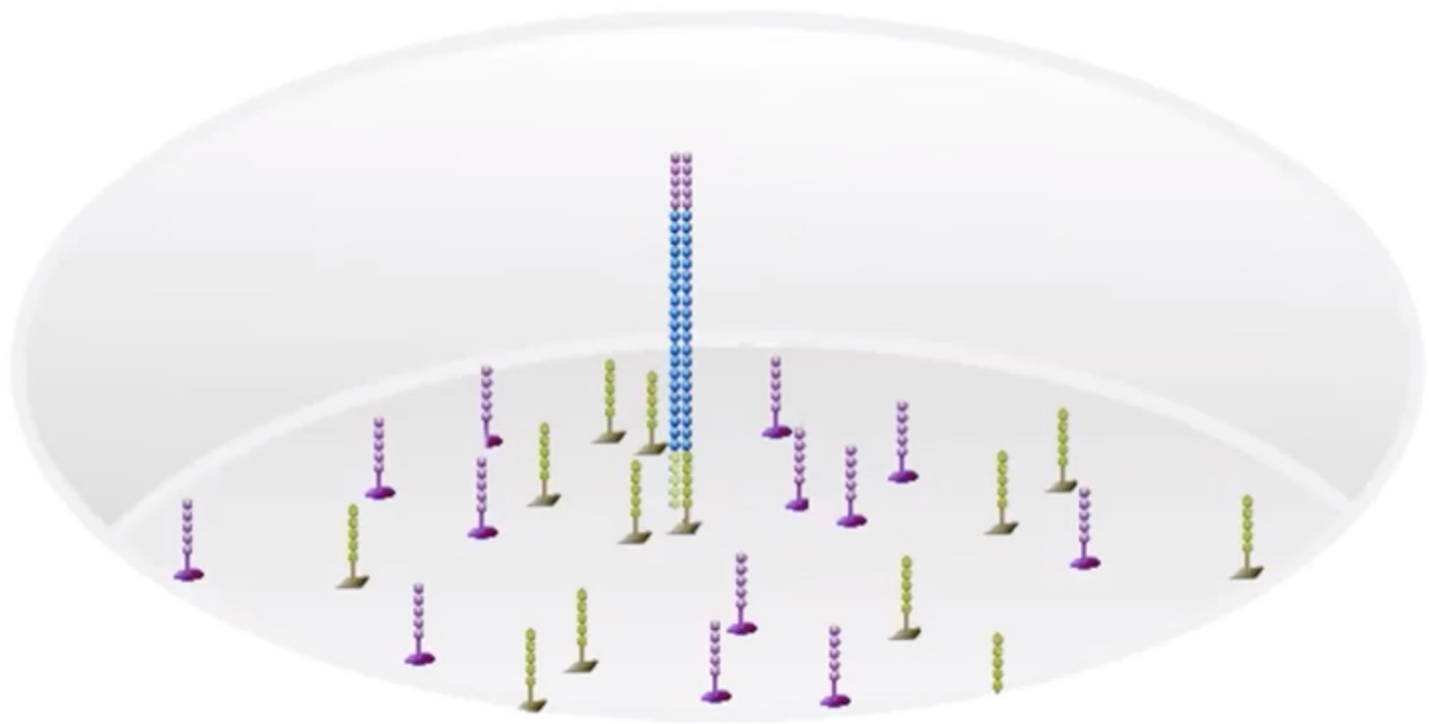
Oligos present inside patterned
microwells



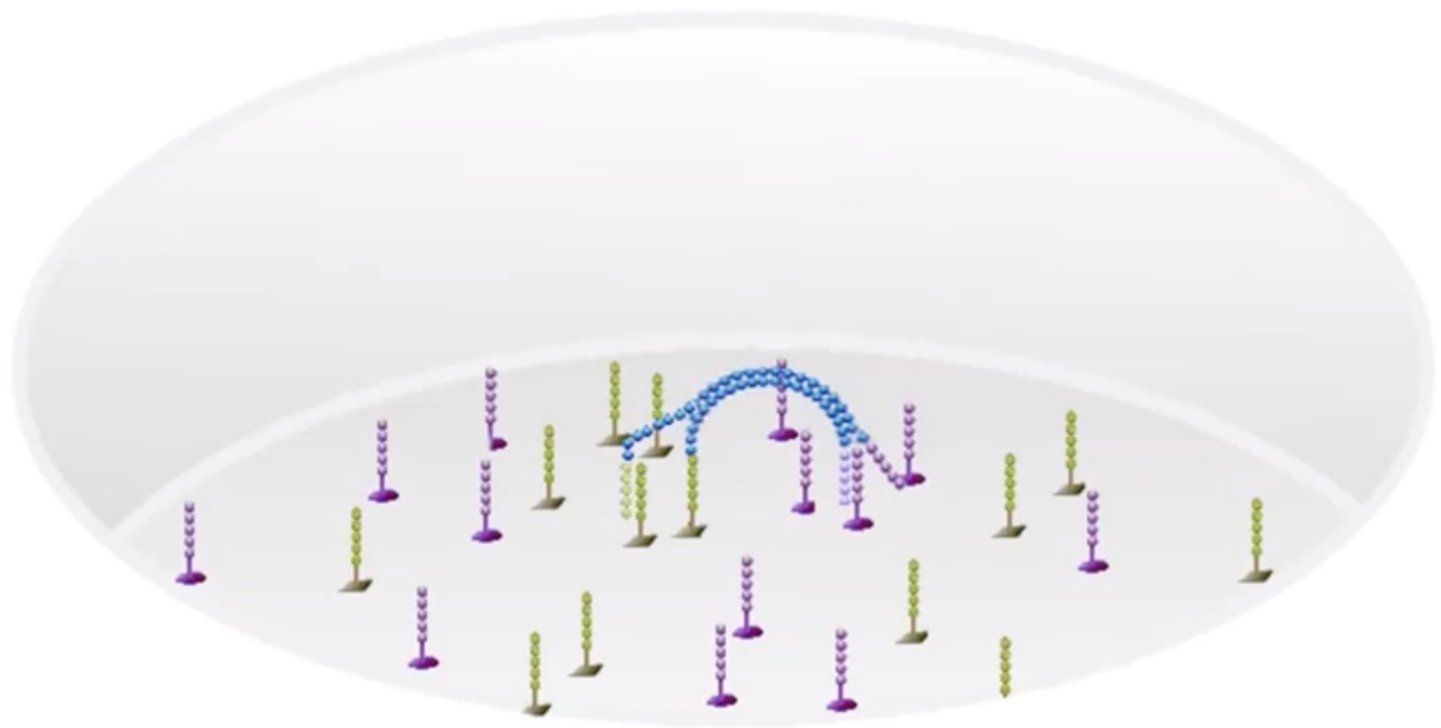
Oligos are not present on surface
between wells to control cluster size



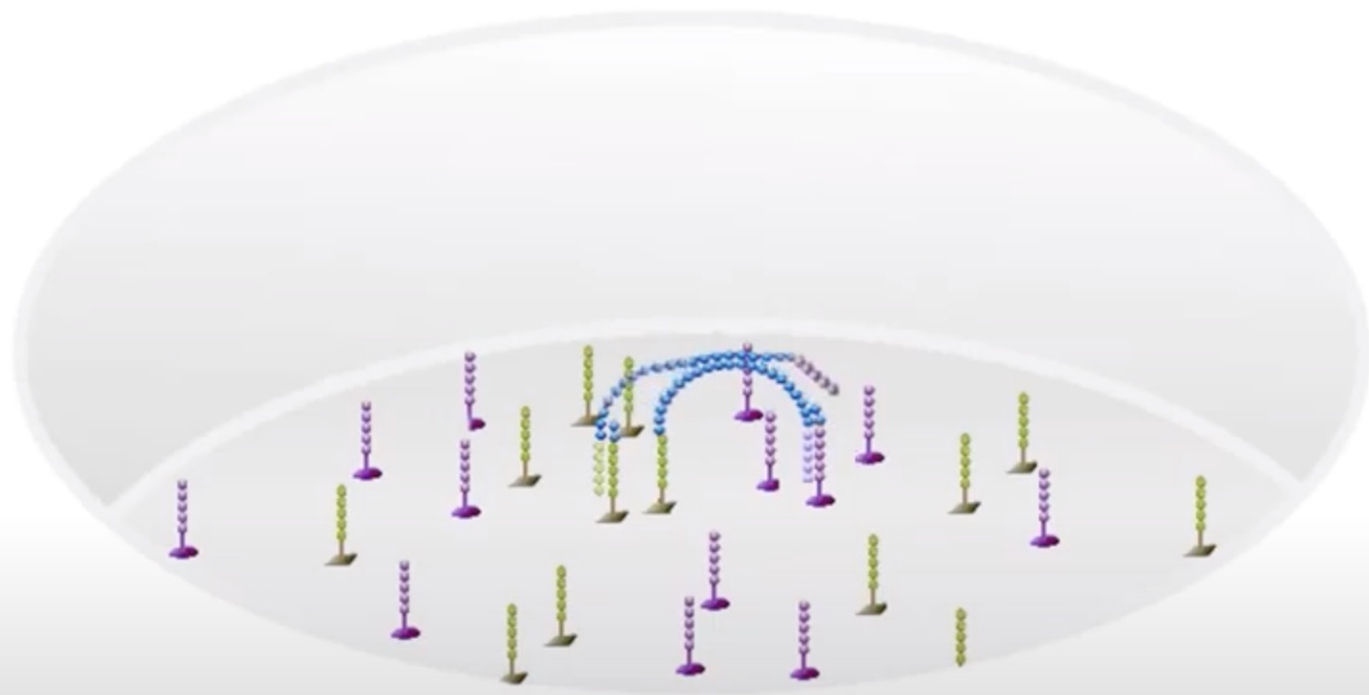
Single stranded DNA is introduced to the flow cell, which hybridizes to an oligo present in the microwell



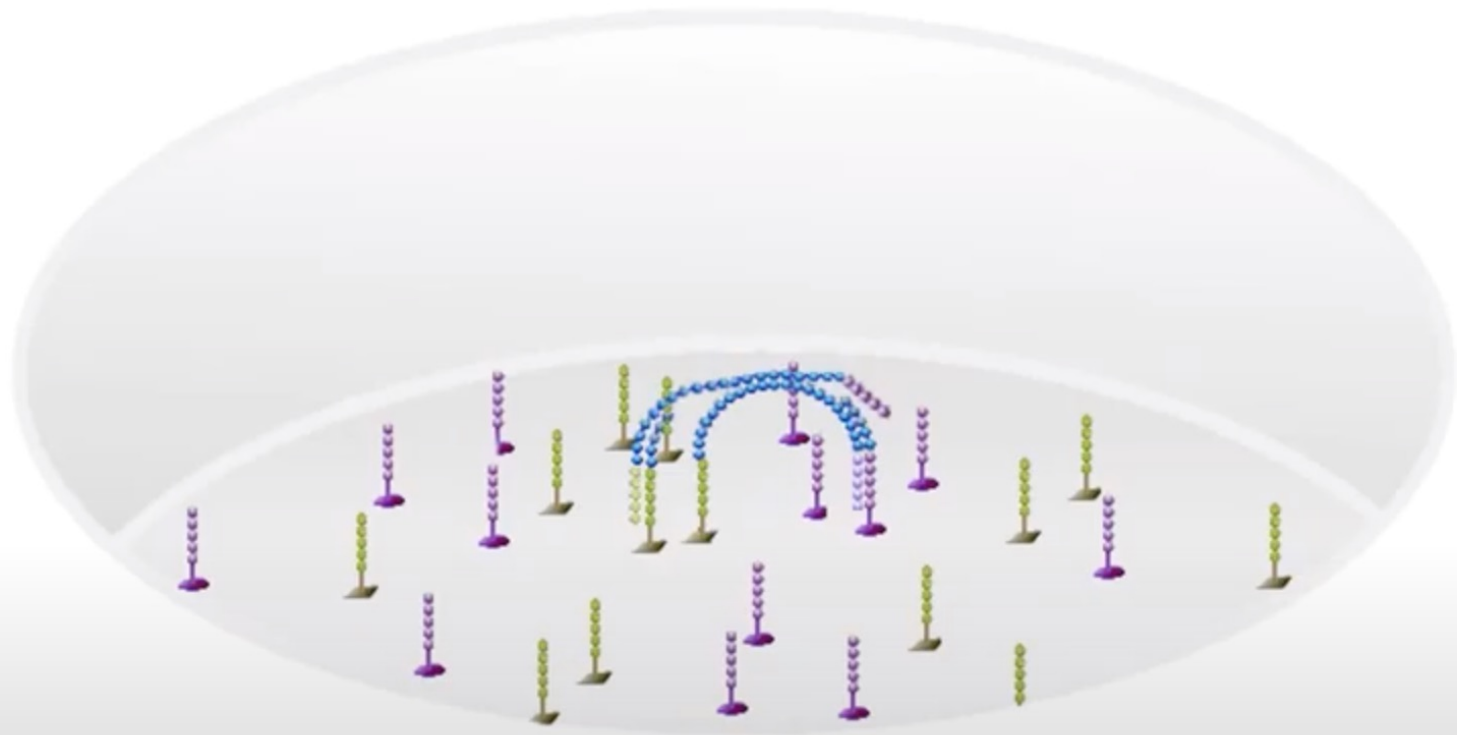
When hybridization occurs amplification immediately begins



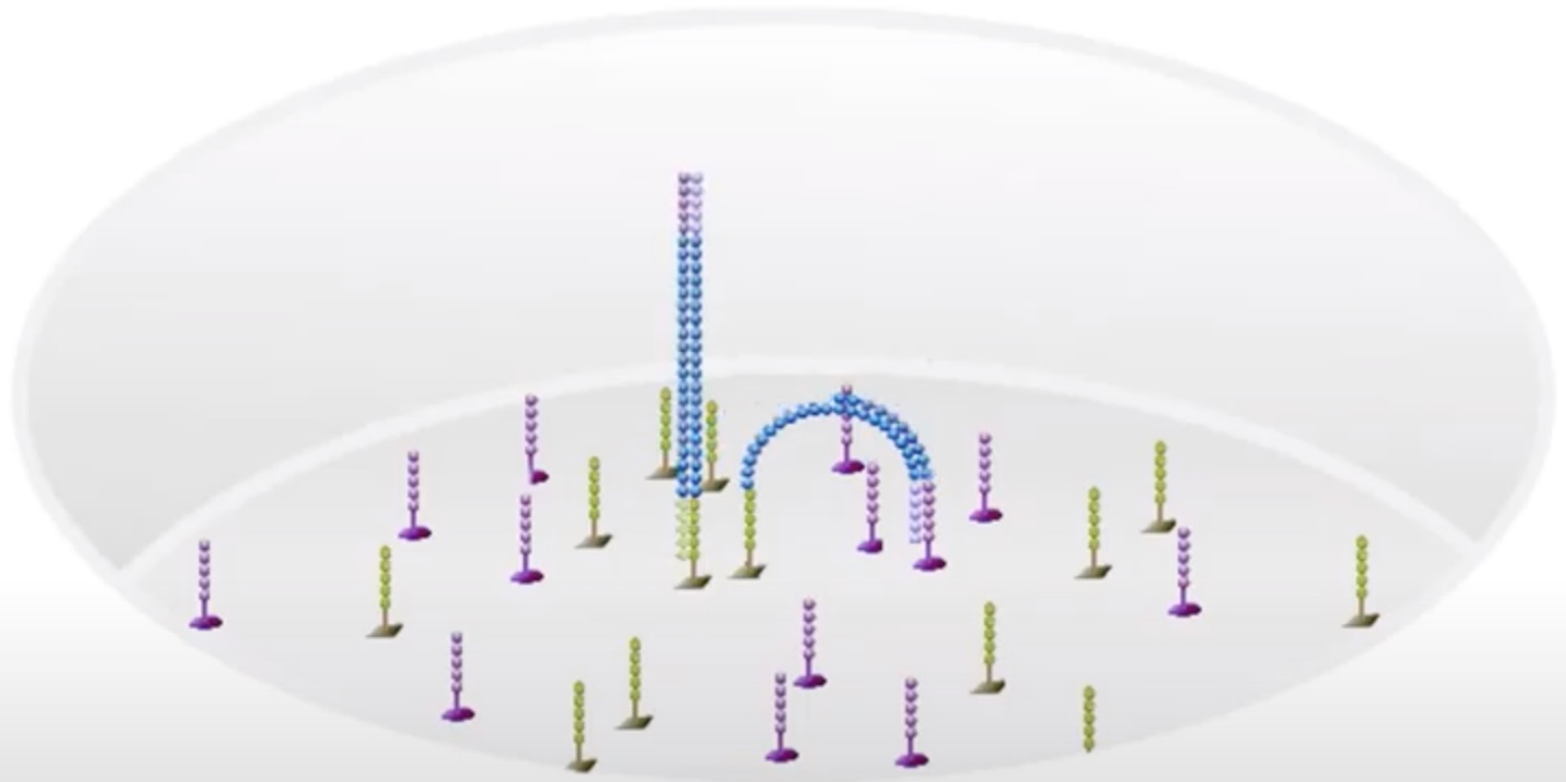
Through a proprietary process, double stranded ends of the DNA are denatured and hybridize to the flow cell



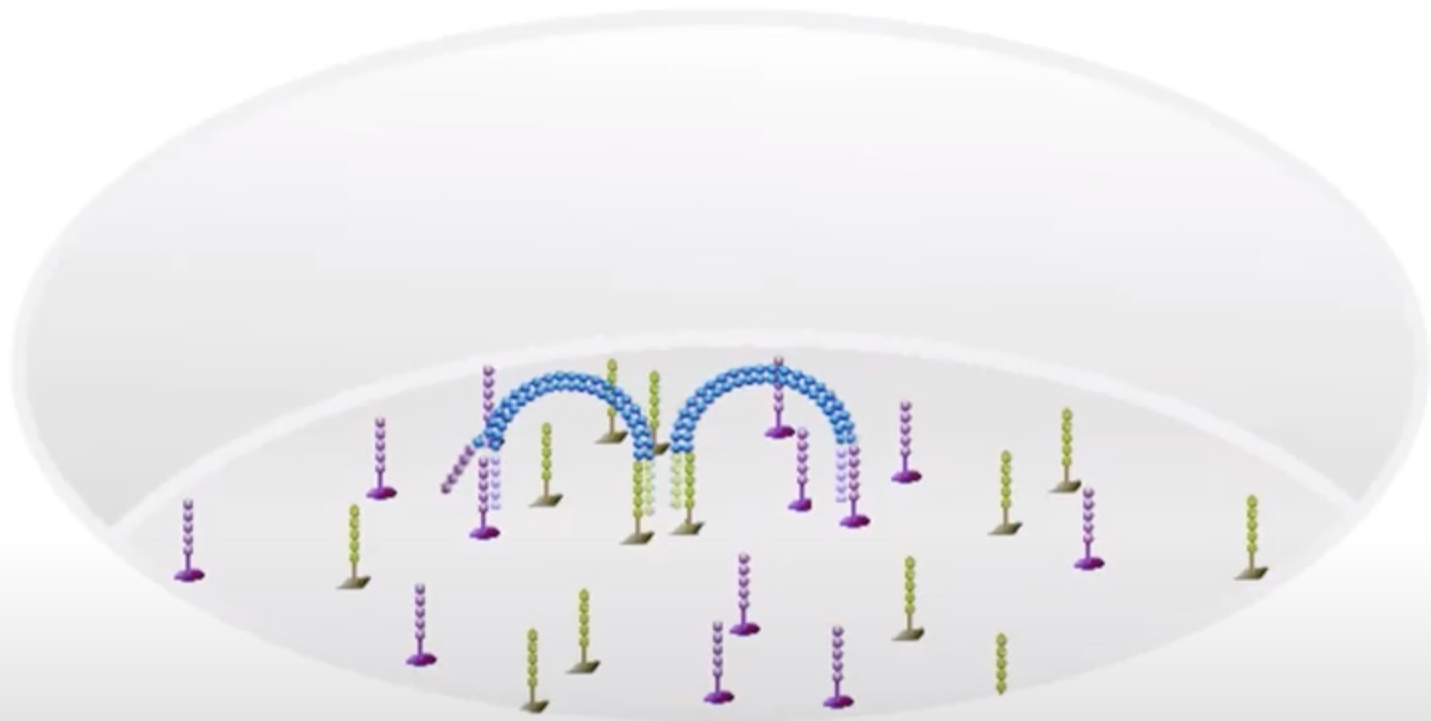
Amplification occurs on both strands simultaneously



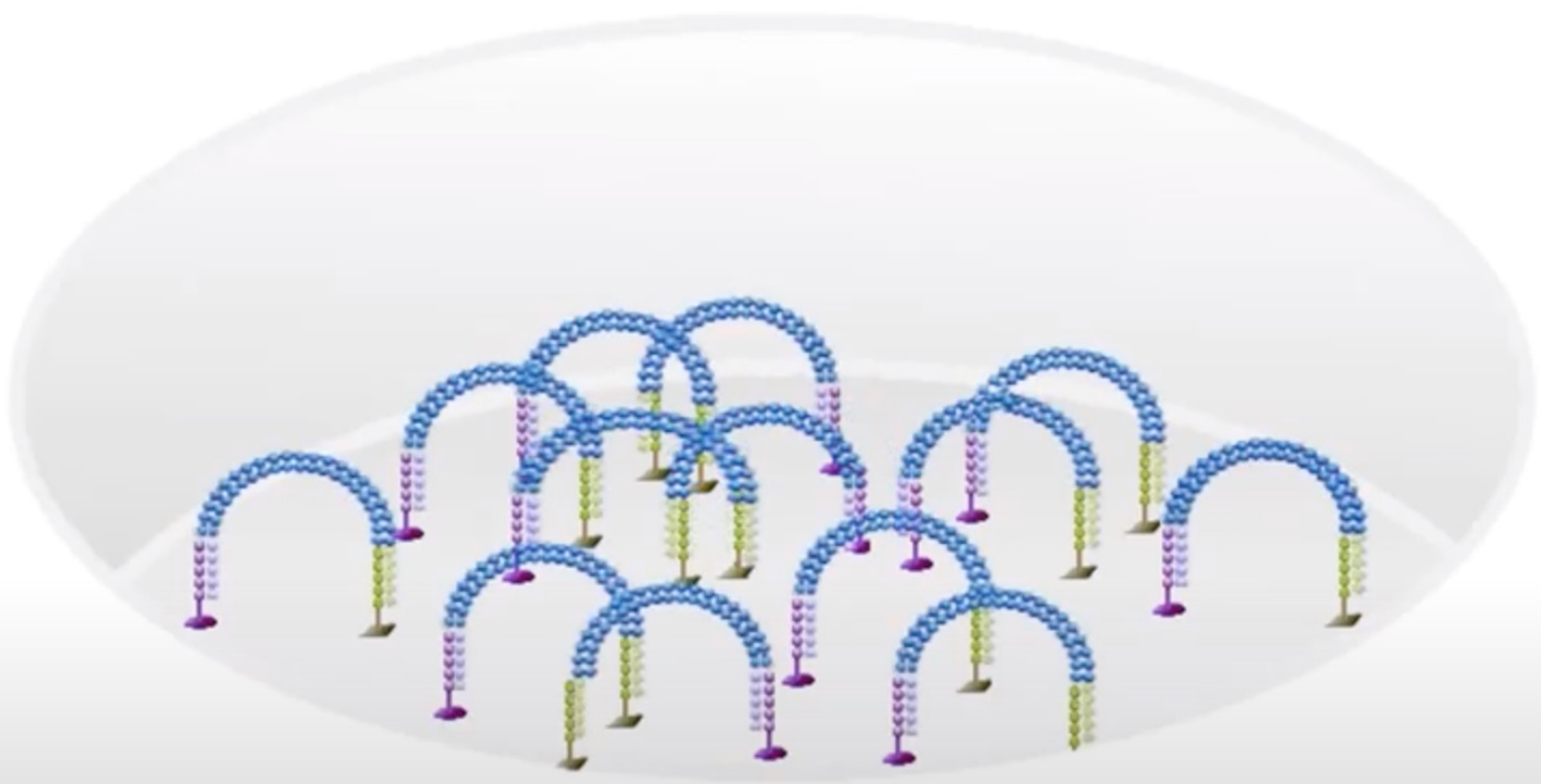
Amplification displaces double stranded
DNA



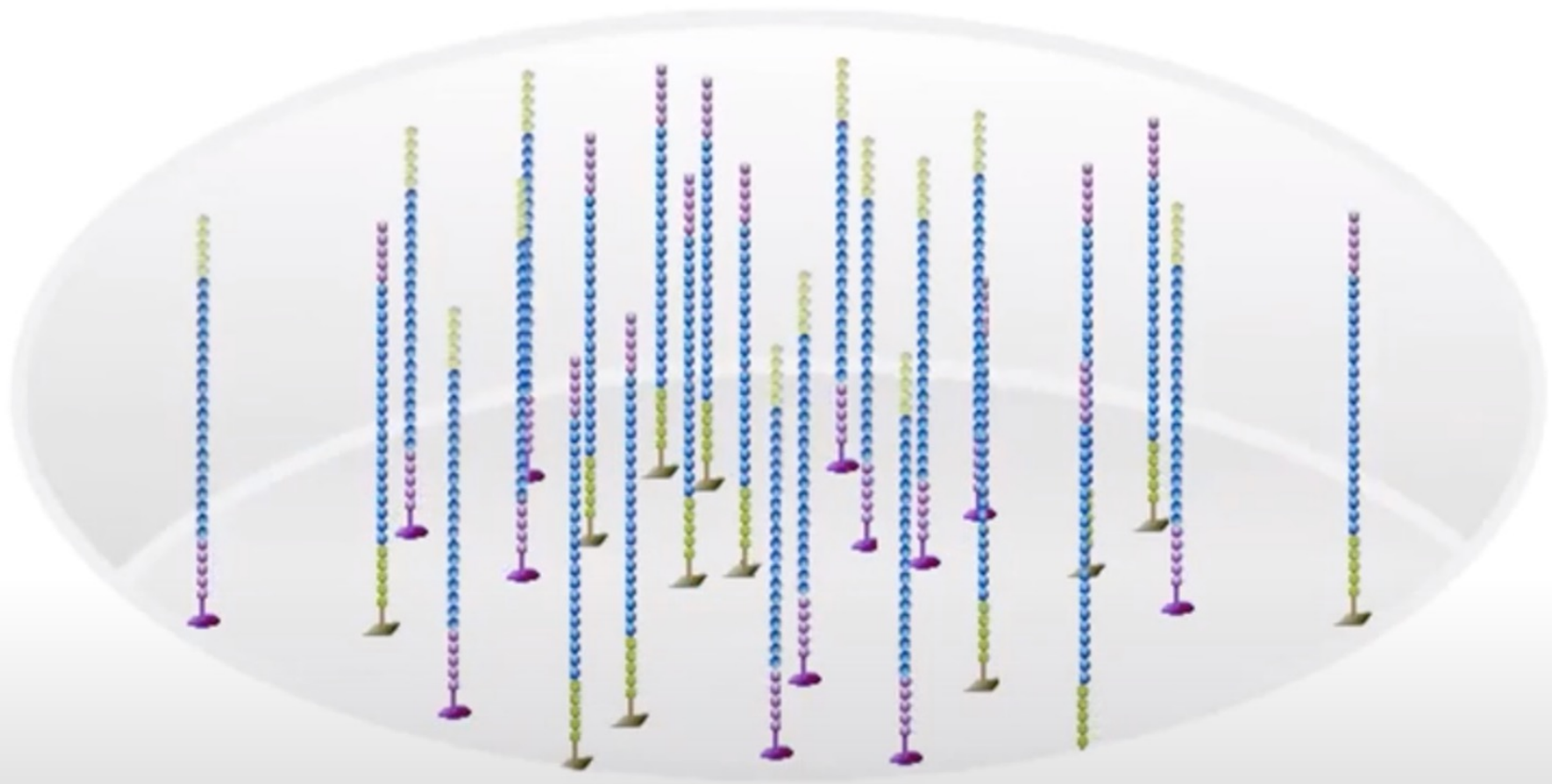
Amplification continues until fully double stranded DNA is formed



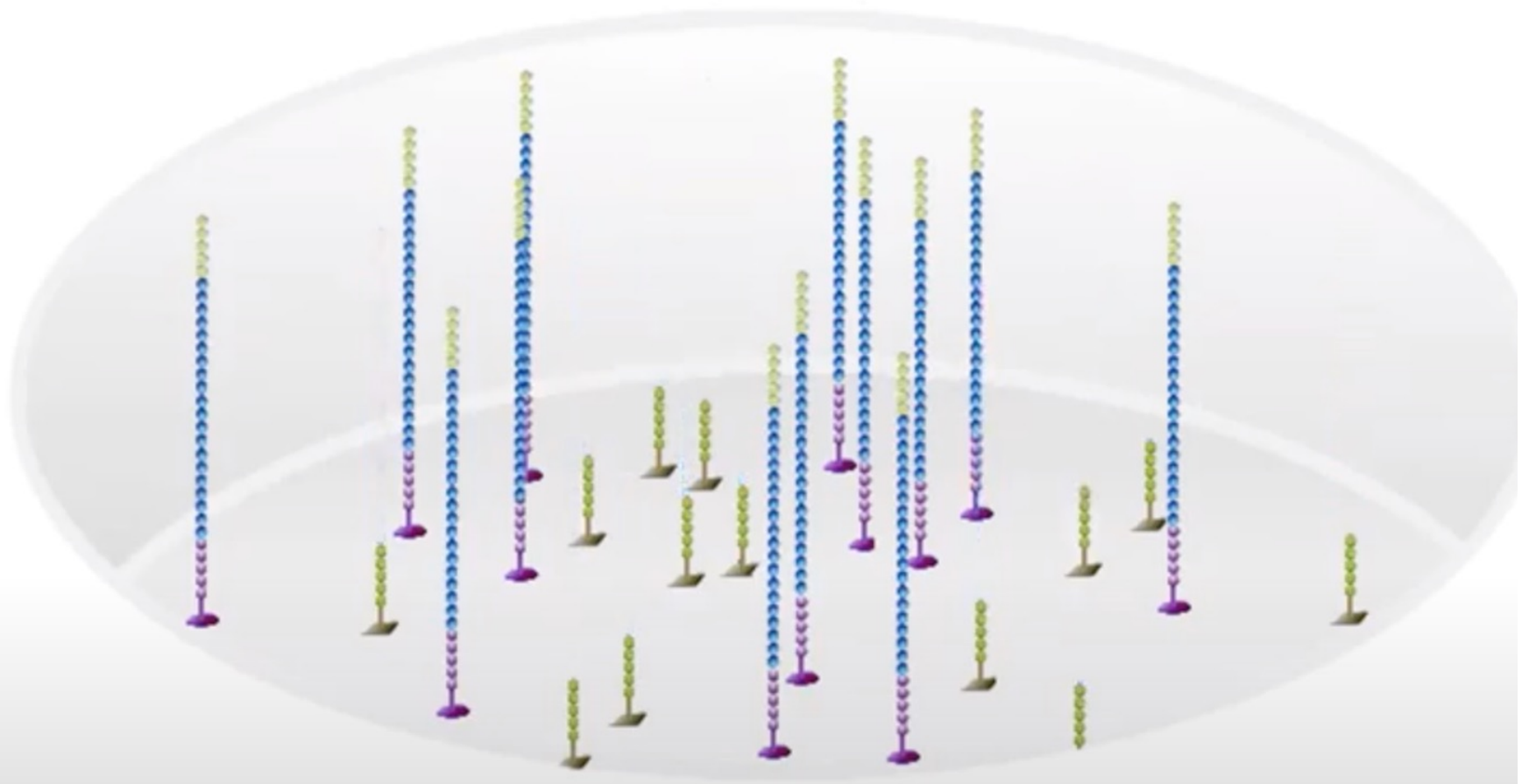
After double stranded DNA is formed, the process repeats itself



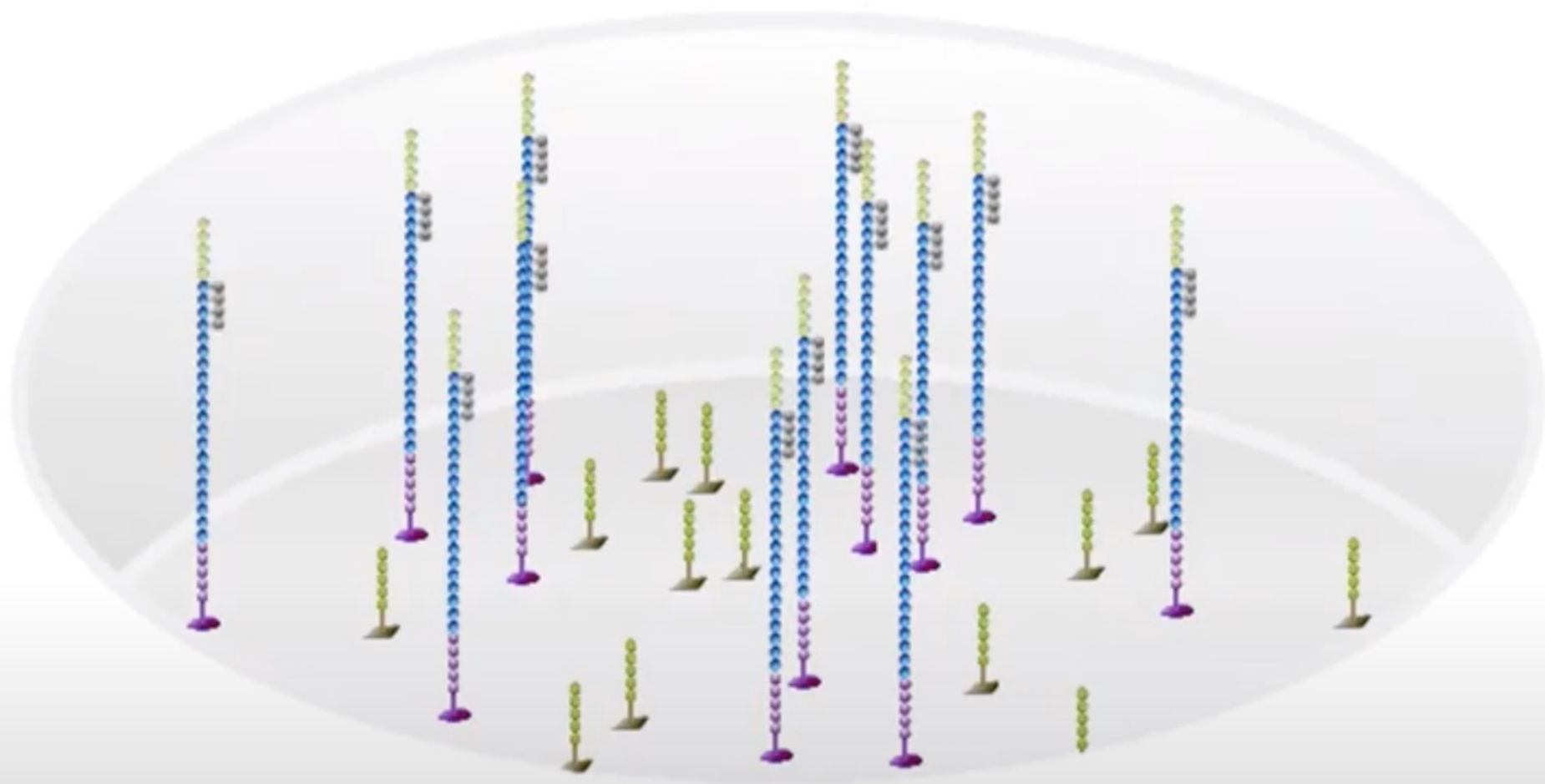
This process continues until no unused oligos are left on the flow cell surface



To prepare the strands for sequencing,
linearization is performed



Reverse strands are cleaved

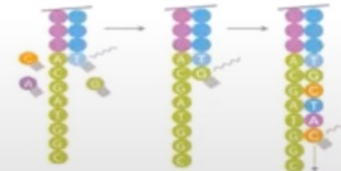


Sequencing primer is hybridized to Read 1 sequencing primer binding site

Illumina Sequencing Workflow

3

Sequencing



Illumina Sequencing Systems

Focused Power



Flexible Power



Production Power



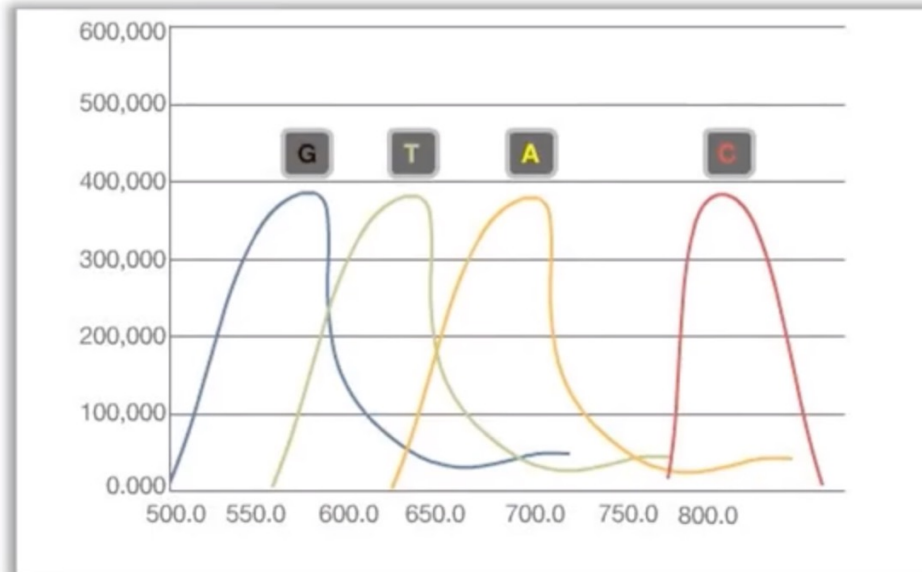
Population Power



For Research Use Only. Not for use in diagnostic procedures.

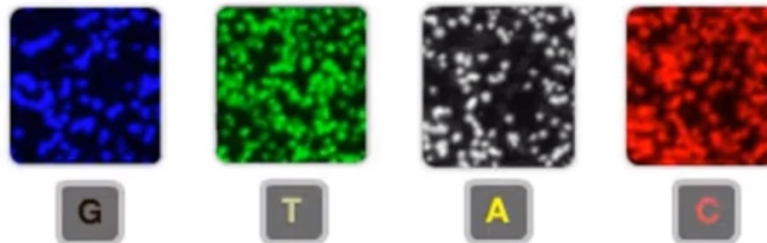
4-Channel SBS Chemistry:

HiSeq, MiSeq



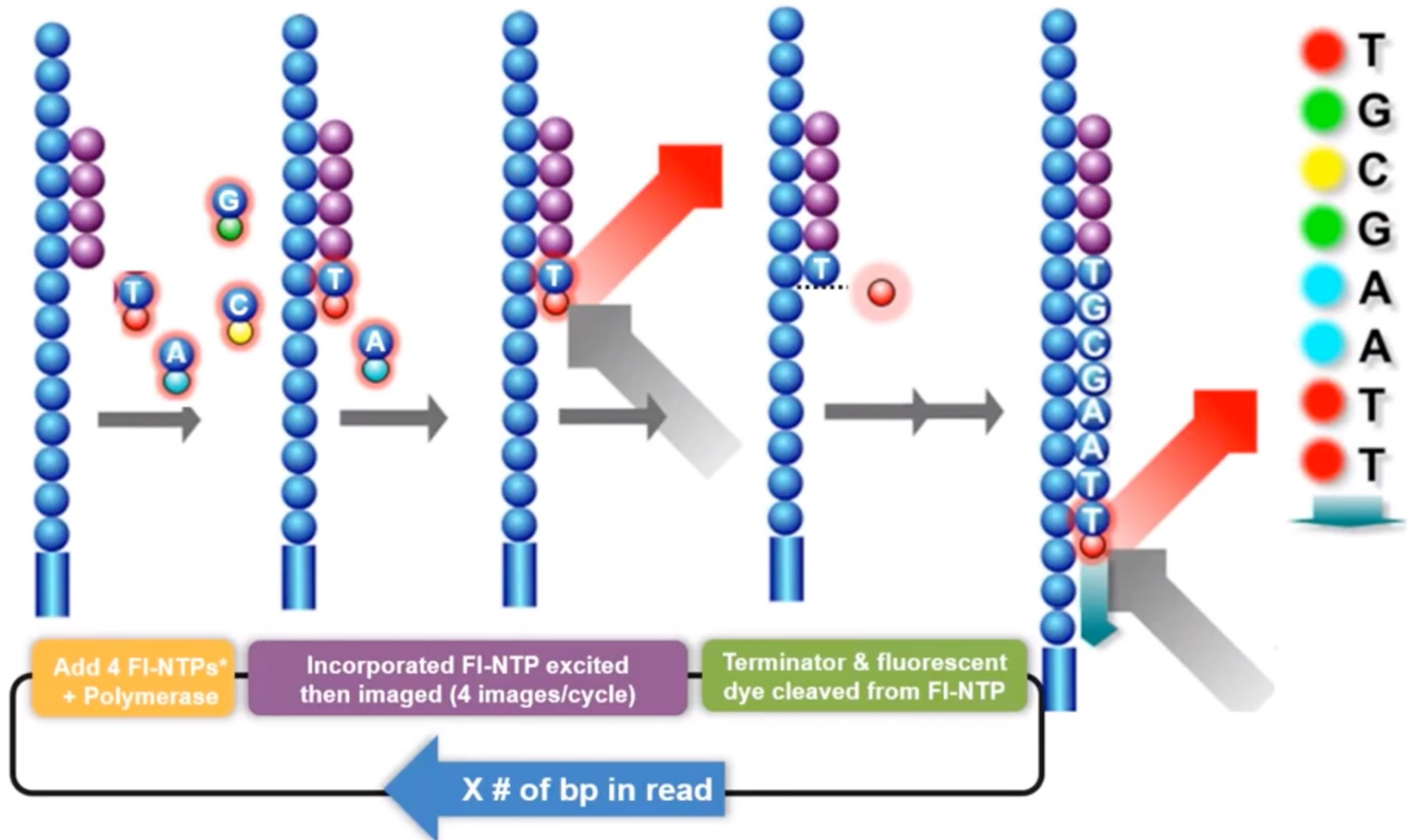
Each of the four DNA bases emits an intensity of a unique wavelength

- Collects four images:
- During each cycle, each cluster appears in only one of four images



A Closer Look At 4-Dye Chemistry

4-channel chemistry

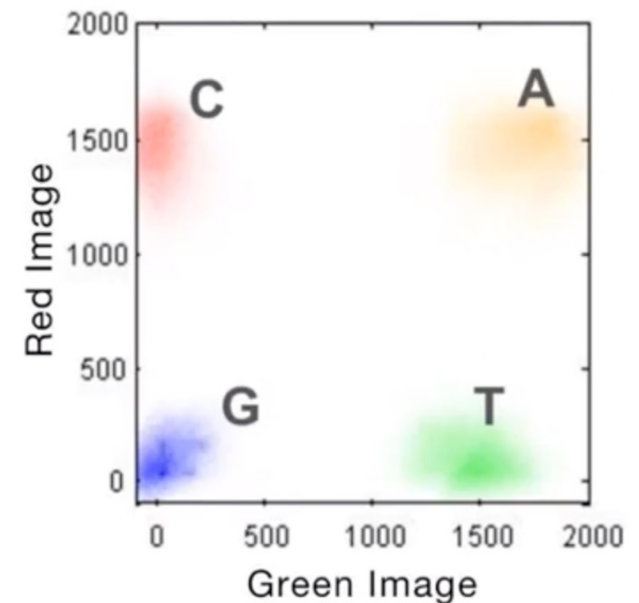
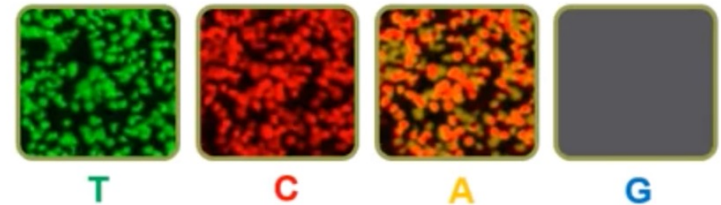


* Fluorescently labelled NTPs

2-Channel SBS Chemistry:

NextSeq 550, MiniSeq, NovaSeq 6000

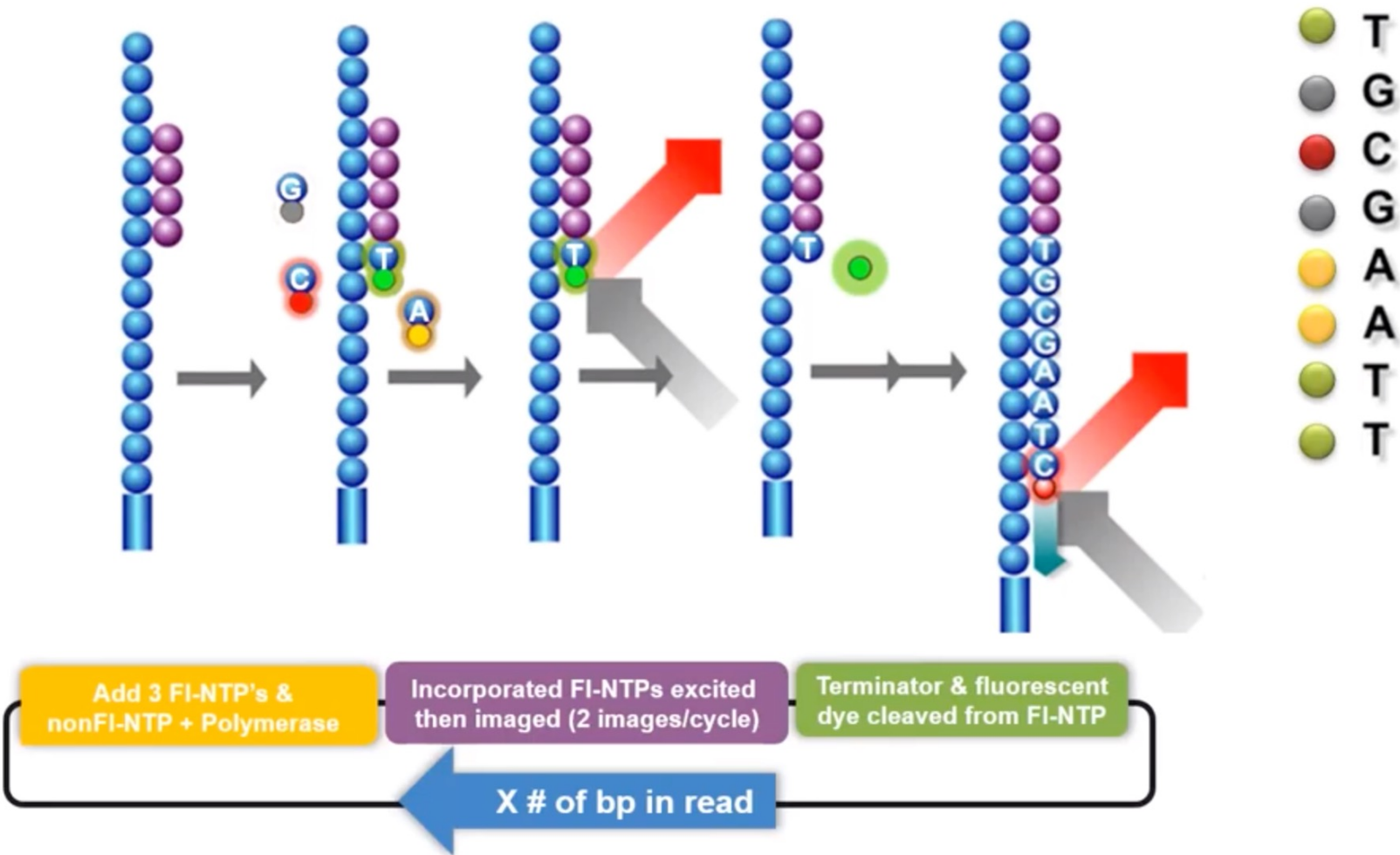
- 2-channel SBS uses two images:
- Clusters appearing in green only are **T**
- Clusters appearing in red only are **C**
- Clusters appearing in both images are **A**
- Clusters not present/dark are **G**
- After imaging, cluster intensities are plotted and bases called accordingly



[Illumina Two-Channel SBS Sequencing Technology Technote](#)

A Closer Look At 2-Dye Chemistry

2-channel chemistry



1-Channel SBS Chemistry:

iSeq 100

SBS chemistry combined with Complementary Metal-Oxide-Semiconductor (CMOS) technology

- The system uses a patterned flow cell with nanowells fabricated over a CMOS chip
- Each sequencing cycle has two chemistry steps in order to determine bases
- Two images are captured within one cycle of sequencing run

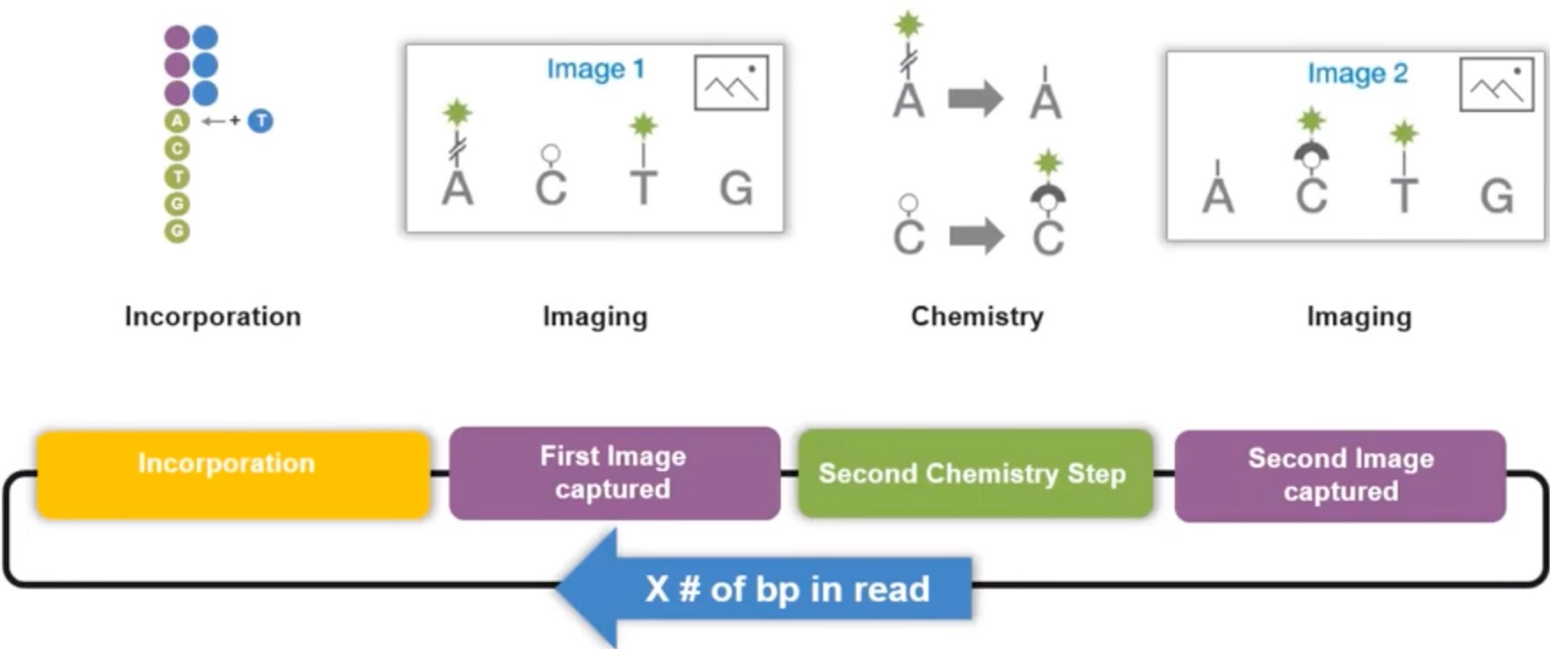
Based on the signal pattern across two images, base calls can be determined

- Intensities extracted from one image and compared to a second image result in four distinct populations, each corresponding to a nucleotide.

A Closer Look At 1-Dye Chemistry

1-channel chemistry

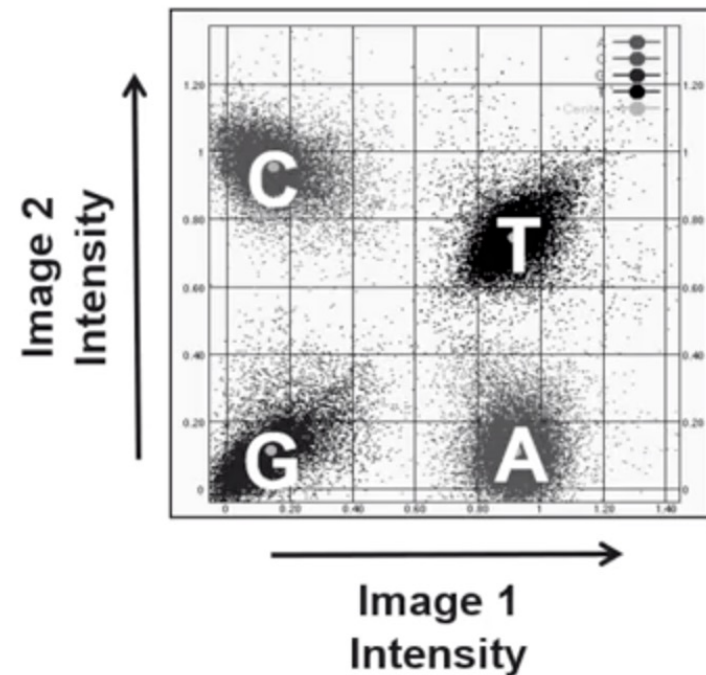
One Sequencing Cycle



Sequencing by synthesis with CMOS detection

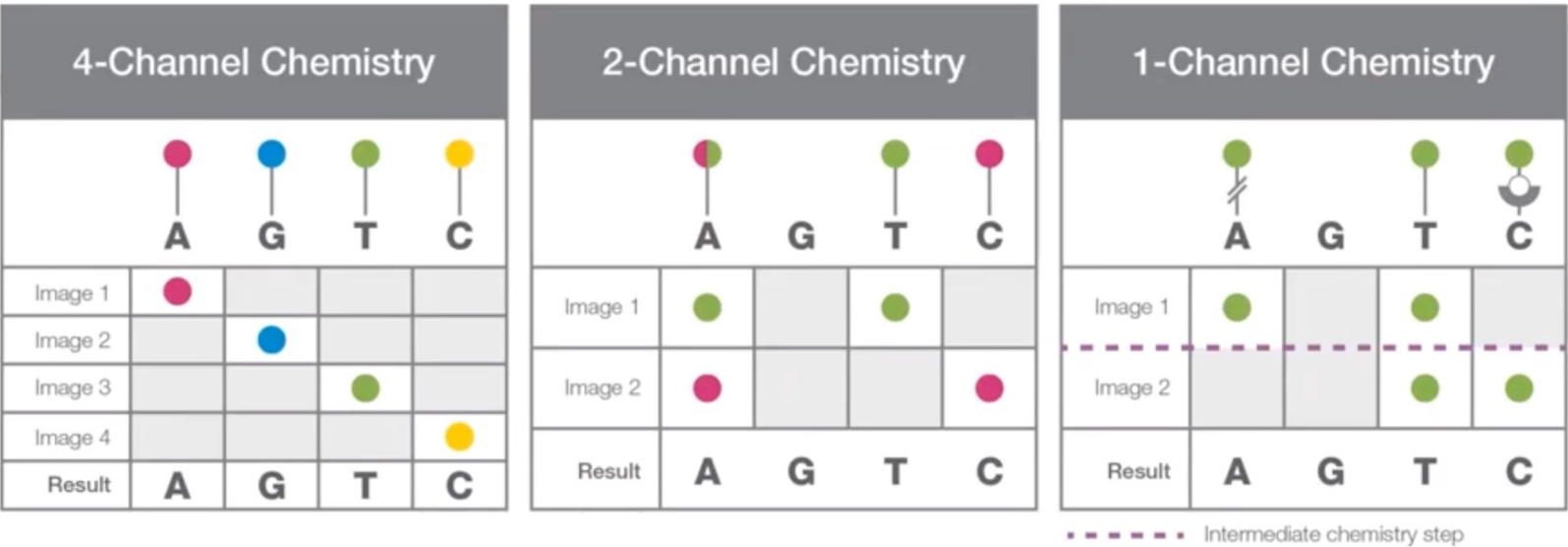
1-channel chemistry

Base	Image 1	Image 2
T	ON	ON
A	ON	OFF
C	OFF	ON
G	OFF	OFF



- The iSeq 100 System uses 1-dye sequencing, which requires one dye and two images to encode data for the four bases

Illumina Chemistry Comparison



4-channel SBS

- Bases are identified using four different fluorescent dyes, one for each base and four images per sequencing cycle

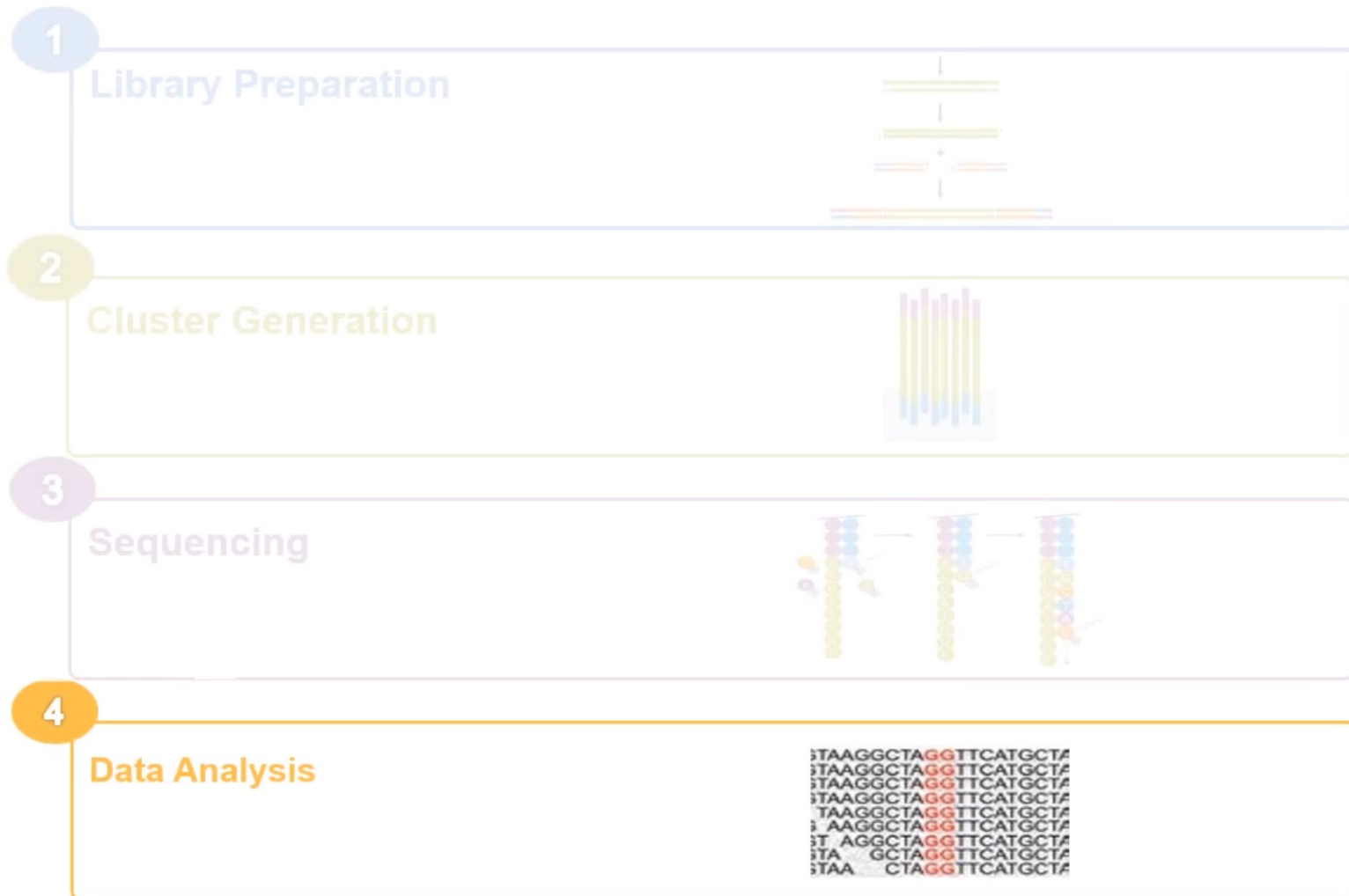
2-channel SBS

- Simplified nucleotide detection by using two fluorescent dyes and two images to determine all four base calls

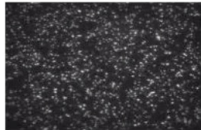
1-channel SBS

- Base calling uses one fluorescent dye and two images, with chemistry step in between, to determine all four base calls

Illumina Sequencing Workflow



Analysis Overview

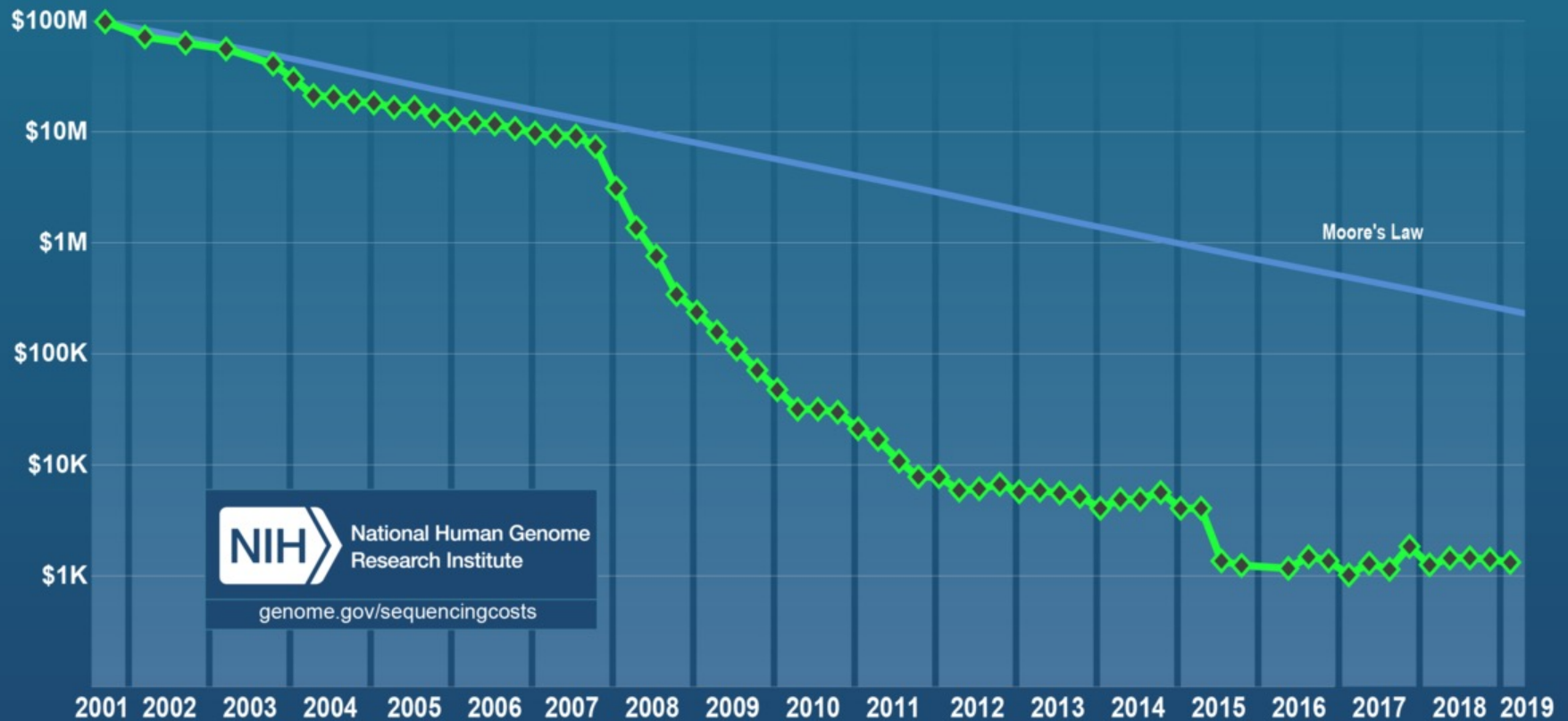
Analysis Type	Software	Outputs
Control Software		 Images, Intensities and Base Calls
Analysis Software	 	 Alignments, Variant Detection
Visualization Software	 	 1,2,3  Annotation, Filtering, Reports

Comparison of Illumina sequencing machine



Sequencing System	iSeq™	MiniSeq™	MiSeq®	NextSeq®	HiSeq®	HiSeq® X	NovaSeq®
					4000	Five/Ten	6000
Output per run	1.2 Gb	7.5 Gb	15 Gb	120 Gb	1.5 Tb	1.8 Tb	1 Tb - 6 Tb ¹
Instrument price	\$19.9K	\$49.5K	\$99K	\$275K	\$900K	\$6M ² /\$10M ²	\$985K
Installed base³	NA	~600	~6,000	~2,400	~2,300 ⁴		~285

Cost per Genome





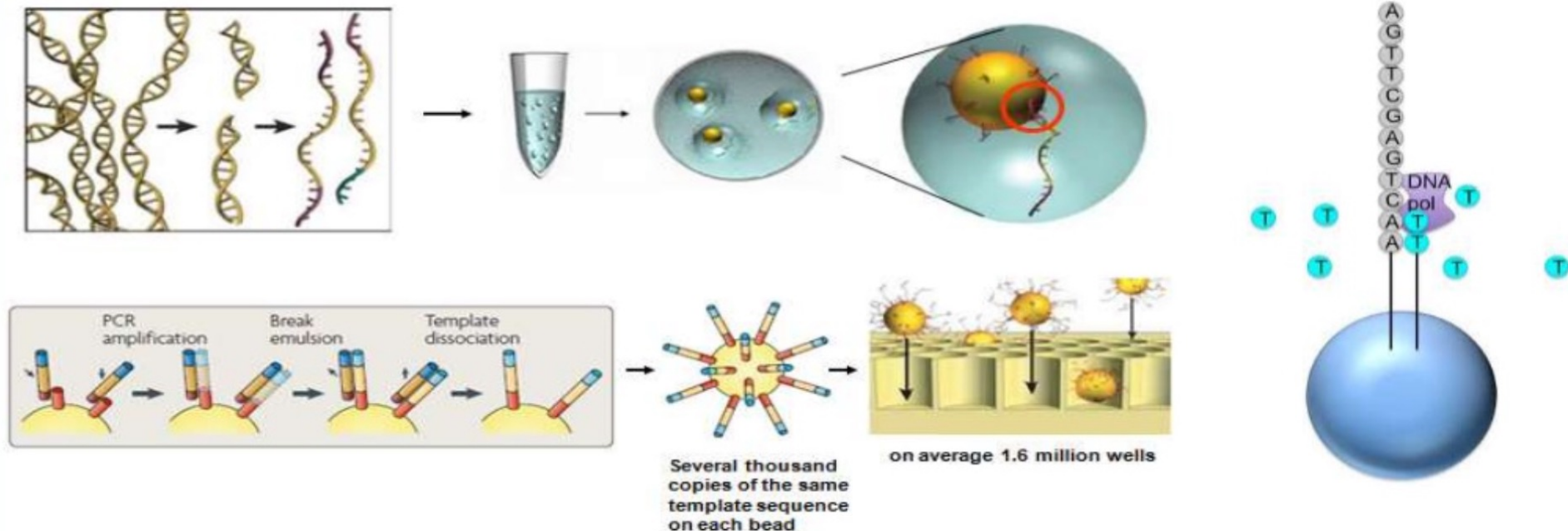
454 SEQUENCING



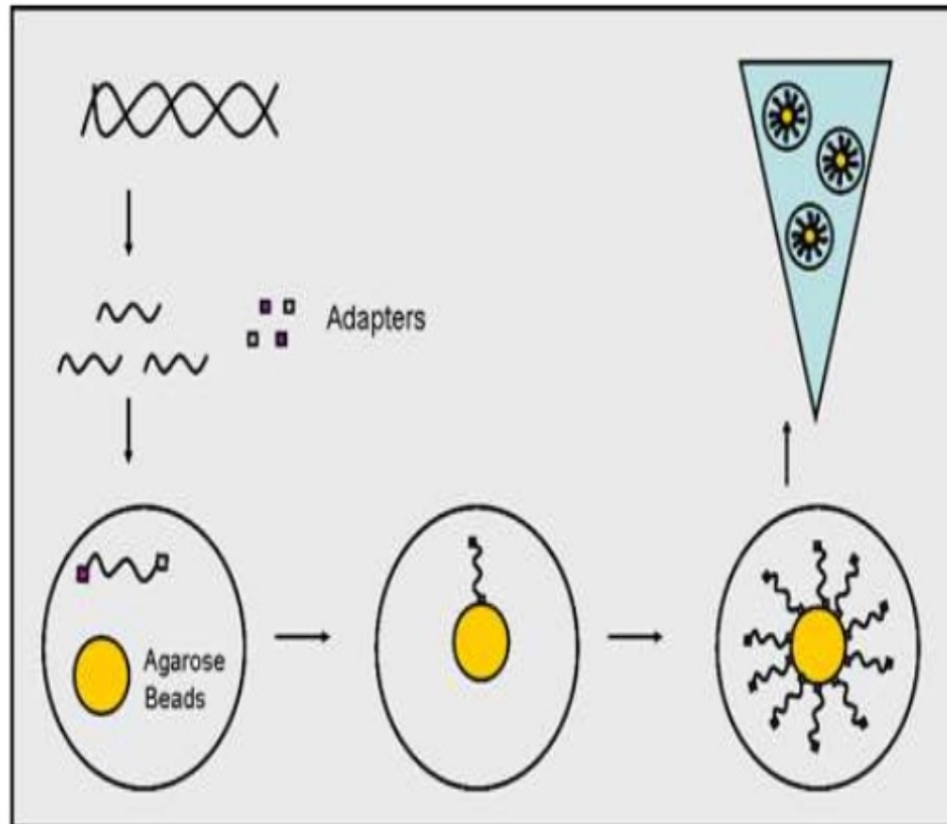
Part Two

ROCHE/454 SEQUENCING

- Sequence much longer reads by sequencing multiple reads at once by reading optical signals as bases are added.
- The DNA or RNA is fragmented into shorter reads up to 1kb.
- Uses Emulsion PCR for Clustal Amplification.
- PYROSEQUENCING as sequencing approach.

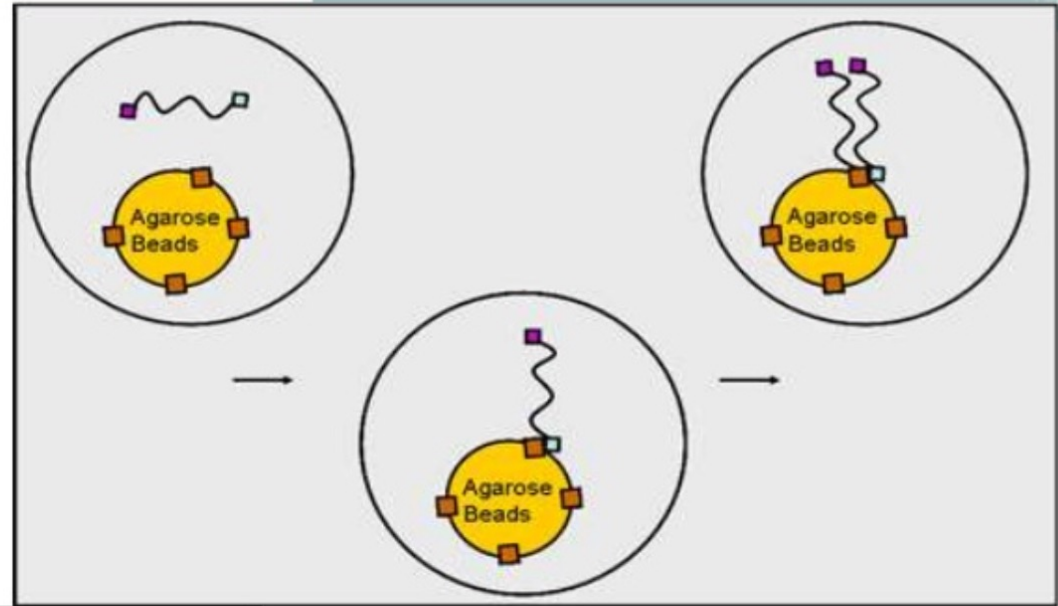


454 (Pyrosequencing)

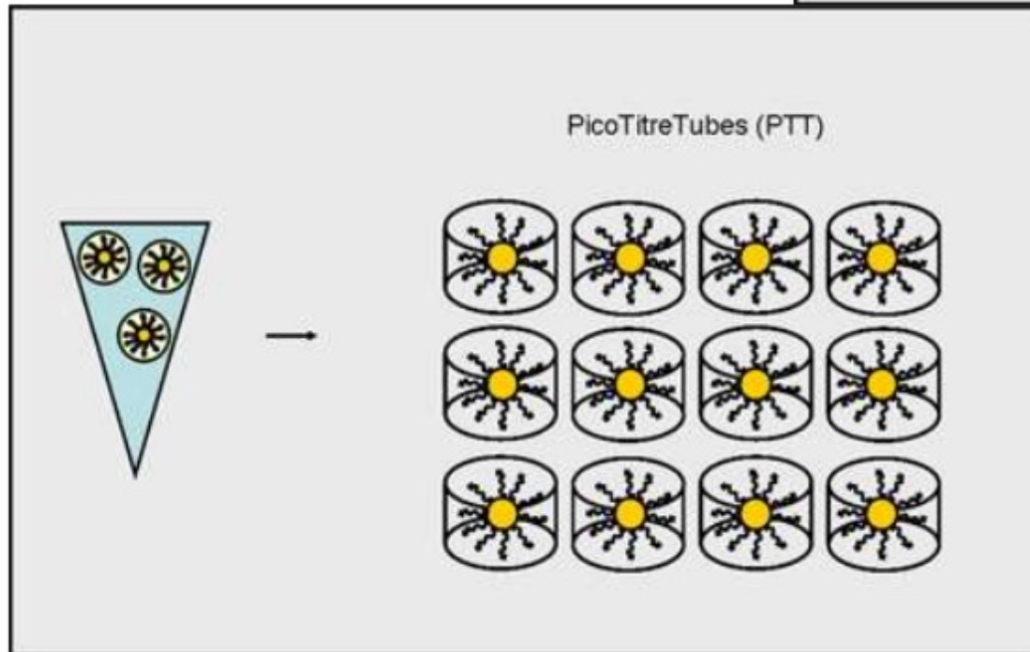


- DNA is fragmented, joined to adapters at either end of the fragmented DNA
- amplified in an **emulsion PCR** (includes agarose bead with complimentary adaptors to fragmented DNA)
- PCR amplified allowing up to 1 million identical fragments around one bead and finally dropped into a PicoTitreTube (PTT)

- Adapter containing the universal priming site are ligated to target ends
- Same primer can be used for amplification

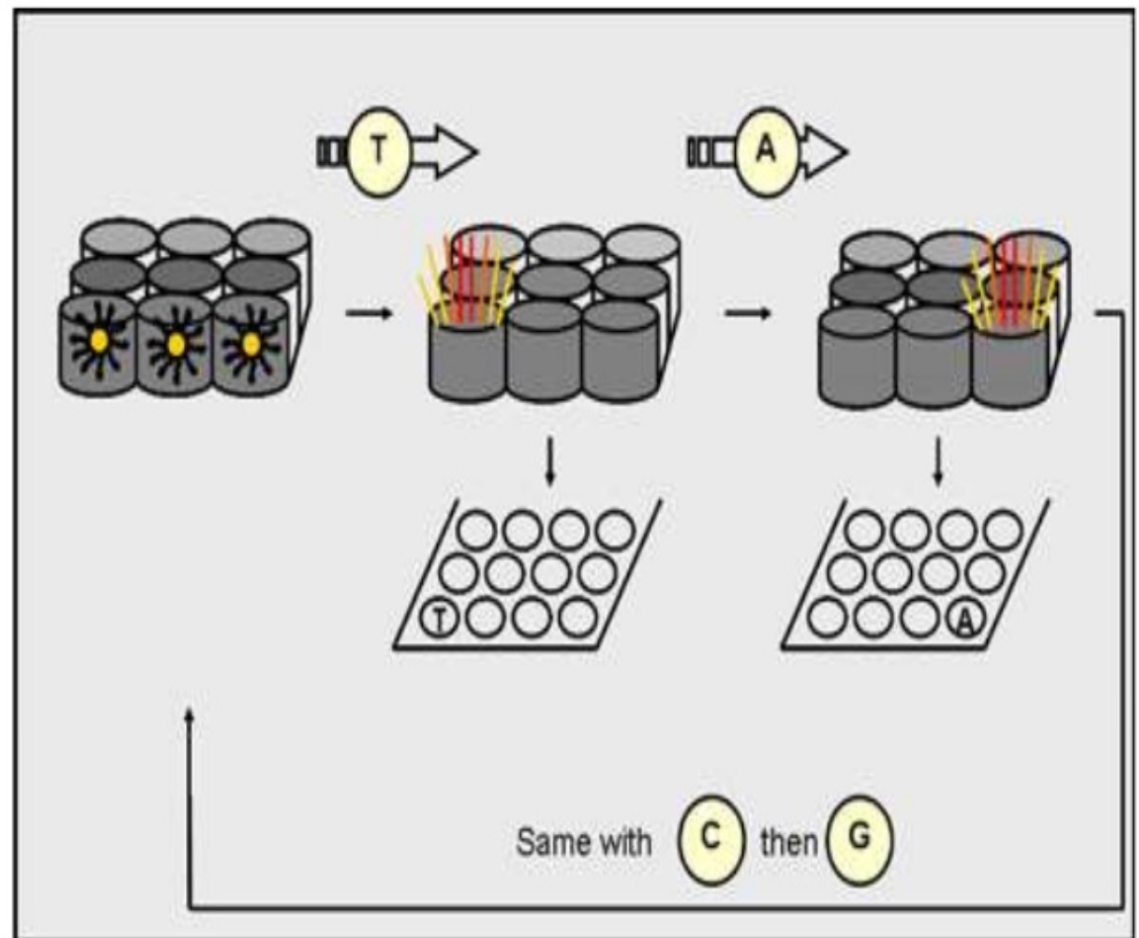


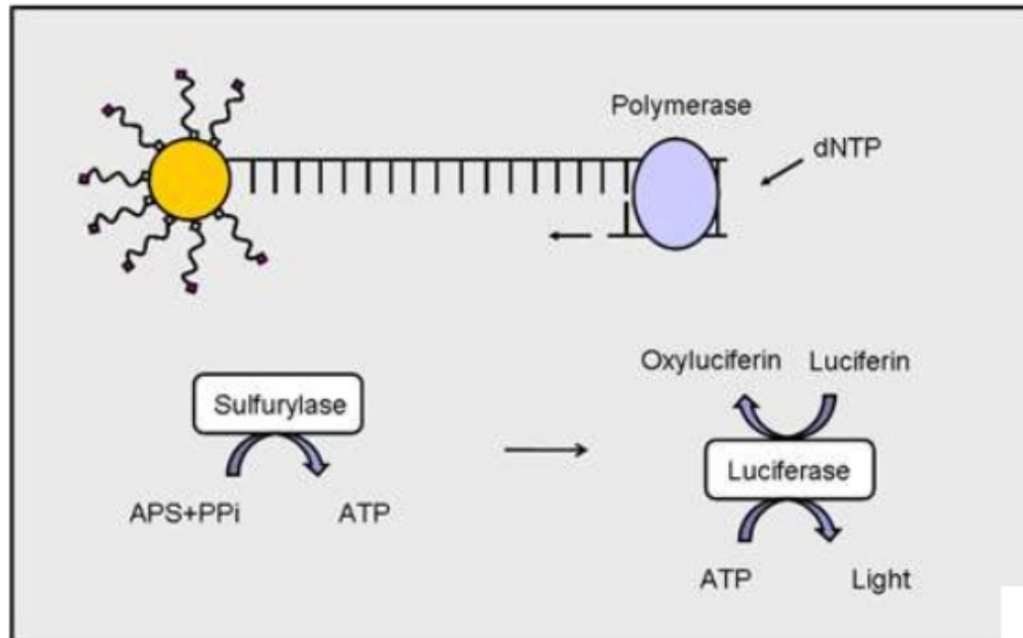
PCR amplification



Pico Titre Tube

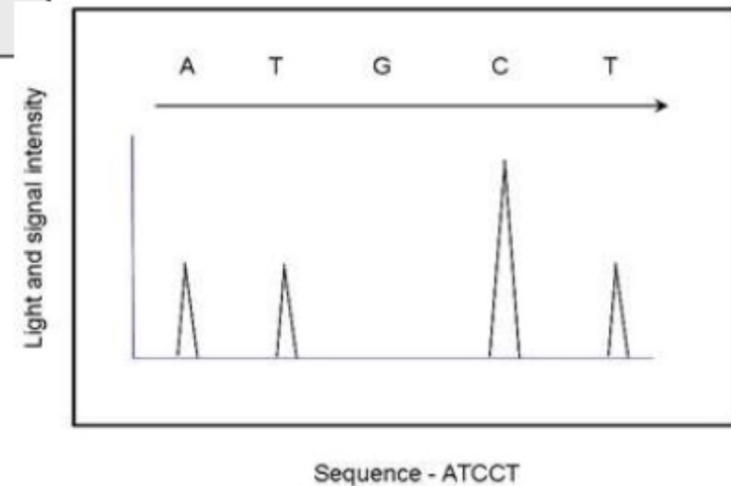
- In Pico titre tube reaction of fluorescence occurs with the addition of nucleotides





Nucleotide addition

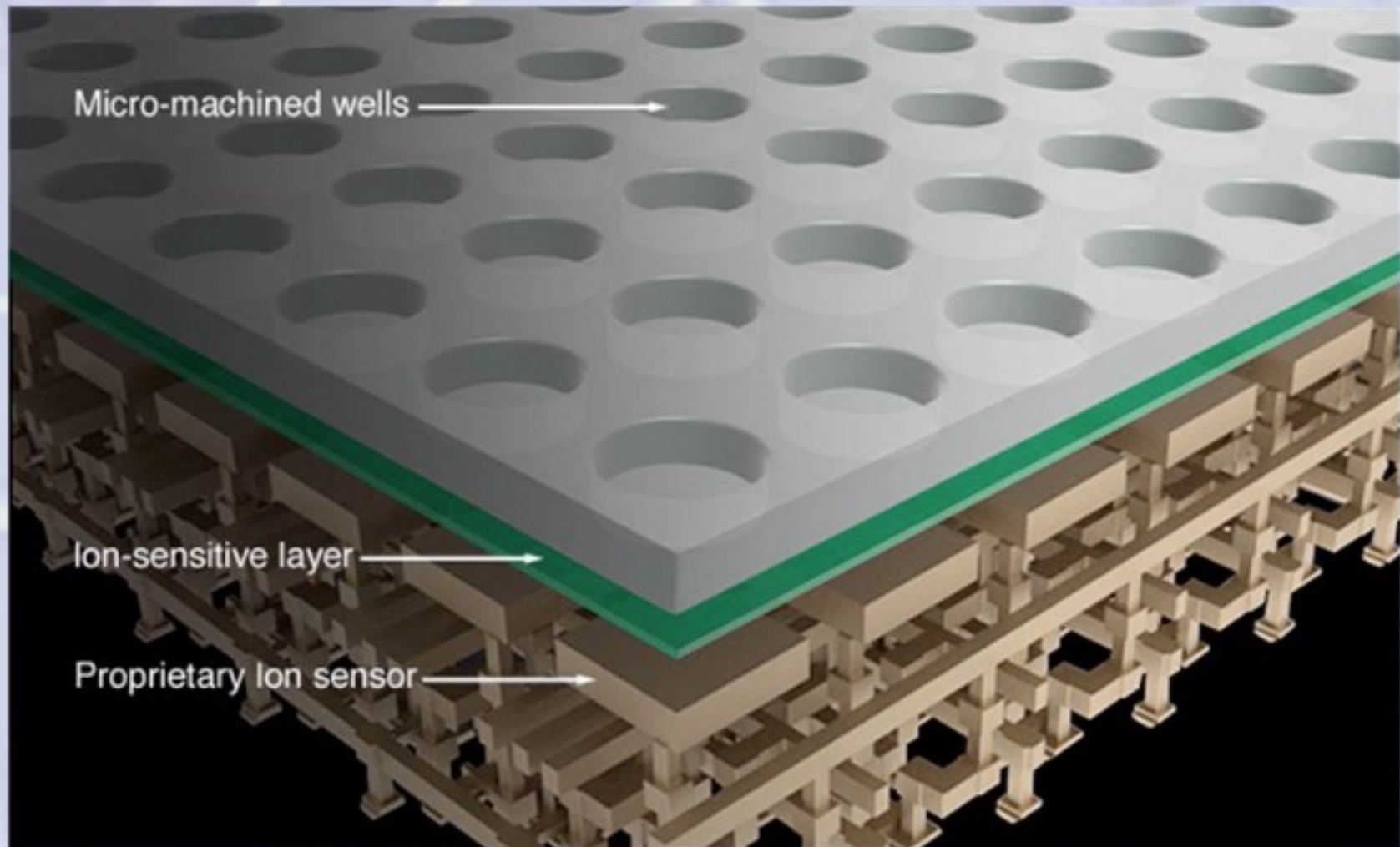
Output





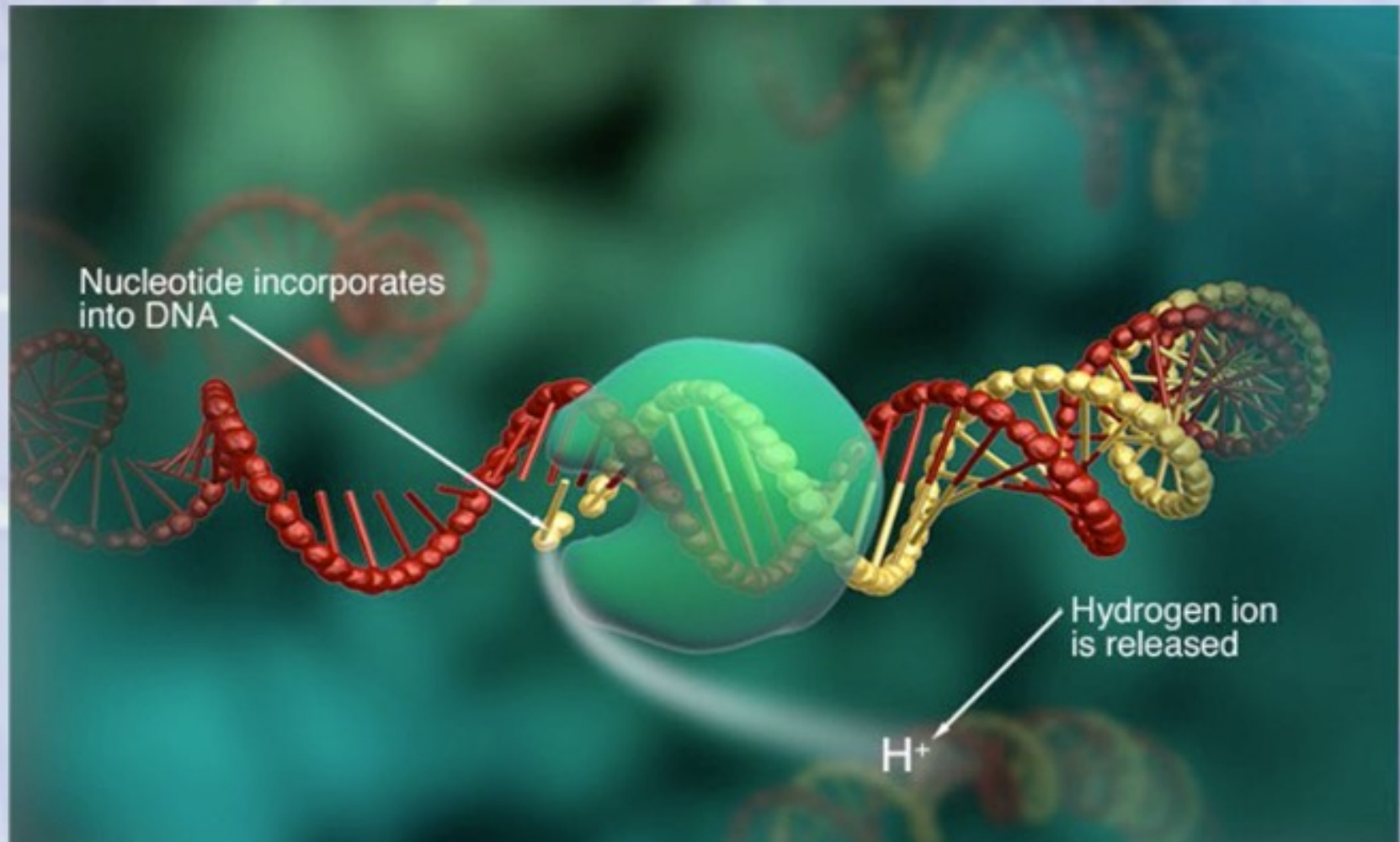
Ion Torrent

An Ion Torrent Chip

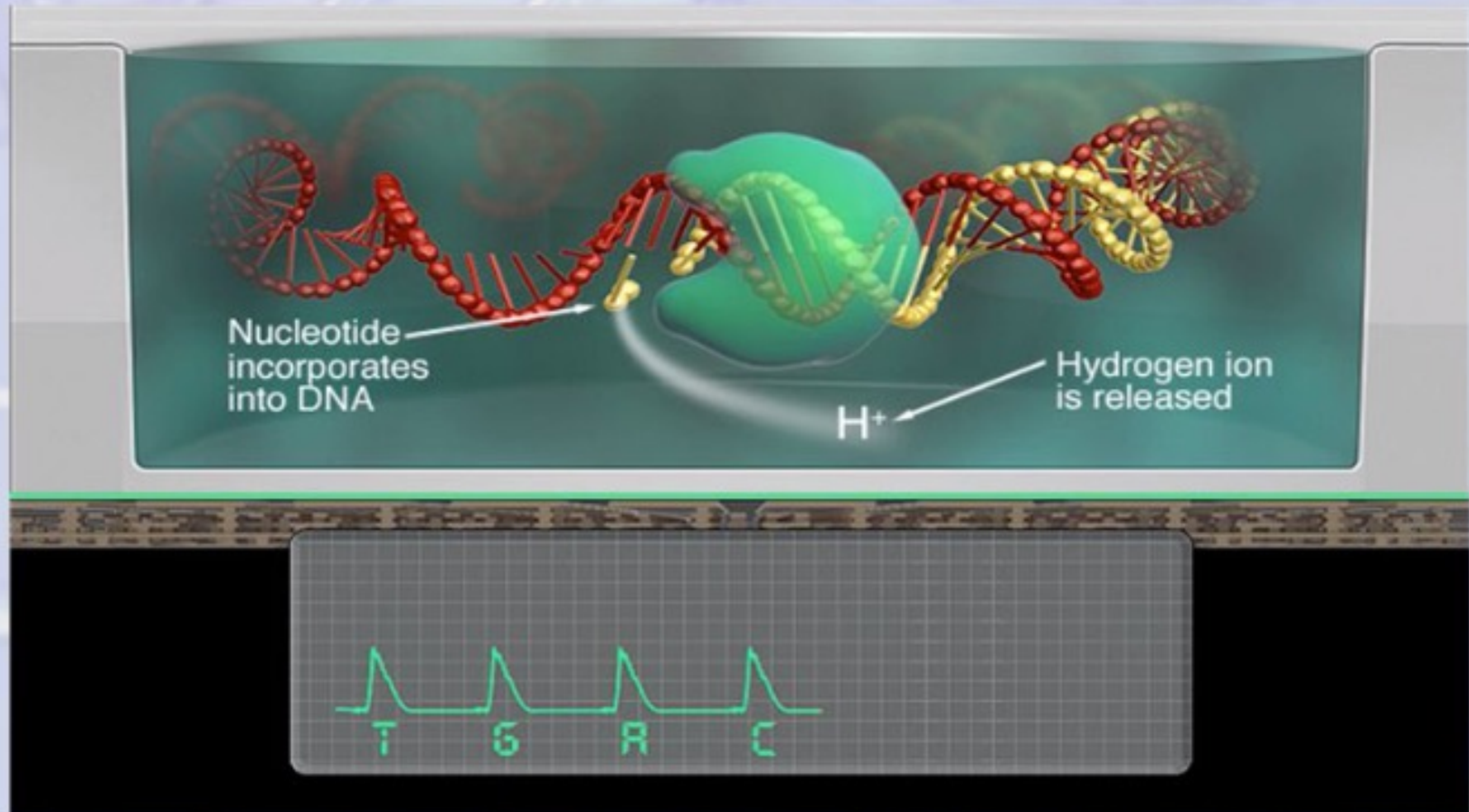


From Ion Torrent promotional material

When a nucleotide is incorporated into a strand of DNA by a polymerase, a hydrogen ion is released



Ion Torrent Sequencing Process



From Ion Torrent promotional material

As in 454, nucleotides are washed over the nascent strand in a prescribed sequence. Each time a nucleotide is incorporated, hydrogen ions are released and detected.

Newest Machine – Ion Proton

- \$150K per machine
- Ion Proton I chip has 165 million sensors
 - Intended for exomes
- Ion Proton II chip has 660 million sensors
 - 50X more than 318 chip
 - Claim \$1K genome this year



SOLID SEQUENCING

Part Four

The SOLiD System: Next-Generation Sequencing

Overview of the SOLiD System –

- **Scalable**
- **Accurate**
Ultra High Throughput
- **Flexible**
- **Mate Pairs**

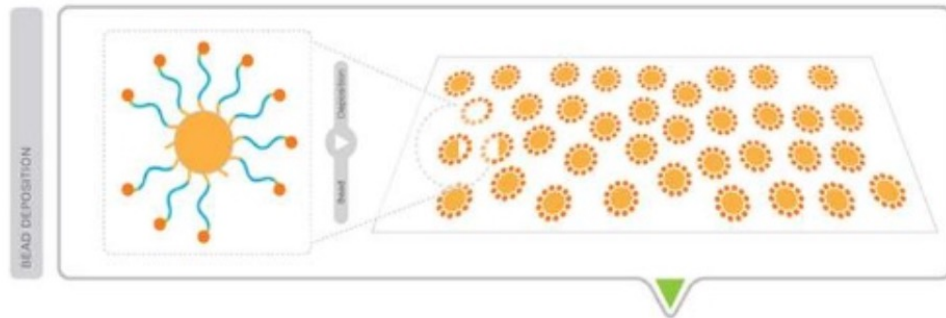
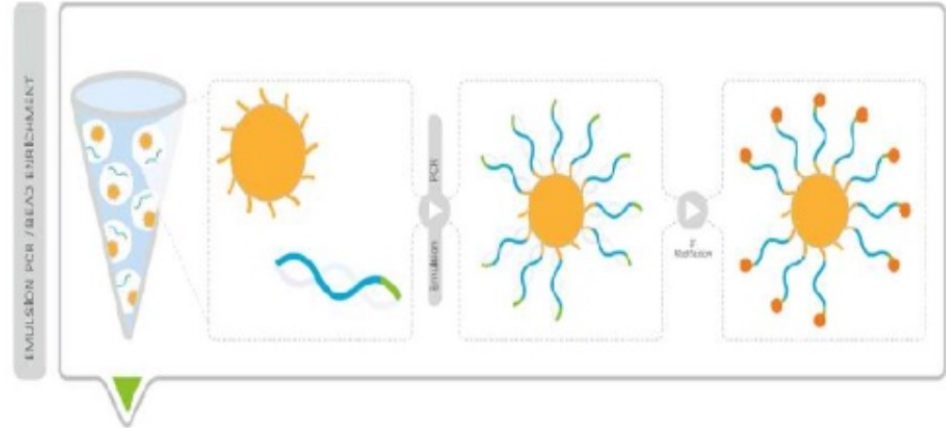
Instruments:

- Applied Biosystems SOLiD System Sequencer
- Covaris S2 Sonicator
- Applied Biosystems 9700 Thermal Cycler
- Hydroshear

SOLiD

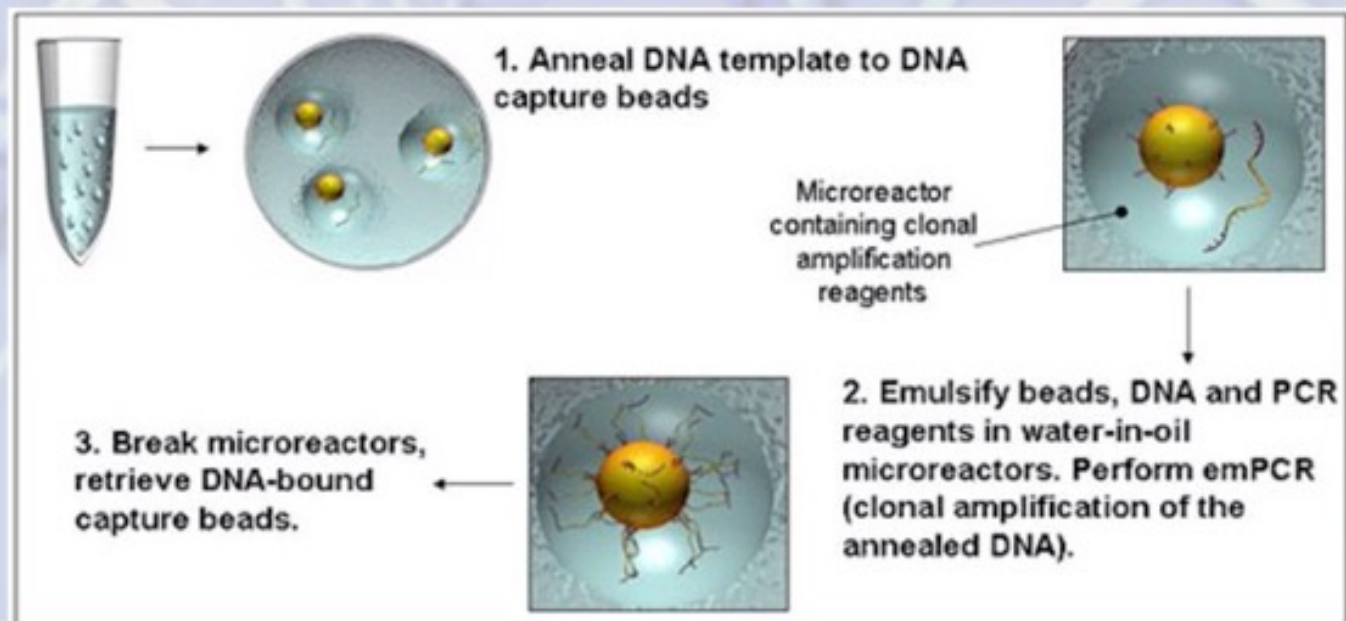
(support oligonucleotides)

- Sequencing by Oligo/Ligation and Detection.
- Steps
 - **Library Preparation**
 - two types of libraries sequencing-fragment or mate-paired are prepared.
 - **Emulsion PCR/Bead Enrichment**
 - amplification of template fragments is done in same manner as 454.
 - **Bead Deposition**
Deposit 3' modified beads onto a glass slide.

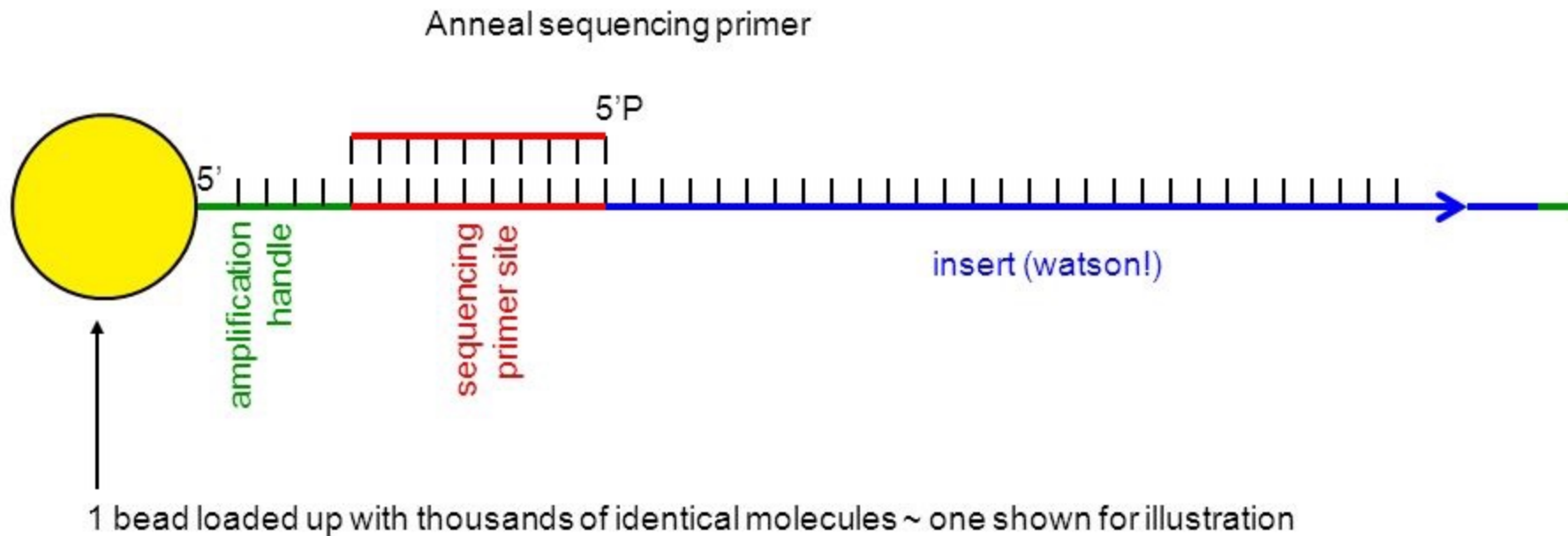


Emulsion PCR

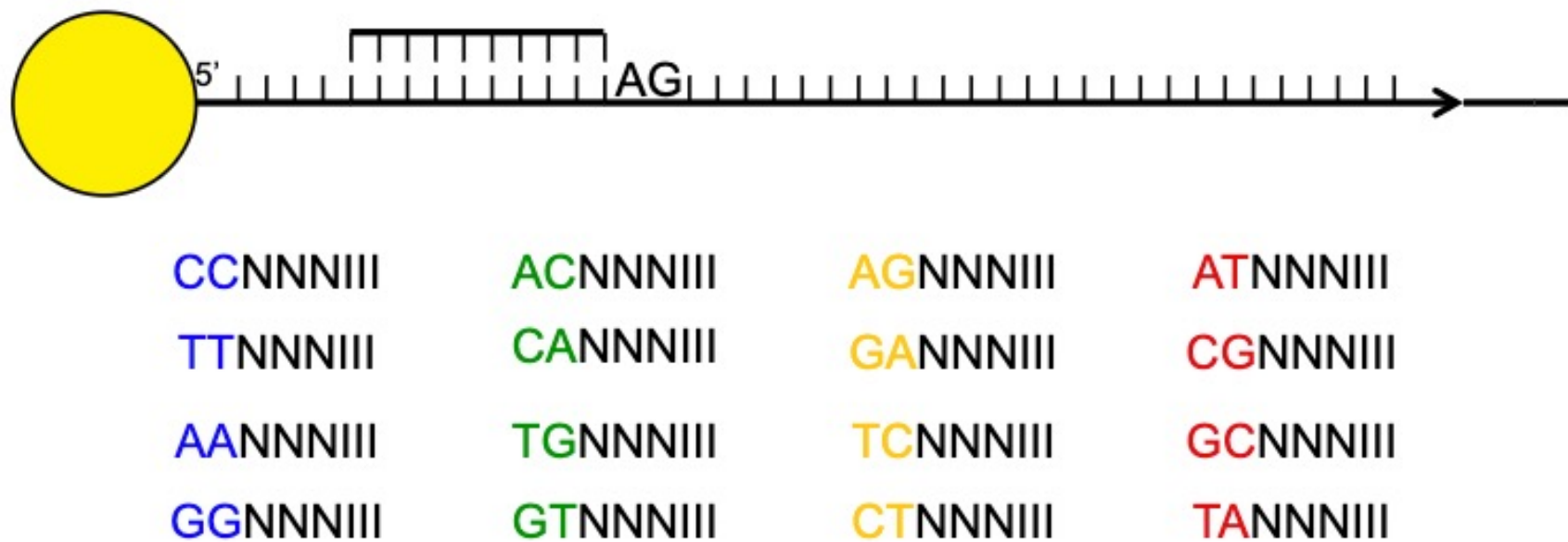
- Emulsion PCR isolates individual DNA molecules along with primer-coated beads in aqueous droplets within an oil phase. A polymerase chain reaction (PCR) then coats each bead with clonal copies of the DNA molecule. The bead is immobilized for sequencing.



SOLiD Sequencing

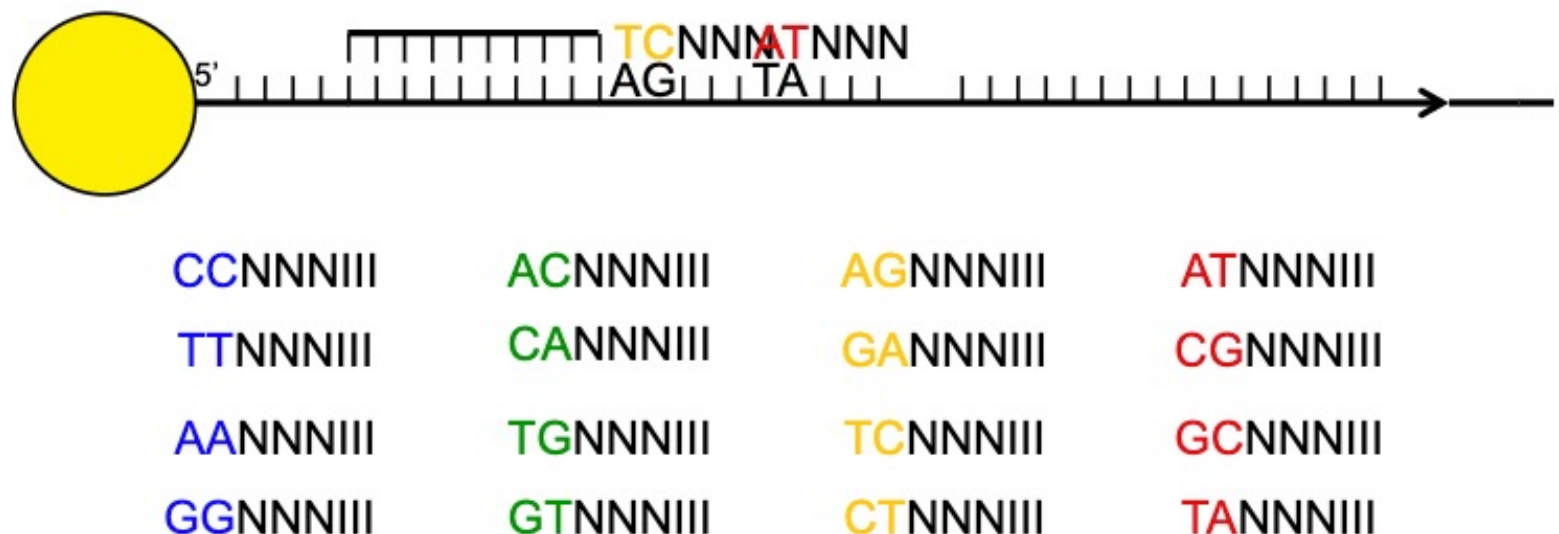


SOLiD Sequencing



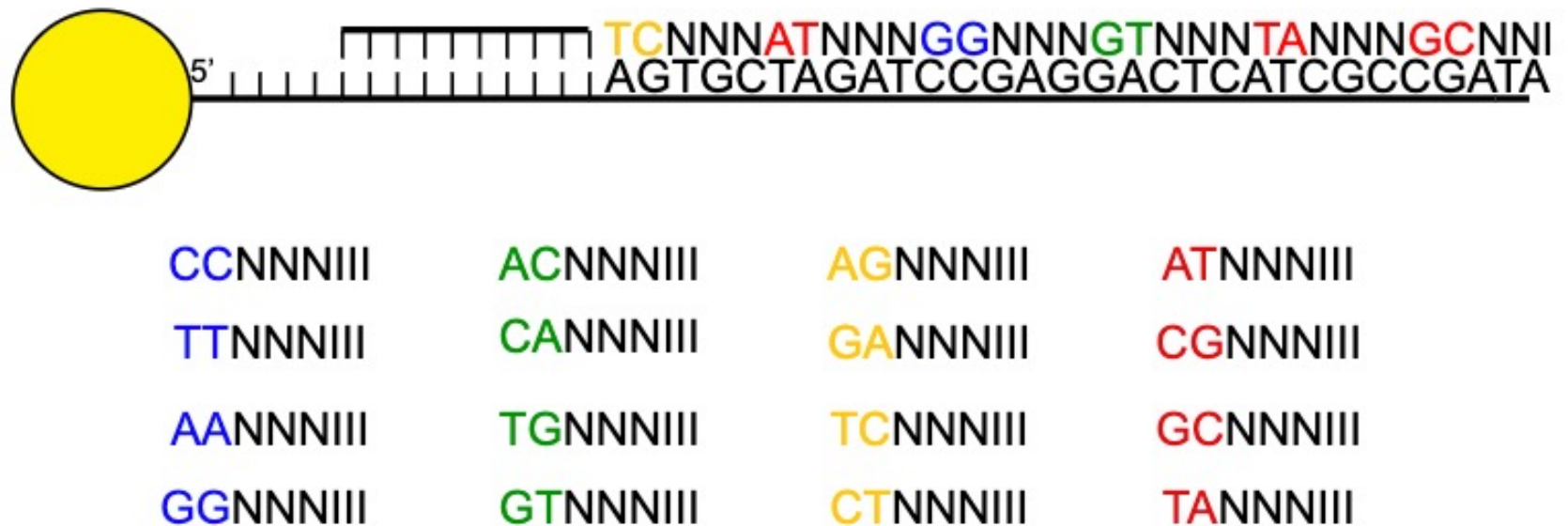
Sixteen probes, but only four colors

SOLiD Sequencing



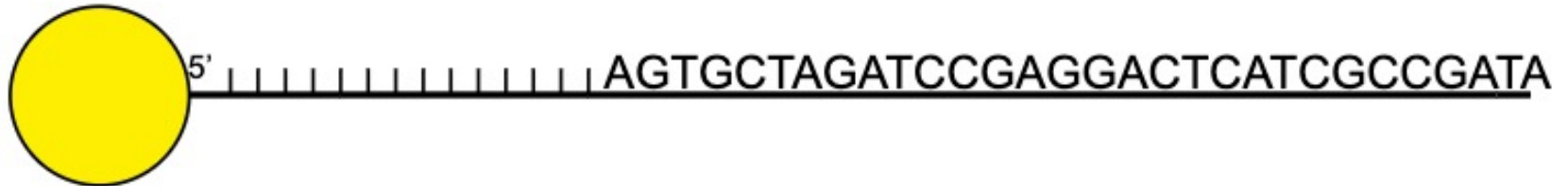
SOLiD Sequencing

After five cycles

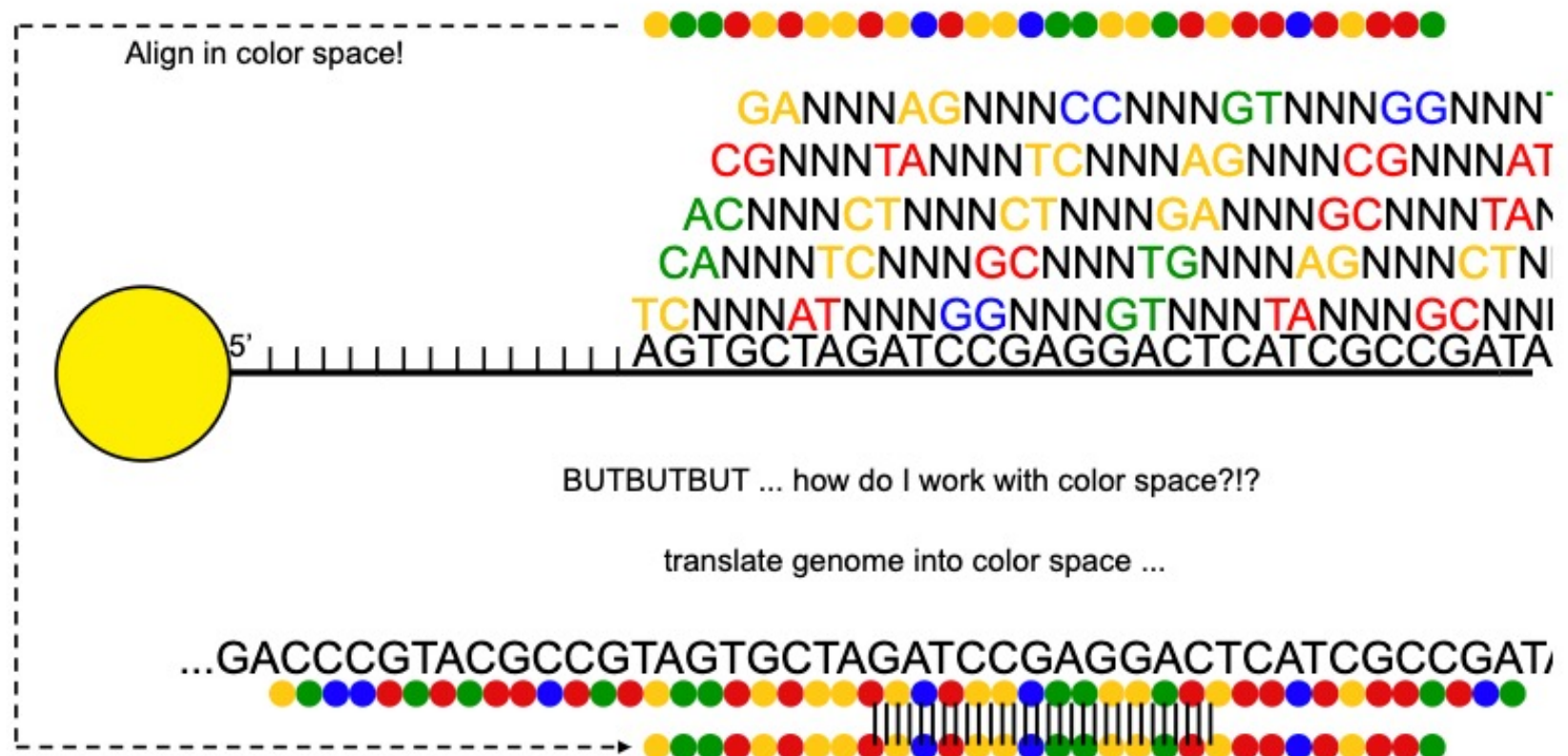


SOLiD Sequencing

“Reset”: Melt off previous extension products
and repeat five cycles with offset of one base



SOLiD Sequencing



Di-Base Colour coding system

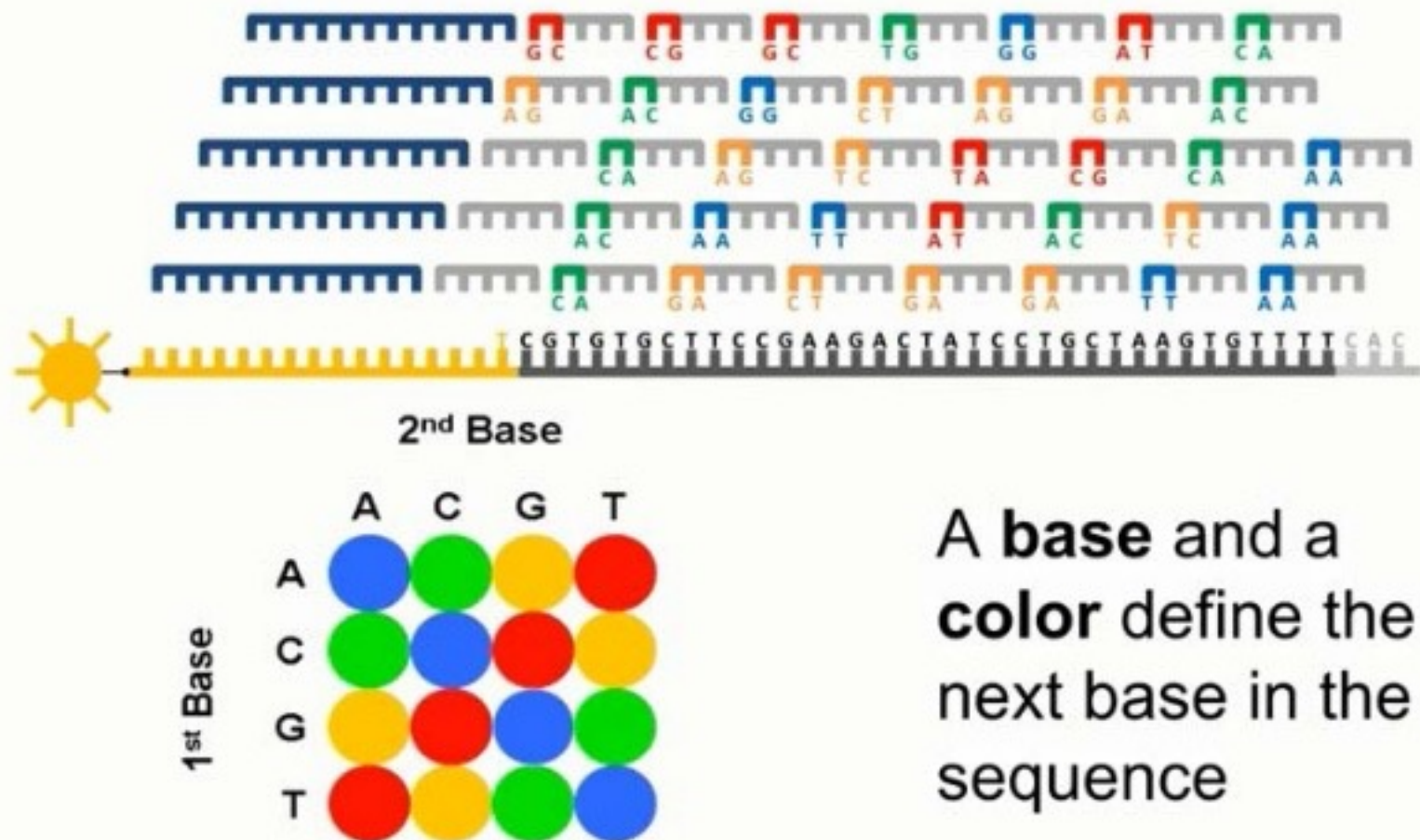
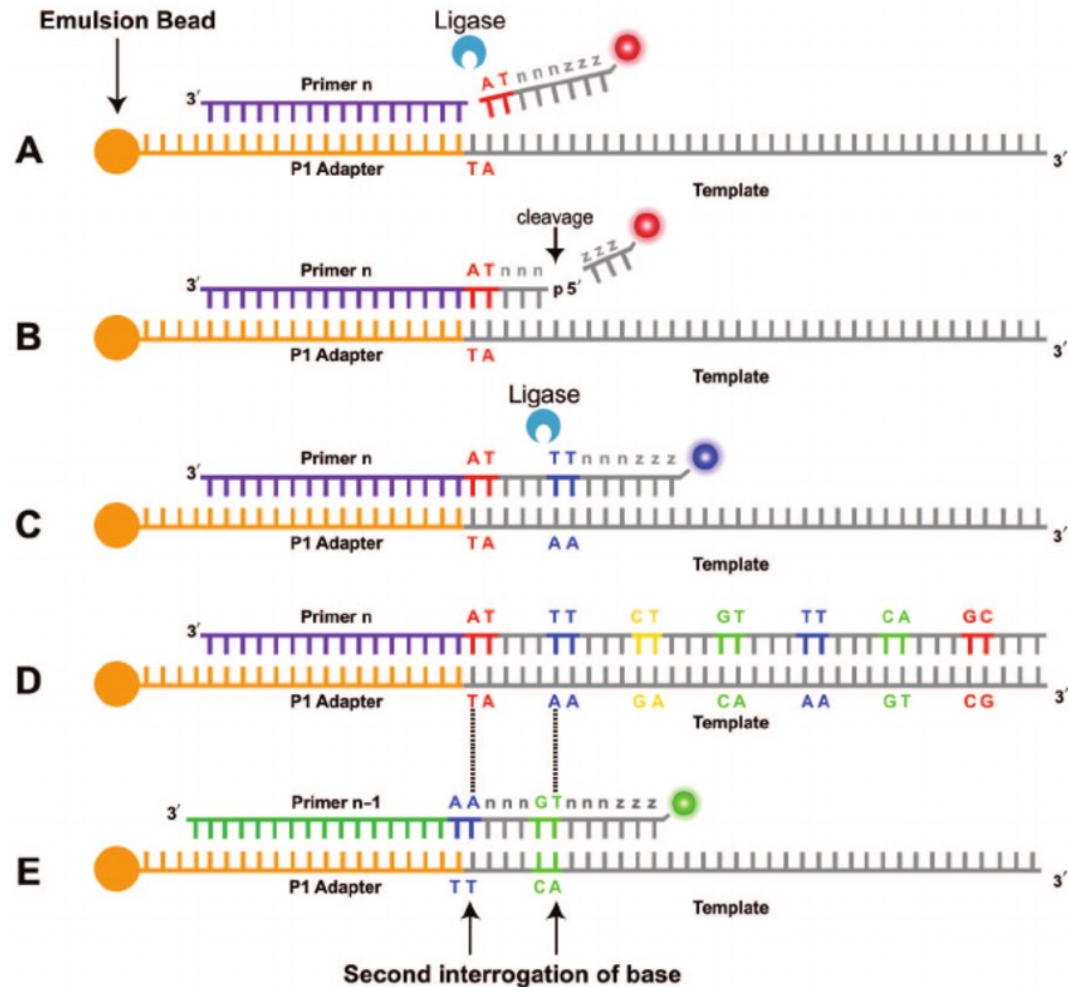
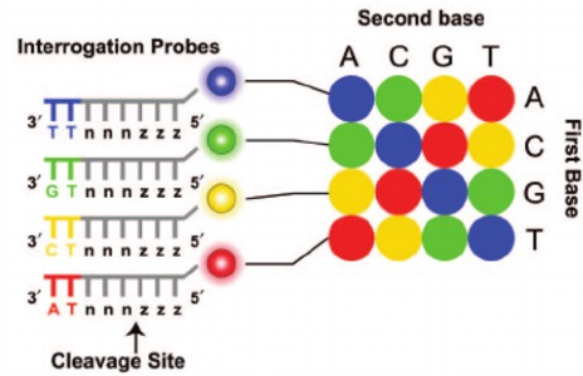
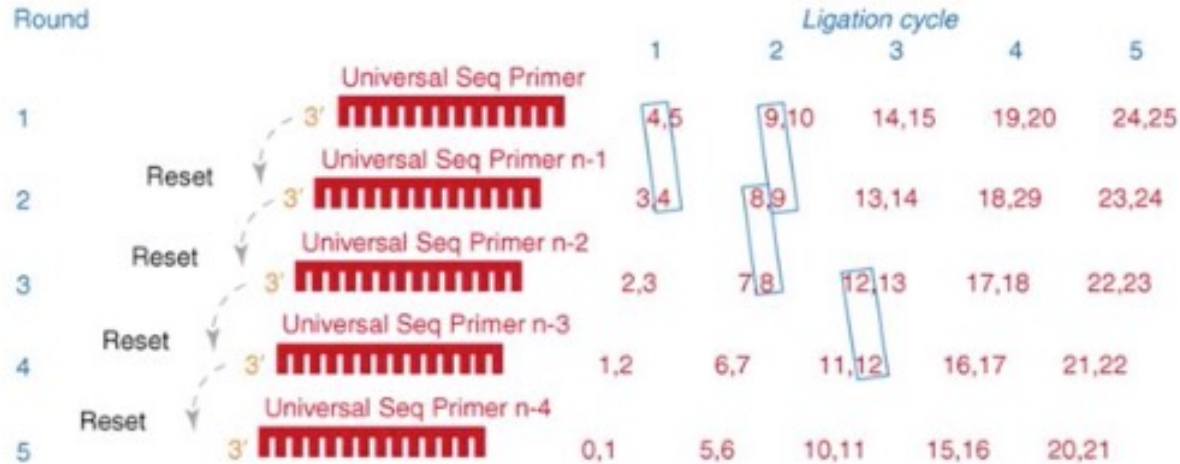
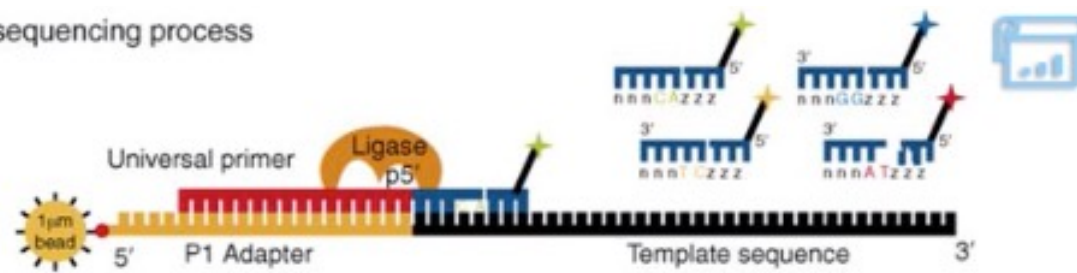


Fig: Di-base colour coding system developed for SOLiD sequencing only [5].

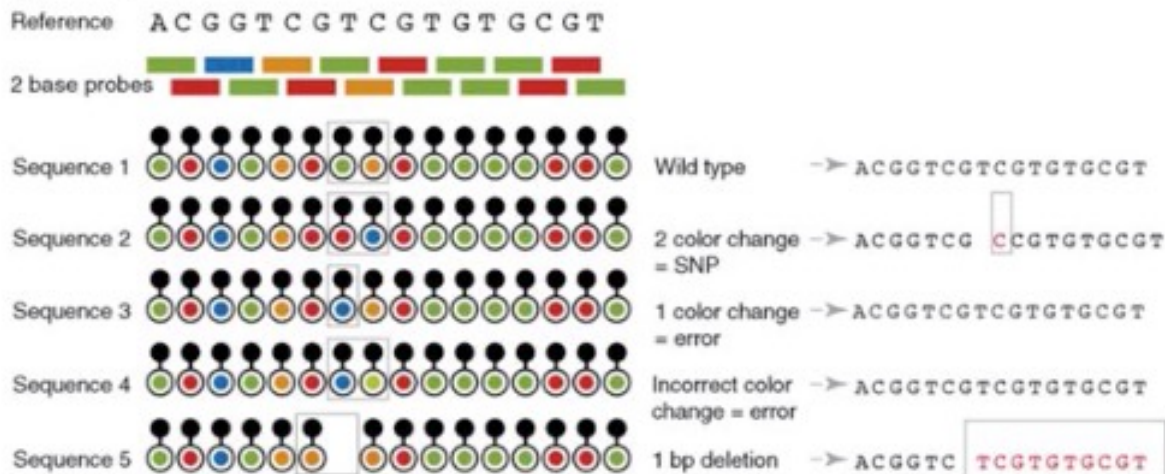
Color-Space Coding

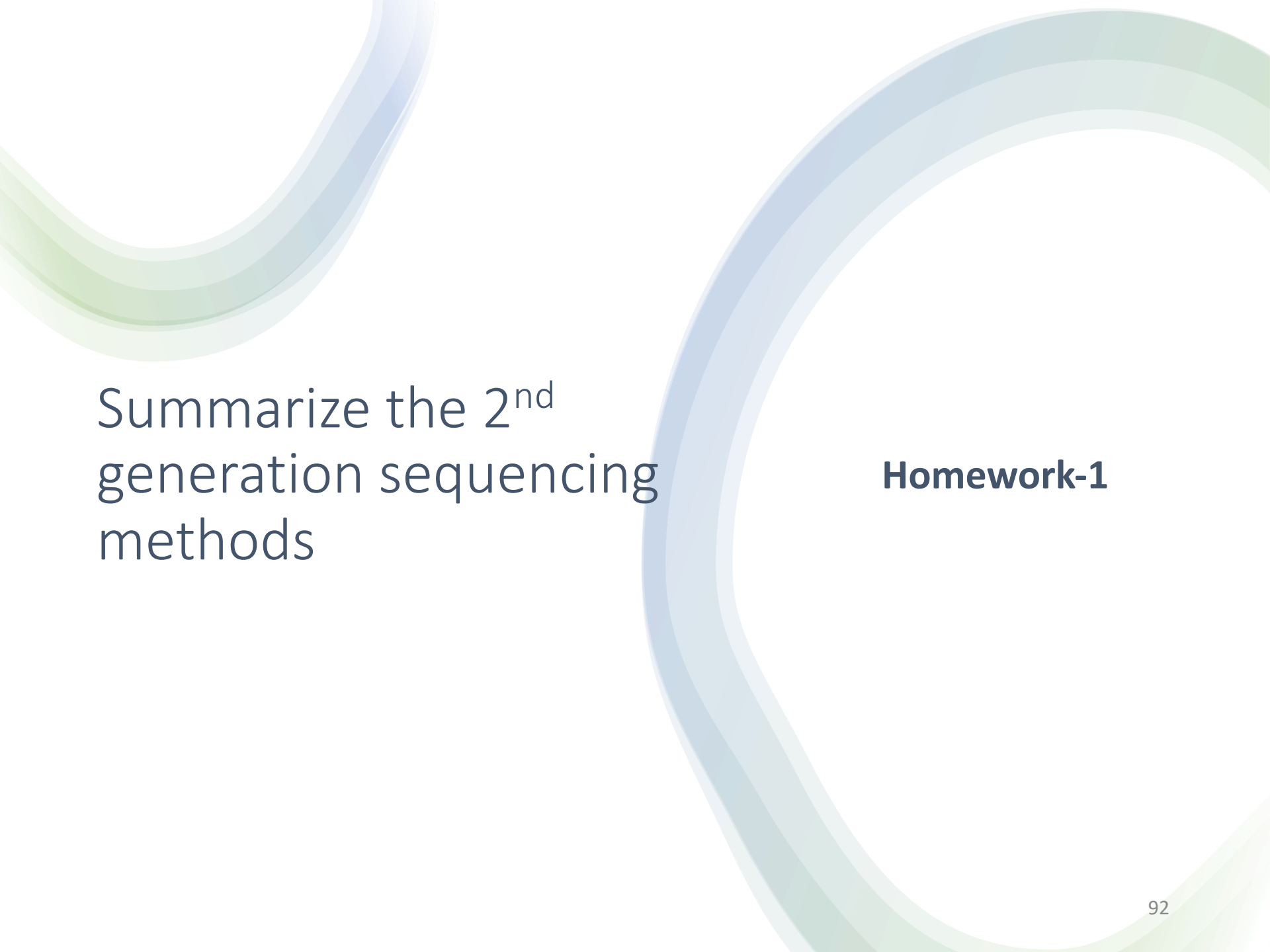


(a) Solid sequencing process



(b) Principles of two base encoding





Summarize the 2nd
generation sequencing
methods

Homework-1